Daily Briefing - October 28, 2026

Your Daily Tech & Programming Digest

Wednesday, October 28, 2026

1000	125116	1155	37
ARTICLES	WORDS	MIN READ	SOURCES

📰 Today's Top Stories

Are meta-scientists ignoring philosophy of science?

👤 noreply@blogger.com (Daniel Lakens) 📅 2025-07-04 📖 8 min 📝 1681 words

TWENTY PERCENT STATISTICIAN

Summary: <p>Are meta-scientists ignoring philosophy of science (PoS)? Are they re-inventing the wheel? A recent panel at the Metascience conference engaged with this question, and the first sentence of the abstract states “Critics argue t...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/07/are-meta-scientists-ignoring-philosophy.html

Type S and M errors as a “rhetorical tool”

👤 noreply@blogger.com (Daniel Lakens)
📅 2025-09-28 📖 17 min 📝 3572 words

TWENTY PERCENT STATISTICIAN

Summary: Update 30/09/2025: I have added a reply by Andrew Gelman below my original blog post. We recently posted a preprint criticizing the idea of Type S and M errors (https://osf.io/2phzb_v1). From our abstract: “While these concepts have been pr...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/09/type-s-and-m-errors-as-rhetorical-tool.html

Why we should stop using statistical techniques that have not been adequately vetted by experts in psychology

👤 noreply@blogger.com (Daniel Lakens)
📅 2025-10-29 📖 27 min 📝 5516 words

TWENTY PERCENT STATISTICIAN

Summary: In a recent post on Bluesky, where Richard Morey reflects on a paper he published with Clintin Davis-Stober that points out concerns with the p-curve method (Morey & Davis-Stober, 2025), he writes: <p cla...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/10/why-we-should-stop-using-statistical.html

ERP Decoding for Everyone: Software and Webinar

👤 Steve Luck 📅 2023-06-23 📖 2 min 📝 420 words ERP BOOT CAMP

Summary: You can access the recording h strong>.
You can access the final PDF of the slides <a href="https://ucdavis.box.com/s/f...

🔗 **Read full article:**
https://erpinfo.org/blog/2023/6/23/decoding-webinar

Important Changes to the 2024 ERP Boot Camp

👤 Steve Luck 📅 2024-03-05 📖 2 min 📝 444 words ERP BOOT CAMP

Summary: We are disappointed to announce that we will not be holding a regular 10-day ERP Boot Camp this summer. We have held Boot Camps nearly every summer since 2007, supported by a series of generous grants from NIMH that allowed us to provide scholarships for all attendees. Unf...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/5/changes-to-the-2024-erp-boot-camp

Standardized Measurement Error now available in BrainVision Analyzer and MNE-Python

👤 Steve Luck 📅 2025-12-19 📖 1 min 📝 253 words ERP BOOT CAMP

Summary: <p class="">A few years ago, we developed a universal metric of data quality for averaged ERPs called the standardized measurement error (click here for more information or watch our <a href="https://www.youtube.com/watch?v=tEKsx0p53rs" t...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/12/19/sme-now-available-in-bva-and-mne-python

Transcranial Focused Ultrasound: A Transformative Tool for Intracranial Ablation, Drug Delivery, and Neuromodulation

📅 2025-11-11 📖 1 min 📝 136 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Transcranial focused ultrasound (tFUS) is an emerging neuromodulation and therapeutic technology offering noninvasive, submillimeter precision for targeting deep brain structures. Unlike transcranial magnetic stimulation (TMS) and transcranial electric stimulation (tES), which are limited by depth-f...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11241049

Network Models of Neurodegeneration: Bridging Neuronal Dynamics and Disease Progression

📅 2026-01-16 📖 1 min 📝 196 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Neurodegenerative diseases are characterized by the accumulation of misfolded proteins and widespread disruptions in brain function. Computational modeling has advanced our understanding of these processes, but efforts have traditionally focused on either neuronal dynamics or the biological processe...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11357276

Optogenetics: Pinpoint Light on Precise Neuromodulation

📅 2025-11-06 📖 1 min 📝 137 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Optogenetics has emerged as a pivotal tool in neuroscience, enabling intricate modulation of targeted neurons within the nervous system. Despite its transformative potential, achieving high spatiotemporal resolution in neuromodulation remains a significant challenge, particularly in free-behaving an...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11230544

Emerging Noninvasive Approaches for the Suppression of Pathological Tremor

📅 2025-12-16 📖 1 min 📝 215 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Pathological tremor affects over 40 million people worldwide, significantly impairing daily activities and quality of life. Pharmacological treatments show limited efficacy, with up to 30% discontinuation rates, while surgical interventions like deep brain stimulation achieve significant tremor redu...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11301615

Assistive Trajectory Planning for Lower Limb Exoskeletons: Strategies From Laboratory-Optimized Gait to Environmentally-Adaptive Locomotion Through Multimodal Parameter Awareness

📅 2025-12-30 📖 1 min 📝 182 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Lower limb assistive exoskeletons (LLEs) show great potential to reduce metabolic cost, improve walking performance, and correct abnormal gait patterns. Among their core control architectures, assistive trajectory planning is key to determining system responsiveness and effectiveness under varying l...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11319164

Join the Mailing List

👤 dziura 📅 2025-12-10 📖 1 min 📝 13 words EMBS

> **Summary:** The post Join the Mailing List appeared first on IEEE EMBS.

🔗 **Read full article:**
https://embc.embs.org/2026/#new_tab

EMBS MentorLink: Fostering Careers Together

👤 Deidre Artis 📅 2026-08-18 📖 1 min 📝 14 words EMBS

> **Summary:** The post EMBS MentorLink: Fostering Careers Together appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/embs-mentorlink-fostering-careers-together/

Alterations in the populations of peripheral immune cells in the blood of a mouse model of progressive myoclonus epilepsy of Lafora type

📖 1 min 📝 17 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): L. Baños-Carrión, A. Campos-Rodriguez, M.A. Garcia-Gimeno, P. Sanz

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004580?dgcid=rss_sd_all

Addressing the between-group comparison problem: Detecting differences between correlation matrix populations due to single-variable perturbations for resting state fMRI

📖 1 min 📝 16 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Itamar Faran, Michael Peer, Shahar Arzy, Yuval Benjamini

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926005549?dgcid=rss_sd_all

A comparison of continuous-wave fNIRS with quantitative fMRI-derived indices of brain function in the visual cortex

📖 1 min 📝 42 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Giulia Rocco, Lucie Chalet, Elizabeth J. Fear, Sara Pomante, Francesca Graziano, Davide Di Censo, Manuela Carriero, Édouard Delaire, Fabrizio Esposito, Mauro Gianni Perrucci, Cosimo Del Gratta, David P...

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006051?dgcid=rss_sd_all

Circadian and sleep-wake modulation of EEG functional connectivity across brain states and Its associations with cognition in humans

📖 1 min 📝 24 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Alpar S Lazar, Zsolt I Lazar, Nayantara Santhi, June C Lo, John A Groeger, Derk-Jan Dijk

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006166?dgcid=rss_sd_all

Cortical pyramidal tract neurons constrain pathological dopaminergic motor responses

👤 Atudorei, M., Olsson, G. E. D., del Agua Villa, C., Corcea, C., Gether, U., Cenci, M. A., Siebner, H., Rickhag, M.

📅 2026-10-28 📖 1 min 📝 148 words BIORXIV NEUROSCIENCE

Summary: Pyramidal tract (PT) neurons constitute a major cortical output pathway to subcortical structures, yet their contribution to parkinsonian and dyskinetic motor states remains unclear. Here, we show that chemogenetic activation of motor cortical PT neurons selectively confines pathological dopaminergi...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.27.758468v1?rss=1

Activity dependent regulation of Angiomotin modulates dendritic complexity in hippocampal neurons

👤 Krueger, S. R., Fawcett, J. P., Wigerius, M.

📅 2026-10-28 📖 1 min 📝 151 words

BIORXIV NEUROSCIENCE

Summary: Neuronal activity alters synaptic protein composition, which is important for maintaining dendritic complexity. Multiple signaling networks have been implicated in these processes, but how increased neuronal activity regulates their expression remains incompletely understood. The long isoform of Ang...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.761988v1?rss=1

Post-intervention high-beta synchronization tracks stable-manifold alignment during stick-balancing learning

👤 Shimokado, H., Hayashi, K., Nomura, T.
📅 2026-10-28 📖 1 min 📝 226 words

BIORXIV NEUROSCIENCE

Summary: Learning to stabilize an unstable object may depend not only on the accuracy of active corrective interventions but also on when to suspend intervention and allow the object's intrinsic dynamics to evolve. We investigated learning-related changes in control dynamics and associated beta-band electroe...

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https://www.biorxiv.org/content/10.64898/2026.10.27.762123v1?rss=1

Claustrum Morphometry Reveals Spatially Heterogeneous Alterations in 22q11DS

👤 Althauser, A. K., Delavari, F., Latreche, C., Van De Ville, D., Eliez, S.
📅 2026-10-28 📖 1 min 📝 228 words

BIORXIV NEUROSCIENCE

Summary: The human claustrum is a small, irregular, and complexly positioned structure, yet it carries long range projections to most cortical regions responsible for executive functioning. Prior structural MRI research has largely relied on global volume as the metric of its morphology. Such measures may ob...

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https://www.biorxiv.org/content/10.64898/2026.10.27.762130v1?rss=1

Variability and Landmark Quality in Anatomical Co-Registration in Neuronavigation for Brain Stimulation

👤 Ebrahimi, Z., Farahmandrad, M., Seyedi, B., Xie, Z., Di Martino, M., Peterchev, A. V., Kammer, T., Goetz, S. M.

📅 2026-10-28 📖 1 min 📝 212 words BIORXIV NEUROSCIENCE

Summary: Accurate neuronavigation depends not only on the technical performance of the tracking system but also on the reproducibility with which anatomical landmarks are physically localized. We quantified operator- and landmark-specific variability in repeated facial and cranial landmark localization in 19...

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https://www.biorxiv.org/content/10.64898/2026.10.24.761558v1?rss=1

Birth weight, cognitive reserve and outcomes after first-episode psychosis

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s44220-026-00751-0

Cellular-resolution three-photon ultra-deep imaging (TrUDI) of neuronal population activity at 2-mm depth in the mammalian brain

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41592-026-03251-2

ADP-ribose induces neuronal ferroptosis through ENT2-mediated nucleotide rewiring

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41589-026-02355-2

The grand plan to transform psychiatry — by personalizing mental-health care

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/d41586-026-03391-7

Resting-state electroencephalographic characteristics in patients with atopic dermatitis accompanied by chronic pruritus

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-76915-w

Computational modeling and preclinical evaluation support targeting somatic instability for Huntington's disease treatment

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-78769-2

Editorial: Women in interacting minds and brains II

👤 Anna Ilona Roberts 📅 2026-10-26 📖 1 min 📝 0 words FRONTIERS HUMAN NEUROSCIENCE

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.2001078

Imaging findings and clinical outcomes in neonates with isolated subependymal echogenicity on cranial ultrasound: a retrospective cohort study

Xuehua Zhang 2026-10-28 1 min 255 words FRONTIERS HUMAN NEUROSCIENCE

Summary: ObjectiveTo describe clinical and imaging findings in neonates with isolated subependymal echogenicity on cranial ultrasound and examine associations with recorded mortality and cerebral palsy.MethodsThis retrospective single-center cohort included 317 neonates admitted between January 2015 and Dece...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1945002

Effects of non-invasive brain stimulation combined with exercise training on pain and physical function in knee osteoarthritis: a systematic review and meta-analysis

Zhaoxiang Meng 2026-10-28 1 min 345 words FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundNon-invasive brain stimulation (NIBS) combined with exercise training (ET) is a promising treatment for knee osteoarthritis (KOA), but its effects on pain and physical function lack systematic evaluation.PurposeTo evaluate the effects of NIBS combined with ET on pain and physical function ...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1902298

Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-16 1 min 222 words FRONTIERS NEUROROBOTICS

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1925089

Is the oxytocin system a keystone in human wellbeing?

Hans P. Nazarloo 2026-10-28 1 min 149 words FRONTIERS NEUROSCIENCE

Summary: It has also taken over a century for science to fully realize that oxytocin is far more than a “female reproductive hormone.” Both the measurement and interpretation of the biochemical and functional properties of oxytocin have raised controversy and remain works in progress. However, oxytocin is no...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1989082

Asymmetric encoder-decoder U-net with hybrid attention and residual dilated convolution for brain tumor segmentation

Ganqin Du 2026-10-28 1 min 187 words FRONTIERS NEUROSCIENCE

Summary: InroductionThe accurate and automatic segmentation of brain tumors from magnetic resonance imaging is regarded as essential for clinical diagnosis and surgical planning. Nevertheless, the concurrent extraction of both local fine details and global contextual cues remains challenging, primarily due t...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1999080

Effects of auditory sensory stimulation on consciousness and physiological parameters among individuals with disorders of consciousness: a systematic review and meta-analysis

Takamichi Taniguchi 2026-10-28 1 min 339 words FRONTIERS NEUROSCIENCE

Summary: ObjectiveTo systematically evaluate the effects of auditory sensory stimulation (SS) on consciousness levels and physiological parameters in patients with disorders of consciousness (DOC). Furthermore, this study aimed to explore intervention characteristics based on the frequency, intensity, time, ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1879145

Source-resolved electroencephalogram microstates reveal opposing prefrontal-posterior dynamics and a candidate temporal-to-precuneus pathway in disorders of consciousness

Hua Yuan 2026-10-28 1 min 149 words FRONTIERS NEUROSCIENCE

Summary: IntroductionObjective and circuit-level biomarkers of consciousness are urgently needed for disorders of consciousness.MethodsWe recorded high-density electroencephalography from 33 patients with disorders of consciousness (17 vegetative state/unresponsive wakefulness syndrome, 16 minimally consciou...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.1995232

De novo POLR2A p.(L1127F) variant presenting with neurodevelopmental regression and dystonia, with favorable response to low-dose dopamine therapy: a case report and literature review

Xulai Shi 2026-10-28 1 min 247 words FRONTIERS NEUROSCIENCE

Summary: BackgroundNeurodevelopmental disorder with hypotonia and variable intellectual and behavioral abnormalities (NEDHIB) is a rare autosomal-dominant condition caused by pathogenic POLR2A variants. It typically presents in infancy with hypotonia, developmental delay, and multisystem manifestations. Fewe...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.1997676

Development and feasibility testing of the taVNS adverse events questionnaire: a multistage methodological study

Karsten Witt 2026-10-28 1 min 301 words FRONTIERS NEUROSCIENCE

Summary: Transcutaneous auricular vagus nerve stimulation (taVNS) is a non-invasive neuromodulatory therapy gaining interest for its neural mechanisms and therapeutic effects. Although taVNS is considered safe, adverse event (AE) reporting remains inconsistent, lacking standardized procedures for assessment ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1974496

Psychosocial distress is a stronger correlate of psychiatric symptom burden than cannabis dependence in young adults: a functional connectome and structural equation modeling study

Jacob Levman 2026-10-28 1 min 343 words FRONTIERS NEUROSCIENCE

Summary: IntroductionCannabis is widely used among young adults and often viewed as a primary driver of adverse psychiatric and neural outcomes, yet the psychosocial and socioeconomic context in which use occurs is rarely modeled jointly with neural and behavioral measures.MethodsUsing the Human Connectome P...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1922823

Assessing consciousness by near-nerve electrostimulation in intensive rehabilitation

Piergiuseppe Liuzzi, Sara Ballanti, Andrea Mannini, Bahia Hakiki, Rachele Burali, Tanita Toci, Maria Chiara Carrozza, Calogero Maria Oddo and Antonello Grippo

2026-10-15 1 min 172 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Consciousness diagnosis in prolonged disorders of consciousness (pDoC) relies on the Coma Recovery Scale–Revised, yet consciousness or its potential for recovery may escape behavioural detection. We tested whether near-nerve stimulation (NNS) of the median nerve, compared to standard surf...

Read full article:
https://iopscience.iop.org/article/10.1088/1741-2552/aea645

Prioritized nonlinear modeling of shared dynamics across neural populations

Trisha Jha, Dong Song and Maryam M Shanechi

2026-10-15 1 min 284 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Advances in neural recording technology enable simultaneous measurements across multiple brain regions, providing new opportunities to study inter-regional interactions. However, several challenges arise when developing nonlinear dynamical models of cross-region interactions. In particula...

Read full article:
https://iopscience.iop.org/article/10.1088/1741-2552/ae9bee

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028064527&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 2025-06-26 1 min 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 2025-08-13 1 min 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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https://pubmed.ncbi.nlm.nih.gov/40800958/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028064527&v=2.20.1

Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800964/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028064527&v=2.20.1

ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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https://pubmed.ncbi.nlm.nih.gov/41358599/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028064527&v=2.20.1

Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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https://pubmed.ncbi.nlm.nih.gov/41676644/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028064527&v=2.20.1

Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41678848/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028064527&v=2.20.1

The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41703344/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028064527&v=2.20.1

EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

🔗 **Read full article:**
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Fidelity-oriented pelvic CBCT-to-CT synthesis using an anatomical information-attention diffusion model

Kunyuan Xu 2026-10-27 1 min 63 words LOW VISION

Summary: Cone-beam computed tomography (CBCT) is widely used in image-guided radiotherapy and adaptive radiotherapy, but its direct quantitative use is limited by scatter artifacts, low soft-tissue contrast, and inaccurate Hounsfield unit (HU) values. Diffusion models provide stable image generation capabili...

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Poros: Perception-Enabled Interactive Assistance for Appliance Use by Blind and Low Vision Users

Wei Liang 2026-10-27 1 min 62 words LOW VISION

Summary: Modern home appliances pose significant accessibility barriers for individuals with visual impairments, driven by increasing functional complexity and minimalist design trends. Drawing on formative interviews with blind and low vision (BLV) participants, we propose Poros, a novel system providing pe...

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DELUGE: Decomposed Entropy-coded Live Unstructured Geometry Exchange for Real-time Particle Streaming

👤 Takefumi Hiraki 📅 2026-10-27 📖 1 min 📝 74 words LOW VISION

Summary: Particle-based physics simulations, including fluids, smoke, and granular media, are fundamental to visual realism in immersive VR and AR. With the growing adoption of social VR and digital twins, demand is increasing for shared experiences in which multiple users interact with the same simulation i...

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A multi-attention mechanism-improved ResNet18 approach for lithology identification and classification

👤 Ruifeng Li 📅 2026-10-27 📖 1 min 📝 61 words LOW VISION

Summary: Lithology identification is vital for improving geological exploration accuracy, optimizing petroleum exploration schemes, and ensuring construction engineering safety. Traditional identification methods suffer from low efficiency, high cost, and heavy dependence on domain expertise. This study prop...

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An open-source platform for measuring optomotor responses in mice

👤 Shogo Soma 📅 2026-10-27 📖 1 min 📝 64 words LOW VISION

Summary: The optomotor response (OMR) is a valuable behavioral index for assessing visual function in basic research; however, commercially available OMR evaluation equipment is expensive and lacks flexibility, as proprietary software and fixed hardware geometry restrict custom stimulus protocols and system ...

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Communication patterns in home medication review for individuals with vision impairment: A qualitative study

👤 Woojin Park 📅 2026-10-27 📖 1 min 📝 37 words LOW VISION

Summary: CONCLUSIONS: In these consultations, professional dialogue supported verification and reassurance, while direct regimen review revealed unverified beliefs and gaps in medication knowledge. The findings suggest a role for accessible counseling that combines information provision with professional ver...

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Global, regional, and national burden of unintentional poisoning, 1990-2023: results from the Global Burden of Disease Study 2023

👤 GBD 2023 Unintentional Poisoning Collaborators

📅 2026-10-27 📖 1 min 📝 65 words

LOW VISION

Summary: BACKGROUND: Unintentional poisoning is a preventable cause of injury-related morbidity and mortality, yet global patterns, trends, and inequalities remain incompletely characterised. Within the Global Burden of Disease (GBD) framework, unintentional poisoning includes poisoning by carbon monoxide an...

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Directing Ru-based acidic oxygen evolution catalysts toward a universal dual-site pathway

Chuangang Hu 2026-10-28 1 min 62 words LOW VISION

Summary: The oxide path mechanism offers vision possibility from kinetics for the anodic oxygen evolution reaction, yet its universal activation remains elusive due to the difficulty of constructing globally appropriate interatomic distances. Here, we address this challenge in the integrated-screened Y(0.1)I...

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Biomimetic computational modelling of retinal ganglion cell responses to contact lens designs with refractive and opaque non-refractive features

Praveen K Bandela 2026-10-28 1 min 67 words LOW VISION

Summary: CONCLUSIONS: In this computational model, both BF and AR lens designs produced measurable changes in simulated RGC temporal activity relative to SV, with AR inducing greater and more configuration-independent modulation. These findings must be considered hypothesis-generating only; they are consiste...

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Growth, nutritional status, and physical fitness trajectories among primary and secondary school students and their associations with poor vision: a school-based study in Chongqing, China

👤 Yong Zhao
📅 2026-10-28 📖 1 min 📝 42 words

LOW VISION

Summary: CONCLUSION: From 2022 to 2026, growth and development indicators among primary and secondary school students in Yuzhong District improved, while poor vision remained prevalent. BMI and VCI trajectories were associated with poor vision, highlighting the importance of integrated monitoring and targete...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux 📅 2026-07-18 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

Xiaobing Yan 2026-09-09 1 min 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Combined Transcranial Direct Current Stimulation and Transcutaneous Cervical Vagus Nerve Stimulation for Anxiety and Depression: An Exploratory Pilot Study

👤 Maria-Manuela Apostol
📅 2026-10-27 📖 1 min 📝 60 words

TDCS TACS TRNS

Summary: Anxiety and depressive disorders are among the leading causes of disability worldwide, yet current treatments remain limited by modest efficacy and restricted accessibility; non-invasive neuromodulation techniques such as transcranial direct current stimulation (tDCS) and transcutaneous cervical vag...

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Non-Invasive Brain Stimulation Techniques and Effects on Neuromodulation and Athletic Performance in Combat Sports: A Systematic Review

👤 Bianca Miarka 📅 2026-10-27 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: CONCLUSIONS: tDCS may influence selected cognitive, neuromuscular, and fatigue-related outcomes in combat sport athletes; however, the direction and magnitude of these effects are inconsistent across studies. Given the substantial methodological heterogeneity and the low-to-very-low certainty of evi...

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Non-Invasive Brain Stimulation for Central Neuropathic Pain: An Exploratory Randomized Sham-Controlled Crossover Trial

👤 Eunhee Sohn 📅 2026-10-27 📖 1 min 📝 64 words TDCS TACS TRNS

Summary: Background/Objectives: Central neuropathic pain (CNP) is difficult to manage, and comparative evidence for non-invasive brain stimulation is limited. We explored the short-term efficacy and safety of repetitive transcranial magnetic stimulation (rTMS) and transcranial direct current stimulation (tDC...

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Noninvasive Neuromodulation for Cognitive Dysfunction in Parkinson's Disease: A Scoping Review

Reginald B O'Hara 2026-10-27 1 min 47 words TDCS TACS TRNS

Summary: CONCLUSIONS: Evidence specific to PD-MCI is limited and mixed. Several studies pairing stimulation with cognitive training reported favorable outcomes, but this pattern was inconsistent and does not establish superiority over stimulation alone. No VNS trial enrolled participants with confirmed PD-MC...

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Comparative Plastid Phylogenomics of 29 Newly Sequenced Bougainvillea Cultivars Resolves the Phylogenetic Placement of Double-Bracted Cultivars

Huiqi Zhao 2026-10-27 1 min 59 words TDCS TACS TRNS

Summary: Bougainvillea (Nyctaginaceae) cultivars are horticulturally prized worldwide for their vivid bracts; however, repeated interspecific hybridization and somatic mutations have obscured their maternal origins, particularly for double-bracted forms. Here, we sequenced and assembled the complete plastome...

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Anodal Transcranial Direct Current Stimulation Reduces Neuroinflammation and Cholinergic Dysregulation After Young Adult Traumatic Brain Injury

👤 Guven Akcay
📅 2026-10-27 📖 1 min 📝 61 words

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Summary: Traumatic brain injury (TBI) during childhood and adolescence can result in persistent motor and cognitive dysfunction due to secondary injury cascades involving neuroinflammation and disruption of neurotransmitter homeostasis. Although transcranial Direct Current Stimulation (tDCS) is a promising n...

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Assessing brain-behaviour coupling in non-invasive brain stimulation: Insights from reliable change indices and Monte Carlo simulations

👤 Alexander D Tang
📅 2026-10-27 📖 1 min 📝 64 words

TDCS TACS TRNS

Summary: Non-invasive brain stimulation (NiBS) studies frequently report exploratory brain-behaviour correlations following the application of NiBS. However, these analyses are typically underpowered and rely on ratio-based change scores with known statistical limitations. We addressed these limitations by p...

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The application of brain-computer interface combined with neuromodulation for stroke upper-limb rehabilitation: a systematic review and network meta-analysis

Zhongjian Luo
2026-10-28 1 min 42 words TDCS TACS TRNS

Summary: CONCLUSION: Some non-invasive MI-BCI combined interventions like neuromodulation, ROB, or FES might improve upper-limb function and daily activities in stroke patients. However, rankings don't imply clinical importance. Due to small samples, limited evidence, and differences, these findings need con...

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Comparison of the efficacy of neuromodulation techniques based on corticospinal excitability in improving exercise-induced fatigue in athletes: a systematic review and network meta-analysis

👤 Li Qing 📅 2026-10-28 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: CONCLUSION: TEAS combined with acupuncture may be associated with the most favorable SUCRA ranking, but this should be interpreted as exploratory for improving EIF in athletes. However, the overall quality of current evidence is low, and SUCRA rankings should be interpreted as exploratory. Future hi...

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Across established non-invasive neuromodulation and emerging deep-targeting strategies: a comparative bibliometric analysis of TI, TMS, tDCS, and tFUS/TUS/LIFU

👤 Shuang Wu
📅 2026-10-28 📖 1 min 📝 61 words

TDCS TACS TRNS

Summary: Non-invasive brain stimulation (NIBS) is expanding from conventional cortical modulation toward approaches capable of engaging deeper brain targets, but the literature structures of major modalities have rarely been compared within a common framework. We conducted a comparative bibliometric mapping ...

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Neural-Discursive Dynamics in Collaborative Spatial Problem Solving: Evidence from fNIRS Hyperscanning and Accountable Talk

👤 Xingguang Guo
📅 2026-10-27 📖 1 min 📝 65 words FNIRS

Summary: High-quality collaboration cannot be inferred simply from how much students talk or how strongly their brain activity is synchronized during joint work. This study examined relations among accountable talk, inter-brain synchrony (IBS), and group task performance in fifth-grade students' collaborativ...

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Probing Executive Function in the Toddler Data Desert: A Touchscreen fNIRS Study of Working Memory and ADHD-Related Symptoms

Caitie Busch 2026-10-27 1 min 60 words FNIRS

Summary: Attention-deficit/hyperactivity disorder (ADHD) is associated with early differences in attentional control and working memory (WM), yet relatively few paradigms are developmentally feasible across the toddler-to-preschool period while also distinguishing spatial from featural information use. Addre...

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Mapping the Global Landscape and Thematic Evolution of fNIRS-Gait Research: A Systematic Bibliometric and Network Visualization Analysis

Naime Uluğ 2026-10-27 1 min 20 words FNIRS

Summary: CONCLUSIONS: Establishing standardized preprocessing pipelines and short-separation regressions may prove pivotal for transitioning mobile neuroimaging into objective, closed-loop clinical neurorehabilitation.

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Self-Selected Versus Assigned Emotion Regulation in Pain Expectancy: Behavioral and fNIRS Evidence

👤 Ping Shi 📅 2026-10-27 📖 1 min 📝 63 words FNIRS

Summary: Background: Expectation is a powerful cognitive modulator of pain, but how different types of emotion regulation approaches interact with expectancy to influence pain processing remains unclear. The present study compared self-selected habitual strategies (cognitive reappraisal or deep breathing) wi...

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Comparative Effects of Acute HIIE and MICE on Executive Function, Prefrontal Oxyhaemoglobin Responses, and Related Biomarkers in Sedentary Young Adults: A Randomized Crossover Trial

👤 Xinyue Yang
📅 2026-10-27 📖 1 min 📝 34 words FNIRS

Summary: CONCLUSIONS: Both exercise sessions were followed by faster reaction times on selected tasks. The small sample, multiple analyses, and potential systemic contributions to fNIRS signals preclude firm conclusions about exercise-mode superiority or biological mediation.

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Cerebral Hemodynamic Patterns in People with Subjective Cognitive Decline: A Systematic Review Based on Functional Near-Infrared Spectroscopy

👤 Israel Contador
📅 2026-10-27 📖 1 min 📝 62 words FNIRS

Summary: Background/Objectives: Subjective cognitive decline (SCD) has been proposed as an early symptomatic manifestation of dementia/Alzheimer's disease. However, its associated brain patterns are not yet fully characterized. In recent years, functional near-infrared spectroscopy (fNIRS) has emerged as a c...

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Resting-State Functional Brain Connectivity and Neurodevelopmental Profiles in 3-Year-Olds with Congenital Heart Disease

👤 Anne Gallagher
📅 2026-10-27 📖 1 min 📝 67 words FNIRS

Summary: Background/Objectives: Congenital heart disease (CHD) is associated with altered brain development and an increased risk of neurodevelopmental difficulties, particularly in more severe forms of the condition. Although alterations in functional brain connectivity have been reported before and after t...

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Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan
📅 2026-10-27 📖 1 min 📝 60 words FNIRS

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

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Multimodal Sensing and Digital Biomarkers in Virtual Reality-Based Cognitive Rehabilitation: Clinical Validation, Adaptive Systems, and Real-World Translation

👤 Raffaele Ferri
📅 2026-10-27 📖 1 min 📝 58 words FNIRS

Summary: Virtual reality (VR) is evolving from a platform for cognitive exercises into an instrumented rehabilitation environment capable of simultaneously delivering treatment and quantifying behavior. This critical narrative review examines how behavioral, ocular, kinematic, physiological, and neurophysiol...

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Unpredictable human screams elicit anticipatory and phasic cortical responses that affect subsequent interpersonal space regulation

Alessandro Grecucci 2026-10-27 1 min 59 words FNIRS

Summary: Unpredictable threats can sustain vigilance states that bias perception, cognition, and behaviour beyond immediate stimulus processing. Human screams, as evolutionarily salient alarm signals, are especially effective in eliciting defensive responses, yet their cortical dynamics and consequences for ...

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EOG Artifact Removal from EEG Using a Gated WGAN with Multi-Scale and Multi-Domain Feature Extraction

Hao Tang 2026-10-27 1 min 61 words BRAIN COMPUTER INTERFACE

Summary: Background: Electroencephalogram (EEG) signals provide important information for brain-computer interfaces (BCIs) and the diagnosis of neurological disorders but are susceptible to electrooculographic (EOG) artifacts. Traditional methods often rely on multichannel data or relatively long windows, li...

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EEG Features of Cortical Intent for Urination and Defecation in Healthy Aging and Spinal Cord Injury: A Descriptive Feasibility Study for BCI Neurorehabilitation

👤 Jie Jia 📅 2026-10-27 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: Background/Objectives: Systematically elucidating the neural mechanisms of bowel and bladder control is crucial for developing closed-loop brain-computer interface (BCI) neurorehabilitation for patients with spinal cord injury (SCI). This study aims to characterize the spatiotemporal Electroencephal...

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The Development and Evolution of Substrate Materials for Brain-Computer Interfaces

👤 Yen Wei 📅 2026-10-27 📖 1 min 📝 58 words BRAIN COMPUTER INTERFACE

Summary: Brain-Computer Interfaces (BCIs) hold tremendous potential for neurological therapeutics and neuroprosthetics; however, their long-term clinical viability is fundamentally limited by chronic interfacial failure. This failure originates from the profound biomechanical mismatch between synthetic micro...

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Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan 📅 2026-10-27 📖 1 min 📝 60 words BRAIN COMPUTER INTERFACE

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

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Agentic EEG Pipeline Synthesis Using Large Language Models and the Model Context Protocol

👤 Katy Tarrit 📅 2026-10-27 📖 1 min 📝 56 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography analysis typically requires multistage processing pipelines for which configuration depends on recording characteristics and the intended analytical objective. Although automated pipelines improve reproducibility, they generally implement predefined processing sequences. We pr...

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Overt and Covert Speech-State Classification from Tripolar Concentric Ring Electrode EEG: A 4 mm and 10 mm Configuration Study

👤 Shoaib Memon 📅 2026-10-27 📖 1 min 📝 76 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) speech-state classification is a noninvasive method to identify the brain's overt and covert behavior. The EEG achieved through the scalp is contaminated by artifacts and volume conduction and suffers from poor spatial resolution. We present the first application of smal...

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The compositional encoding of hand-eye coordinated movements for single neurons in the posterior parietal cortex

👤 Richard A Andersen 📅 2026-10-27 📖 1 min 📝 77 words BRAIN COMPUTER INTERFACE

Summary: Human posterior parietal cortex (PPC) is known to play a role in hand-eye (H-E) coordination, yet the underlying encoding mechanisms remain uncertain. We recorded 412 single neurons across 11 sessions from motor cortex (MC; n = 251) and PPC (n = 161) in a single human participant performing an H-E c...

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NCEL: Convolution-Enhanced Lightweight Neural Architecture for Robust EEG Signal Decoding in Noninvasive Brain-Computer Interfaces

👤 Liangliang Hu
📅 2026-10-27 📖 1 min 📝 57 words

BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) is a cornerstone of brain-computer interfaces (BCIs) and clinical neuro-diagnostics due to its non-invasiveness and high temporal resolution. However, robust EEG decoding remains challenging, particularly for resource-constrained deployments. While lightweight architectu...

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DSGTANet: A semi-supervised dual-stream graph-temporal adaptation network for cross-subject continuous EEG emotion recognition

👤 Wang-Yang Xu
📅 2026-10-27 📖 1 min 📝 61 words

BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: Continuous emotion recognition using electroencephalogram (EEG) is a critical topic in brain-computer inter-faces (BCIs), with applications in mental disorder regulation and human-computer interaction. However, its practical appli-cation is significantly constrained by several key issues,...

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The application of brain-computer interface combined with neuromodulation for stroke upper-limb rehabilitation: a systematic review and network meta-analysis

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Psychomusicology: A resounding closing cadence.

2024-01-22 1 min 256 words PSYCHOMUSICOLOGY

Summary: From 2012 to 2023, the American Psychological Association served as publisher of Psychomusicology: Music, Mind, and Brain. Annabel Cohen and Mark Schmuckler were the successive editors-in-chiefs during this time. As the journal is ceasing publication, the two editors reflect on the developm...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000305

Suggestibility and psychedelics: From therapeutics to social context.

📅 2024-12-12 📖 1 min 📝 175 words CLINICAL NEUROSCIENCE

Summary: Suggestibility is an important aspect of the psychedelic experience that merits greater attention in the expanding field of psychedelic research. This article provides a comprehensive road map for exploring the critical questions and issues at the intersection of suggestion, suggestibility, and psyc...

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http://doi.org/10.1037/cns0000412

Evaluating dispositional emotion-related factors in the effect of acceptance on affect reactivity: A secondary analysis of an experimental study.

📅 2024-12-30 📖 1 min 📝 257 words CLINICAL NEUROSCIENCE

Summary: Acceptance as a self-regulation strategy has beneficial effects on affect reactivity in response to stressors via reducing negative affect and state anxiety as well as improving positive affect. However, dispositional emotion-related factors, namely, experiential avoidance, emotional dysregulation, ...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000416

Open monitoring meditation defined as the nonreactive meta-awareness of mind wandering.

📅 2024-12-12 📖 1 min 📝 128 words CLINICAL NEUROSCIENCE

Summary: Open monitoring meditation is generally defined as the nonreactive, meta-aware observation of moment-by-moment phenomenal experiences associated with mental contents without focusing on predetermined mental or physical objects. However, in the current literature, neither general definitions nor more...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000411

Multisensory interplay and theories of consciousness.

📅 2024-11-25 📖 1 min 📝 174 words CLINICAL NEUROSCIENCE

Summary: Sensory interplay typically refers to cases where the content of one sensory modality affects the content of another sensory modality. Although there is much evidence today of such interplay, it is unknown how to interpret them. Is it a one-way causal function between two separate states (e.g., a vi...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000407

How ghosting and stress impact vulnerable narcissism and maladaptive daydreaming: The role of mindfulness and rumination.

📅 2024-09-12 📖 1 min 📝 204 words CLINICAL NEUROSCIENCE

Summary: The present study aims to explore the influence of ghosting and stress on maladaptive daydreaming and vulnerable narcissism within a multivariate framework, using mindfulness and rumination as mediators. We employed a cross-sectional design and conducted it among 301 university students (M<...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000403

The conscious experience of thinking aloud: Are positive feelings associated with better think-aloud performance?

📅 2024-12-30 📖 1 min 📝 216 words CLINICAL NEUROSCIENCE

Summary: The think-aloud interview is a psychological procedure used to measure an individual’s cognition in solving a task. However, very little is known about how an individual’s conscious experience of the interview might be associated with their performance during the interview. This is consequential as ...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000410

Mystical experiences in virtual reality.

📅 2024-04-04 📖 1 min 📝 176 words CLINICAL NEUROSCIENCE

Summary: Virtual reality (VR) has shown promise as a psychological means of inducing mystical experiences. Here, we compared the subjective responses of 48 undergraduate volunteers to script-guided multiuser perceptual experiences in VR (VR perception [VR-P]) with their responses to a control condit...

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http://doi.org/10.1037/cns0000396

Effect of a peer-led meditation club on adolescents' well-being, empathy, and compassion.

📅 2024-09-09 📖 1 min 📝 197 words CLINICAL NEUROSCIENCE

Summary: Many studies have examined the effect of meditation sessions led by experienced teachers on personal benefits for adults. The present study investigated the effect of broad-based mindfulness practices with an adolescent peer facilitator on prosocial outcomes in addition to individual gains. The 20 s...

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http://doi.org/10.1037/cns0000400

Prefrontal cortex activity of adolescents with problematic smartphone use in response to gains and losses: A functional near-infrared spectroscopy study.

📅 2026-09-10 📖 1 min 📝 237 words BEHAVIORAL NEUROSCIENCE

Summary: Problematic smartphone use (PSU) has become a widespread problem that is highly prevalent in adolescent population. Previous studies suggest that PSU is associated with impairment in reward processing system. Thus, increased sensitivity to rewards could make individuals more vulnerable to abusive be...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000668

Topiramate and anterior cingulate cortex high-frequency stimulation mitigate aggression and promote sociability in socially isolated male mice.

📅 2026-08-10 📖 1 min 📝 230 words BEHAVIORAL NEUROSCIENCE

Summary: Excessive aggression induced by social isolation (SI) has been closely linked to dysfunction in prefrontal circuits, particularly the anterior cingulate cortex (ACC). Topiramate, an antiepileptic drug known to enhance GABAergic and inhibit glutamatergic transmission, has demonstrated antiaggressive ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000663

Is memory spared in monkeys with selective lesions of the hippocampus?

📅 2026-08-03 📖 1 min 📝 239 words BEHAVIORAL NEUROSCIENCE

Summary: Normal explicit/declarative memory requires a system of anatomically related structures in the medial temporal lobe that includes the hippocampus and the entorhinal, perirhinal, and parahippocampal cortex. Although extensive medial temporal lobe damage in primates causes robust anterograde amnesia, ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000658

Distal pup cues attenuate defensive reactivity and fear-potentiated startle magnitude in postpartum rats without altering sensorimotor gating.

📅 2026-07-09 📖 1 min 📝 241 words BEHAVIORAL NEUROSCIENCE

Summary: Using a novel behavioral paradigm that integrates fear-potentiated startle (FPS) and prepulse inhibition testing, this study examined how motherhood and infant cues impact innate defensive reactivity (as measured by the magnitude of basal startle reflex), attentional processing (as measured by prepu...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000661

A novel rat spatial inference task reveals rapid deduction of transitive relationships using schemas and deliberation.

📅 2026-07-13 📖 1 min 📝 264 words BEHAVIORAL NEUROSCIENCE

Summary: Inferential reasoning is a vital cognitive ability that enables animals to navigate novel situations by leveraging existing relational knowledge of memory schema, with hypothesized roles of prefrontal cortical—hippocampal circuits. Transitive inference (TI) tasks test the ability of subjects to infe...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000660

Keep Android Open 🔓

📅 2026-02-26 📖 3 min 📝 625 words FMHY

Summary: <h3 id="android-will-become-a-locked-down-platform-in-less-than-200-days" tabindex="-1">Android will become a locked-down platform in less than 200 days. </h3> <p>In August 2025, Google announced th...

🔗 **Read full article:**
https://fmhy.net/posts/KeepAndroidOpen

Monthly Updates [April]

📅 2026-04-01 📖 4 min 📝 885 words FMHY

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https://fmhy.net/posts/april-2026

Monthly Updates [May]

📅 2026-05-01 📖 4 min 📝 892 words FMHY

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https://fmhy.net/posts/may-2026

Monthly Updates [October]

📅 2026-10-01 📖 5 min 📝 1097 words FMHY

> **Summary:** <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   Re-ordered Live Sports based on the <a href="https://challonge.com/ajqvt0wo/fullscreen" rel=...

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https://fmhy.net/posts/oct-2026

Hunger beats tenure: Advice for junior engineers

👤 akavi 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://felixrieseberg.com/hunger-beats-tenure-advice-for-junior-engineers/ Comments URL: <a href="https://news.ycombinator.com/item?id=50266719">https://news.ycombinator.com/item?id=5026...

🔗 **Read full article:**
https://felixrieseberg.com/hunger-beats-tenure-advice-for-junior-engineers/

Wealth Inequality in America drawn to scale

👤 haraball 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://dreamagainusa.org/ Comments URL: https://news.ycombinator.com/item?id=50270764 Points: 4 # Comments: 0

🔗 **Read full article:**
https://dreamagainusa.org/

Spain train crash: 'At least 10 dead' in high-speed collision

👤 ultral 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.thetimes.com/world/europe/article/spain-train-crash-madrid-latest-deaths-victims-807kwv23l Comments URL: <a href="https://news.ycombinator.com/item?...

🔗 **Read full article:**
https://www.thetimes.com/world/europe/article/spain-train-crash-madrid-latest-deaths-victims-807kwv23l

Registration is now full for the 2024 ERP Boot Camp

👤 Steve Luck 📅 2024-03-16 📖 1 min 📝 106 words ERP BOOT CAMP

> **Summary:** The demand for the 2024 ERP Boot Camp was far beyond our expectations, and we reached our maximum registration of 30 people within one day. We already have a waiting list of over 30 people, so we have closed the registration site. <...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/15/registration-full

New Paper: Using Multivariate Pattern Analysis to Increase Effect Sizes for ERP Amplitude Comparisons

👤 Steve Luck 📅 2024-06-10 📖 2 min 📝 521 words ERP BOOT CAMP

> **Summary:** Carrasco, C. D., Bahle, B., Simmons, A. M., & Luck, S. J. (2024). Using multivariate pattern analysis to increase effect sizes for event-related potential analyses. Psychophysiology, 61, e14570. https://doi.org/10.1111/psyp.14570 [<a h...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/10/erp-core-decoding-paper

Call for Student/YP Deputy Editor

Deidre Artis 2026-09-29 1 min 14 words EMBS

Summary: The post Call for Student/YP Deputy Editor appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/call-for-student-yp-deputy-editor/

Time-Dynamic Semantic-Activity Entanglement: Quantifying Communication-Based Synchrony in Collaborative VR

Pooja Pol, Jianing Zhang, Qing Huang 2026-10-28 1 min 187 words ARXIV CS HC

Summary: arXiv:2610.37427v1 Announce Type: new Abstract: Collaborative virtual reality (VR) produces rich multimodal traces of team behavior, yet practical methods for characterizing time-varying team communication remain limited. We present time-dynamic semantic-activity entanglement (TSAE), inspired by th...

🔗 **Read full article:**
https://arxiv.org/abs/2610.37427

Comparing Temporal Deep Learning Architectures for Discrete Cross-User Myoelectric Control

Ethan Eddy, Evan Campbell, Scott Bateman, Erik Scheme

2026-10-28 1 min 208 words

ARXIV CS HC

Summary: arXiv:2610.37374v1 Announce Type: new Abstract: The recognition of dynamic gestures via discrete myoelectric control is becoming an attractive input modality for event-based human-machine interaction. While recent work suggests that discrete cross-user models are possible (i.e., no calibration data...

🔗 **Read full article:**
https://arxiv.org/abs/2610.37374

Understanding Human-Robot Collaboration through Joint Action

Chengwen Zhang, Jianyang Wang, Chang Liu, Yukun Wang, Yuan Xu, Zhoutong Ye, Jie Cai, Chun Yu, Yuanchun Shi

2026-10-28 1 min 126 words

ARXIV CS HC

Summary: arXiv:2610.37298v1 Announce Type: new Abstract: Human-robot collaboration requires partners to coordinate interdependent contributions toward a shared goal. Joint Action theory provides concepts for understanding this coordination, but how these concepts are instantiated and combined across empiric...

🔗 **Read full article:**
https://arxiv.org/abs/2610.37298

An Interactive Mobile Augmented Reality System for Efficient Virtual Model-to-Patient Registration in Thoracic Surgery

Emmanuel Ajaero, Pascal Spiegler, George Rakovich, Marta Kersten-Oertal, Yiming Xiao

2026-10-28 1 min 192 words ARXIV CS HC

Summary: arXiv:2610.37271v1 Announce Type: new Abstract: Video-Assisted Thoracoscopic Surgery (VATS) is a minimally invasive procedure, where precise surgical port placement is vital, yet it relies heavily on surgeon experience and mental mapping of pre-operative imaging in the operating room. To address th...

Read full article:
https://arxiv.org/abs/2610.37271

Peers4U: Designing Learner-Centered Virtual Peers for Children's Multimodal Mathematics Learning

Zhi Zheng, Binglin Liu, Weihao Chen, Lan Luo, Yu Mei, Jie Cai, Chun Yu, Yuanchun Shi

2026-10-28 1 min 153 words ARXIV CS HC

Summary: arXiv:2610.37238v1 Announce Type: new Abstract: In early mathematics, children learn abstract concepts through concrete representations, such as manipulating objects, drawing, and speaking. Learning with peers can enrich this process by exposing children to diverse strategies and enabling them to e...

Read full article:
https://arxiv.org/abs/2610.37238

Science-Fiction Transformation as a Design Lens for Visualization Transformation

👤 Abdulhaq Adetunji Salako, Christian Tominski

📅 2026-10-28 📖 1 min 📝 222 words

ARXIV CS HC

Summary: arXiv:2610.37131v1 Announce Type: new Abstract: Science fiction often features characters or objects that undergo radical changes in form while still being recognizable as the same entity. For example, a vehicle might unfold into a robot, or one body might flow or shape-shift into another. We use s...

🔗 **Read full article:**
https://arxiv.org/abs/2610.37131

Externalizing Hand Grip Pressure Through Real-Time Visual Feedback

👤 Andrew L. Huard, Avinash Ajit Nargund, Misha Sra

📅 2026-10-28 📖 1 min 📝 202 words

ARXIV CS HC

Summary: arXiv:2610.36769v1 Announce Type: new Abstract: Grip pressure, including contact force and its distribution across the hand, is central to skilled grasping but is not directly visible. This makes it difficult to teach, learn, and self-correct in grip-dependent motor tasks, where verbal instruction ...

🔗 **Read full article:**
https://arxiv.org/abs/2610.36769

Translating Poetic Emotion into Dynamic Mathematical Visuals: A Strange Attractor-Based Approach

Subhadra Nayak 2026-10-28 1 min 99 words ARXIV CS HC

Summary: arXiv:2610.36698v1 Announce Type: new Abstract: Poetry conveys emotional experiences through language. This project investigates how poetic emotional experiences can be represented through dynamic mathematical visualisations while further developing a pipeline for this purpose. By leveraging a pret...

🔗 **Read full article:**
https://arxiv.org/abs/2610.36698

Beyond the 1D Reality-Virtuality Continuum: A 2D Framework for Seven Extended Reality Paradigms

Rajashekhar Vachiravelu Saminathan 2026-10-28 1 min 160 words ARXIV CS HC

Summary: arXiv:2610.36674v1 Announce Type: new Abstract: The Reality-Virtuality (RV) continuum helps in describing the relationship between physical and computer-generated environments. However, the increasing number of Extended Reality (XR) paradigms introduces new technologies. Their characteristics canno...

🔗 **Read full article:**
https://arxiv.org/abs/2610.36674

Immersive Commons: Designing Ephemeral Spatial Dialogue for Co-Located Mixed Reality Environments

Wuwu Ran, Mohammed Faraj Al-Suwaidi
2026-10-28 1 min 150 words ARXIV CS HC

Summary: arXiv:2610.36014v1 Announce Type: new Abstract: Head-mounted mixed reality (MR) often prioritises individual immersion, potentially drawing attention away from others sharing the room. This paper asks how MR can support co-located spatial dialogue through embodied interaction and activity visible t...

🔗 **Read full article:**
https://arxiv.org/abs/2610.36014

p-Adic statistical field theory of fluctuation effects in hierarchical neural networks

W. A. Z\'u\~niga-Galindo
2026-10-28 1 min 281 words ARXIV QBIO NC

Summary: arXiv:2610.36400v1 Announce Type: cross Abstract: The Buice-Cowan field theory [PRE 75, 051919 (2007)] describes fluctuations and correlations of neural activity through a path integral whose mean-field limit is the Wilson-Cowan equation and whose critical behavior belongs to the directed percolati...

🔗 **Read full article:**
https://arxiv.org/abs/2610.36400

SOMA: A Benchmark for Fine-Grained Egocentric Behavior and Body-Motion Understanding

Matea Tashkovska, Na\"el Ouerghemi, Haozhe Qi, Alexander Mathis

2026-10-28 1 min

242 words

ARXIV QBIO NC

Summary: arXiv:2610.36384v1 Announce Type: cross Abstract: Perceiving human body motion is critical for multimodal systems that understand and assist people. Yet current evaluations of Multimodal Large Language Models (MLLMs) largely focus on objects, scenes, and coarse actions, leaving their ability to rea...

Read full article:
https://arxiv.org/abs/2610.36384

MtReNIRS: A Self-Supervised Foundation Model for Transferable Neurovascular Representations

Tony Chae, Kleanthis Avramidis, Dimitra Emmanouilidou, Ivan J. Tashev

2026-10-28 1 min

200 words

ARXIV QBIO NC

Summary: arXiv:2610.36159v1 Announce Type: cross Abstract: Functional near-infrared spectroscopy (fNIRS) offers a promising approach for monitoring the hemodynamic correlates of neural activity in naturalistic settings. However, existing modeling methods for fNIRS are largely confined to small-scale, single...

Read full article:
https://arxiv.org/abs/2610.36159

Every route counts, but some count more: a max-plus view of the C. elegans brain

Emma Reyner-Fuentes, Sebastian Rudolph, Carmen Pel\'aez-Moreno, Francisco J. Valverde-Albacete

2026-10-28 1 min 156 words ARXIV QBIO NC

Summary: arXiv:2610.37587v1 Announce Type: new Abstract: Brains do not simply add up every signal they receive: they select which routes dominate. Yet connectomes are usually analyzed with tools that sum over all paths and place neural states in Euclidean spaces, blurring the feedback circuits that may sust...

Read full article:
https://arxiv.org/abs/2610.37587

Sufficiency Theorem and Closed-Form Posterior for a Windowed Change Point Detection Task

Payton Hanks, Rishidev Chaudhuri, Timothy D. Hanks

2026-10-28 1 min 206 words

ARXIV QBIO NC

Summary: arXiv:2610.37486v1 Announce Type: new Abstract: To function effectively in a changing world, agents must detect changes and act on them in a timely fashion. We study this in the context of a change point detection task in which clicks arrive from a Poisson process whose rate changes at an unknown t...

Read full article:
https://arxiv.org/abs/2610.37486

Local learning is all you need: brain-aligned predictive coding networks

👤 Dirk G\"utlin, Ryszard Auksztulewicz, Adrien Doerig

📅 2026-10-28 📖 1 min 📝 206 words

ARXIV QBIO NC

Summary: arXiv:2610.37327v1 Announce Type: new Abstract: Predictive Coding (PC) is an influential account of how brain representations emerge through hierarchical, local, self-supervised prediction-based learning, without requiring globally propagated error signals. Despite extensive evidence supporting PC,...

🔗 **Read full article:**
https://arxiv.org/abs/2610.37327

Exploring Neural Dynamics with Time Delays: A Numerical and Bifurcation Analysis Approach

👤 Martin R Nelson, Jonathan J Crofts

📅 2026-10-28 📖 1 min 📝 177 words

ARXIV QBIO NC

Summary: arXiv:2610.36920v1 Announce Type: new Abstract: The Wilson-Cowan model is a classic example of a ``neural mass model'' - a simplified yet powerful way to describe the activity of large networks of neurons. It is particularly interesting because, even in its relatively simple form, it can produce a ...

🔗 **Read full article:**
https://arxiv.org/abs/2610.36920

Neuronal Identity as an Organizational Basis for Analyzing Neural Population Dynamics

Seungjae Han, Joshua Yedam You, Soi Kim, Minho Eom, Jinho Park, Seunghee Park, Byeongwook Lee, Young-Gyu Yoon

2026-10-28 1 min 237 words ARXIV QBIO NC

Summary: arXiv:2610.36745v1 Announce Type: new Abstract: Individual neurons exhibit diverse molecular, morphological, and electrophysiological properties that shape their distinct roles in neural circuits. While these multimodal features are largely shared across individuals, they remain an underutilized so...

🔗 **Read full article:**
https://arxiv.org/abs/2610.36745

The geometry of cognition: from neural populations to neurotechnology

Lucas Bayones, Devika Narain, Roman Rossi-Pool

2026-10-28 1 min 225 words

ARXIV QBIO NC

Summary: arXiv:2610.36541v1 Announce Type: new Abstract: Cognition is implemented by distributed neural populations whose collective activity evolves through structured, low-dimensional state spaces. In this review, we survey the analytical and conceptual tools used to study latent population dynamics in bi...

🔗 **Read full article:**
https://arxiv.org/abs/2610.36541

Temporal smoothness constraints on efficient neurobiological codes imply temporal specialization

John J. Vastola, Samuel J. Gershman
2026-10-28 1 min 170 words ARXIV QBIO NC

Summary: arXiv:2610.36272v1 Announce Type: new Abstract: Neurobiological circuits may be energy efficient in part because there is a division of labor: different subsystems compute or represent different things. In light of this, it is interesting that a major difference between subsystems is the time scale...

🔗 **Read full article:**
https://arxiv.org/abs/2610.36272

Vitamin B6 augments antidepressant activity of venlafaxine by restoring cortical EEG oscillations, behavior and oxidative biomarkers in chronic unpredictable mild stress (CUMS)-induced model of depression-like behavior

1 min 30 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Fashwa Khan Tareen, Sana Javaid, Waseem Ashraf, Syed Muhammad Muneeb Anjum, Rana Muhammad Zahid Mushtaq, Asad Abrar, Faleh Alqahtani, Imran Imran

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004506?dgcid=rss_sd_all

Stage-specific exposure to an activity-permissive medium enhances neuronal activity in oligodendrocyte-enriched cortical organoids

📖 1 min 📝 34 words BRAIN RESEARCH

> **Summary:** Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Clara Chung, Michelle J. Kim, Grace Field, Konstantina Pilarinos, Elizabeth K. Kharitonova, Natalie Baker Campbell, Marie Shi, Christopher V. Gabel, Joseph L. Orofino, Ella Zeldich

🔗 **Read full article:**
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Neurodegenerative diseases in Sub-Saharan Africa: diagnostic invisibility, biological heterogeneity and a precision-neuroscience agenda

📖 1 min 📝 11 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Maulilio John Kipanyula

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226006962?dgcid=rss_sd_all

Investigating fatigability and cortical activation during prolonged foot tapping in multiple sclerosis

📖 1 min 📝 28 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Zuhal Abasiyanik, Lousin Moumdjian, Natalia Skorupa, Daphne Kos, Bart Van Wijmeersch, Miguel D’Haeseleer, Inge Zijdewind, Peter Feys, Felipe Balistieri Santinelli

🔗 **Read full article:**
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Neural dynamics of auditory attention: Interactions between top-down and bottom-up processes

📖 1 min 📝 12 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): S. Graceffo, M. Elhilali

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006002?dgcid=rss_sd_all

Neurovascular coupling in simultaneous EEG-fNIRS studies: A scoping review of analytical frameworks and clinical applications

📖 1 min 📝 18 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Yu Wu, Rajamanickam Yuvaraj, Stéphane Perrey, Hasan Ayaz, Wei-Peng Teo

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006130?dgcid=rss_sd_all

Ameliorating effects and mechanisms of therapeutic electrical stimulation on the recurrent laryngeal nerve injury model in rats

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-58788-1

Genetic mechanisms underlying the associations between brain structure and mental health

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-79115-2

Functional connectivity in cerebral visual impairment: implications for daily visual behaviors

📅 2026-10-28 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-76731-2

Compensation of the effects of temperature on a motor system in the crab, Cancer borealis

👤 Kathleen JacquerieJames M. DiMartinoAnanya DalalJi ZengEve Marderahttps://ror.org/05abbep66Volen Center and Biology Department, Brandeis University, Waltham, MA 02454

📅 2026-10-21 📖 1 min 📝 50 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceCold-blooded animals that live in climates with significant seasonal changes in ambient temperature must have myriad mechanisms to function over a wide range of environmental conditions. We explore...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2625185123?af=R

Action features dominate cortical representation during natural vision

👤 Heejung JungHunar JainXiaochun HanMa FeilongJane HanDeepanshi ShokeenCara Van UdenIsabella HansonAndrew C. ConnollyTor D. WagerJames V. HaxbySamuel A. Nastaseahttps://ror.org/049s0rh22Department of Psychological and Brain Sciences, Dartmouth College, Hanover, NH 03755bIndependent Researcher, Sunnyvale, CA 94085chttps://ror.org/03taz7m60Department of Psychology, Center for Computational Language Sciences, University of Southern California, Los Angeles, CA 90089

📅 2026-10-23 📖 1 min 📝 51 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceHuman cortex is functionally organized to prioritize the most important features of our external world. In perceiving the natural world, however, these features are often overlapping and intermixed...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2625018123?af=R

Translational control of microglial metabolism, inflammatory responses, and survival by the eIF4E-binding proteins

Sara BermudezJung-Hyun ChoiNiaz MahmoodJacob W. VogelHashem AlmousaZiying HuangSung-Hoon KimVivian Yuchan ZhuLinqiao ZhouDiana Casas-AntillonMoein YaqubiDanielle CozachencoJo Anne StrattonOskar HanssonLuke HealyArgel Aguilar-VallesNahum Sonenbergahttps://ror.org/01pxwe438Department of Biochemistry and Goodman Cancer Institute, McGill University, Montreal, QC H3A 1A3, Canadabhttps://ror.org/02wnxgj78Department of Biochemistry, Chungbuk National University, Cheongju 28644, South Koreachttps://ror.org/012a77v79Department of Clinical Sciences, Lund University, Lund 221 84, SwedendSciLifeLab, Lund 221 00, Swedenehttps://ror.org/0227as991Department of Microbiology and Molecular Biology, Chungnam National University, Daejeon 34134, South Koreafhttps://ror.org/03rmrcq20Department of Pathology and Laboratory Medicine, Faculty of Medicine, University of British Columbia, Vancouver, BC V5Z 1M9, CanadagThe Hospital for Sick Children, Toronto, ON M5G 1E8, Canadahhttps://ror.org/01pxwe438Department of Neurology and Neurosurgery, Montreal Neurological Institute, McGill University, Montreal, QC H3A 1A1, Canadaihttps://ror.org/03490as77Institute of Biophysics Carlos Chagas Filho, Federal University of Rio de Janeiro, Rio de Janeiro 21941-170, Braziljhttps://ror.org/02qtvee93Department of Neuroscience, Carleton University, Ottawa, ON K1S 5B6, Canada

📅 2026-10-21 📖 1 min 📝 47 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceThe link between Alzheimer’s disease (AD) risk and microglial function is well-known, yet the mechanisms governing microglial responses remain unclear. Postmortem and experimental studies demonstra...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2610766123?af=R

Naturalistic spatial education engages distinct neural processes for mental rotation, cognitive load, and linguistic demands

Eloise B. GacettaMarissa L. LawsLauren S. AuletTeoman S. OzaydinJonathan S. TsayConrad ZapantaJessica F. Cantlonahttps://ror.org/05x2bcf33Department of Psychology, Carnegie Mellon University, Pittsburgh, PA 15213bhttps://ror.org/01an3r305Center for the Neural Basis of Cognition, Carnegie Mellon University and University of Pittsburgh, Pittsburgh, PA 15213chttps://ror.org/0072zz521Department of Psychological and Brain Sciences, University of Massachusetts Amherst, Amherst, MA 01003dhttps://ror.org/05x2bcf33Neuroscience Institute, Carnegie Mellon University, Pittsburgh, PA 15213ehttps://ror.org/05x2bcf33Department of Biomedical Engineering, Carnegie Mellon University, Pittsburgh, PA 15213

📅 2026-10-21 📖 1 min 📝 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceSpatial visualization skill is a foundational component of human intelligence and a powerful lever for success in science, technology, engineering, and mathematics (STEM). Yet little is known about...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2601206123?af=R

C1ql2 gates stress vulnerability by regulating microglia-mediated synaptic pruning in the BLA-vHPC circuit

👤 Chun-Yan WangWei-Zhu LiuShuang-Mei LiaoJing NiuHan-Qing PanWen-Jie YouKe ZhaoBao MaMing-Fen LuJia-Xin ZouPing HuWen-Hua ZhangBing-Xing Panahttps://ror.org/042v6xz23Jiangxi Institute of Respiratory Diseases and Department of Pathology, The First Affiliated Hospital, School of Basic Medical Sciences and Institute of Biomedical Innovation, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinabhttps://ror.org/042v6xz23Jiangxi Provincial Key Laboratory of Brain Science and Brain Health, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinachttps://ror.org/042v6xz23Postdoctoral Research Station, The First Affiliated Hospital, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinadhttps://ror.org/042v6xz23Institute of Translational Medicine, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinaehttps://ror.org/042v6xz23Department of Neurology, The Second Affiliated Hospital, Jiangxi Medical College, Nanchang University, Nanchang 330006, China

📅 2026-10-21 📖 1 min 📝 43 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceStress-related emotional disorders involve maladaptive synaptic remodeling in the basolateral amygdala (BLA), which is highly heterogeneous across neurons projecting to distinct downstream targets....

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2600854123?af=R

BVLab-analyzer: a unified visual toolbox for terabyte-scale whole-brain cell detection, neuron reconstruction, and vascular network extraction

Tingwei Quan 2026-10-07 1 min 139 words FRONTIERS NEUROINFORMATICS

Summary: BVLab-Analyzer is a standalone graphical software tool for end-to-end analysis of terabyte-scale whole-brain microscopy images. Operating through a zero-code interface, it handles data import (TIFF, BioImageVision, OME-Zarr), deep learning-based semantic segmentation, 3D skeleton reconstruction, and...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1925112

Transformers for fMRI analysis: from spatiotemporal modeling to brain foundation models

Bocheng Wang 2026-10-07 1 min 217 words FRONTIERS NEUROINFORMATICS

Summary: Functional magnetic resonance imaging (fMRI) serves as a pivotal tool for investigating the functional organization and dynamic connectivity of the human brain, indirectly reflecting neural activity through blood oxygen level-dependent (BOLD) signals. However, the high-dimensional spatiotemporal str...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1869687

Natural language processing for suicidal behavior detection in short Spanish clinical notes from first-contact interviews

👤 Ansel Y. Rodríguez-González
📅 2026-10-07 📖 1 min 📝 267 words

FRONTIERS NEUROINFORMATICS

Summary: IntroductionSuicide is a multifactorial phenomenon influenced by social, cultural, biological, and psychological factors, whose early detection remains challenging. According to the World Health Organization, suicide is the third leading cause of death among individuals aged 15–29 years, representin...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1919962

Mechanonociceptive tactile feedback enhances prosthetic grasp control during repeated human-in-the-loop interaction

👤 Elif Hocaoglu
📅 2026-10-15 📖 1 min 📝 215 words

FRONTIERS NEUROROBOTICS

Summary: Upper-limb prosthetic systems continue to face a fundamental challenge in providing sensory feedback that remains functionally effective during repeated interaction. This study investigates innocuous painful feedback (IPF), a controlled mechanonociceptive tactile modality intended to provide salient...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1910158

Brain-inspired predictive visual attention for spatiotemporal semantic understanding of complex film and television videos

Peng Han 2026-10-22 1 min 275 words FRONTIERS NEUROROBOTICS

Summary: Due to the unique characteristics of film and television, such as montage editing and narrative structure jumps, semantically salient targets frequently experience temporal feature drift, making it difficult for existing models to maintain stable semantic understanding output when keyframes are lost...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1934032

Ordered onset of regional desynchronization and a transition in movement-class discriminability during cue-paced motor imagery

Seho Lee 2026-10-21 1 min 298 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Motor imagery recruits a distributed network whose components and large-scale organization are well characterized. The temporal order in which those components are engaged is not: the latency at which each region begins to desynchronize within a trial has not been compared statistically i...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb161

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261028002347&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Intelligent Sensing and Adaptive Control for Low-Loss Soybean Harvesting: A Review of Multi-Scenario Technologies

👤 Zhong Tang 📅 2026-10-27 📖 1 min 📝 60 words LOW VISION

Summary: Soybean harvesting is strongly affected by low pod position, lodging, pod shattering, moisture variation, and changing material flow, making stable low-loss operation difficult across diverse field conditions. This review synthesizes 179 core publications identified through structured literature sea...

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A 3D Vision-Based Localization Method for Automated Loading of Large-Sized Heat Exchangers

👤 Qing Jiang 📅 2026-10-27 📖 1 min 📝 62 words LOW VISION

Summary: Manual loading of large-sized finned-tube heat exchangers has low productivity and imposes substantial physical demands on workers. In addition, industrial 3D vision systems often produce large regions of missing depth data when imaging highly reflective metallic surfaces, which complicates grasp-po...

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Hybrid Vision Perception for Low-Cost Autonomous Vehicle Navigation

👤 Abdelkader El Kamel 📅 2026-10-27 📖 1 min 📝 69 words LOW VISION

Summary: High-performance perception systems for autonomous vehicles (AVs) commonly use LiDAR, RADAR, and multiple cameras. These sensors provide useful information, but their cost and computational requirements can be difficult to justify for small prototypes and low-cost platforms. This work therefore uses...

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Fidelity-oriented pelvic CBCT-to-CT synthesis using an anatomical information-attention diffusion model

👤 Kunyuan Xu 📅 2026-10-27 📖 1 min 📝 63 words LOW VISION

Summary: Cone-beam computed tomography (CBCT) is widely used in image-guided radiotherapy and adaptive radiotherapy, but its direct quantitative use is limited by scatter artifacts, low soft-tissue contrast, and inaccurate Hounsfield unit (HU) values. Diffusion models provide stable image generation capabili...

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Poros: Perception-Enabled Interactive Assistance for Appliance Use by Blind and Low Vision Users

Wei Liang 2026-10-27 1 min 62 words LOW VISION

Summary: Modern home appliances pose significant accessibility barriers for individuals with visual impairments, driven by increasing functional complexity and minimalist design trends. Drawing on formative interviews with blind and low vision (BLV) participants, we propose Poros, a novel system providing pe...

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DELUGE: Decomposed Entropy-coded Live Unstructured Geometry Exchange for Real-time Particle Streaming

Takefumi Hiraki 2026-10-27 1 min 74 words LOW VISION

Summary: Particle-based physics simulations, including fluids, smoke, and granular media, are fundamental to visual realism in immersive VR and AR. With the growing adoption of social VR and digital twins, demand is increasing for shared experiences in which multiple users interact with the same simulation i...

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A multi-attention mechanism-improved ResNet18 approach for lithology identification and classification

Ruifeng Li 2026-10-27 1 min 61 words LOW VISION

Summary: Lithology identification is vital for improving geological exploration accuracy, optimizing petroleum exploration schemes, and ensuring construction engineering safety. Traditional identification methods suffer from low efficiency, high cost, and heavy dependence on domain expertise. This study prop...

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An open-source platform for measuring optomotor responses in mice

Shogo Soma 2026-10-27 1 min 64 words LOW VISION

Summary: The optomotor response (OMR) is a valuable behavioral index for assessing visual function in basic research; however, commercially available OMR evaluation equipment is expensive and lacks flexibility, as proprietary software and fixed hardware geometry restrict custom stimulus protocols and system ...

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Communication patterns in home medication review for individuals with vision impairment: A qualitative study

👤 Woojin Park
📅 2026-10-27 📖 1 min 📝 37 words

LOW VISION

Summary: CONCLUSIONS: In these consultations, professional dialogue supported verification and reassurance, while direct regimen review revealed unverified beliefs and gaps in medication knowledge. The findings suggest a role for accessible counseling that combines information provision with professional ver...

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Global, regional, and national burden of unintentional poisoning, 1990-2023: results from the Global Burden of Disease Study 2023

👤 GBD 2023 Unintentional Poisoning Collaborators

📅 2026-10-27 📖 1 min 📝 65 words

LOW VISION

Summary: BACKGROUND: Unintentional poisoning is a preventable cause of injury-related morbidity and mortality, yet global patterns, trends, and inequalities remain incompletely characterised. Within the Global Burden of Disease (GBD) framework, unintentional poisoning includes poisoning by carbon monoxide an...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

Xiaobing Yan 2026-09-09 1 min 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

👤 Jarred Yasuhara-Bell 📅 2026-10-27 📖 1 min 📝 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Combining Cognitive Training and Brain Stimulation: A Novel Approach to Address Disordered Eating and Antipsychotic-Induced Weight Gain in People With Severe Mental Illness

👤 Ulrike Schmidt
📅 2026-10-26 📖 1 min 📝 69 words

TDCS TACS TRNS

Summary: CONCLUSION: The initiation of antipsychotic therapy is a critical clinical window for early intervention to mitigate the long-term metabolic and psychiatric consequences of antipsychotic-induced weight gain. Combining ABMT + tDCS is acceptable and may help reduce binge eating in individuals on antip...

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High-Intensity Transcranial Alternating Current Stimulation for Post-Stroke Cognitive Impairment: A Triple-Blind Randomized Trial

Xiquan Hu 2026-10-26 1 min 60 words TDCS TACS TRNS

Summary: CONCLUSION: Hi-tACS improved global cognition and working memory in patients with PSCI and was generally well tolerated. Exploratory fNIRS findings suggested possible accompanying changes in cortical activation and network organization but did not establish a neurophysiological mechanism. Importantl...

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Comparative analysis of mitochondrial genomes in Zantedeschia parental lines and their hybrid progeny

Hongzhi Wu 2026-10-27 1 min 31 words TDCS TACS TRNS

Summary: CONCLUSIONS: This study expands current knowledge of mt genome diversity in the Araceae and provides valuable genomic resources for future studies of Zantedeschia mitogenome evolution, genetic resource conservation, and molecular breeding.

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Combined Transcranial Direct Current Stimulation and Transcutaneous Cervical Vagus Nerve Stimulation for Anxiety and Depression: An Exploratory Pilot Study

👤 Maria-Manuela Apostol
📅 2026-10-27 📖 1 min 📝 60 words

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Summary: Anxiety and depressive disorders are among the leading causes of disability worldwide, yet current treatments remain limited by modest efficacy and restricted accessibility; non-invasive neuromodulation techniques such as transcranial direct current stimulation (tDCS) and transcutaneous cervical vag...

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Non-Invasive Brain Stimulation Techniques and Effects on Neuromodulation and Athletic Performance in Combat Sports: A Systematic Review

Bianca Miarka 2026-10-27 1 min 59 words TDCS TACS TRNS

Summary: CONCLUSIONS: tDCS may influence selected cognitive, neuromuscular, and fatigue-related outcomes in combat sport athletes; however, the direction and magnitude of these effects are inconsistent across studies. Given the substantial methodological heterogeneity and the low-to-very-low certainty of evi...

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Non-Invasive Brain Stimulation for Central Neuropathic Pain: An Exploratory Randomized Sham-Controlled Crossover Trial

Eunhee Sohn 2026-10-27 1 min 64 words TDCS TACS TRNS

Summary: Background/Objectives: Central neuropathic pain (CNP) is difficult to manage, and comparative evidence for non-invasive brain stimulation is limited. We explored the short-term efficacy and safety of repetitive transcranial magnetic stimulation (rTMS) and transcranial direct current stimulation (tDC...

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Noninvasive Neuromodulation for Cognitive Dysfunction in Parkinson's Disease: A Scoping Review

Reginald B O'Hara 2026-10-27 1 min 47 words TDCS TACS TRNS

Summary: CONCLUSIONS: Evidence specific to PD-MCI is limited and mixed. Several studies pairing stimulation with cognitive training reported favorable outcomes, but this pattern was inconsistent and does not establish superiority over stimulation alone. No VNS trial enrolled participants with confirmed PD-MC...

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Comparative Plastid Phylogenomics of 29 Newly Sequenced Bougainvillea Cultivars Resolves the Phylogenetic Placement of Double-Bracted Cultivars

Huiqi Zhao 2026-10-27 1 min 59 words TDCS TACS TRNS

Summary: Bougainvillea (Nyctaginaceae) cultivars are horticulturally prized worldwide for their vivid bracts; however, repeated interspecific hybridization and somatic mutations have obscured their maternal origins, particularly for double-bracted forms. Here, we sequenced and assembled the complete plastome...

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Anodal Transcranial Direct Current Stimulation Reduces Neuroinflammation and Cholinergic Dysregulation After Young Adult Traumatic Brain Injury

👤 Guven Akcay
📅 2026-10-27 📖 1 min 📝 61 words

TDCS TACS TRNS

Summary: Traumatic brain injury (TBI) during childhood and adolescence can result in persistent motor and cognitive dysfunction due to secondary injury cascades involving neuroinflammation and disruption of neurotransmitter homeostasis. Although transcranial Direct Current Stimulation (tDCS) is a promising n...

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Assessing brain-behaviour coupling in non-invasive brain stimulation: Insights from reliable change indices and Monte Carlo simulations

👤 Alexander D Tang
📅 2026-10-27 📖 1 min 📝 64 words

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Summary: Non-invasive brain stimulation (NiBS) studies frequently report exploratory brain-behaviour correlations following the application of NiBS. However, these analyses are typically underpowered and rely on ratio-based change scores with known statistical limitations. We addressed these limitations by p...

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Neural-Discursive Dynamics in Collaborative Spatial Problem Solving: Evidence from fNIRS Hyperscanning and Accountable Talk

👤 Xingguang Guo
📅 2026-10-27 📖 1 min 📝 65 words FNIRS

Summary: High-quality collaboration cannot be inferred simply from how much students talk or how strongly their brain activity is synchronized during joint work. This study examined relations among accountable talk, inter-brain synchrony (IBS), and group task performance in fifth-grade students' collaborativ...

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Probing Executive Function in the Toddler Data Desert: A Touchscreen fNIRS Study of Working Memory and ADHD-Related Symptoms

👤 Caitie Busch 📅 2026-10-27 📖 1 min 📝 60 words FNIRS

Summary: Attention-deficit/hyperactivity disorder (ADHD) is associated with early differences in attentional control and working memory (WM), yet relatively few paradigms are developmentally feasible across the toddler-to-preschool period while also distinguishing spatial from featural information use. Addre...

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Mapping the Global Landscape and Thematic Evolution of fNIRS-Gait Research: A Systematic Bibliometric and Network Visualization Analysis

👤 Naime Uluğ 📅 2026-10-27 📖 1 min 📝 20 words FNIRS

Summary: CONCLUSIONS: Establishing standardized preprocessing pipelines and short-separation regressions may prove pivotal for transitioning mobile neuroimaging into objective, closed-loop clinical neurorehabilitation.

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Self-Selected Versus Assigned Emotion Regulation in Pain Expectancy: Behavioral and fNIRS Evidence

👤 Ping Shi 📅 2026-10-27 📖 1 min 📝 63 words FNIRS

Summary: Background: Expectation is a powerful cognitive modulator of pain, but how different types of emotion regulation approaches interact with expectancy to influence pain processing remains unclear. The present study compared self-selected habitual strategies (cognitive reappraisal or deep breathing) wi...

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Comparative Effects of Acute HIIE and MICE on Executive Function, Prefrontal Oxyhaemoglobin Responses, and Related Biomarkers in Sedentary Young Adults: A Randomized Crossover Trial

👤 Xinyue Yang
📅 2026-10-27 📖 1 min 📝 34 words FNIRS

Summary: CONCLUSIONS: Both exercise sessions were followed by faster reaction times on selected tasks. The small sample, multiple analyses, and potential systemic contributions to fNIRS signals preclude firm conclusions about exercise-mode superiority or biological mediation.

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Cerebral Hemodynamic Patterns in People with Subjective Cognitive Decline: A Systematic Review Based on Functional Near-Infrared Spectroscopy

Israel Contador
2026-10-27 1 min 62 words FNIRS

Summary: Background/Objectives: Subjective cognitive decline (SCD) has been proposed as an early symptomatic manifestation of dementia/Alzheimer's disease. However, its associated brain patterns are not yet fully characterized. In recent years, functional near-infrared spectroscopy (fNIRS) has emerged as a c...

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Resting-State Functional Brain Connectivity and Neurodevelopmental Profiles in 3-Year-Olds with Congenital Heart Disease

👤 Anne Gallagher
📅 2026-10-27 📖 1 min 📝 67 words FNIRS

Summary: Background/Objectives: Congenital heart disease (CHD) is associated with altered brain development and an increased risk of neurodevelopmental difficulties, particularly in more severe forms of the condition. Although alterations in functional brain connectivity have been reported before and after t...

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Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan
📅 2026-10-27 📖 1 min 📝 60 words FNIRS

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

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Multimodal Sensing and Digital Biomarkers in Virtual Reality-Based Cognitive Rehabilitation: Clinical Validation, Adaptive Systems, and Real-World Translation

👤 Raffaele Ferri
📅 2026-10-27 📖 1 min 📝 58 words FNIRS

Summary: Virtual reality (VR) is evolving from a platform for cognitive exercises into an instrumented rehabilitation environment capable of simultaneously delivering treatment and quantifying behavior. This critical narrative review examines how behavioral, ocular, kinematic, physiological, and neurophysiol...

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Unpredictable human screams elicit anticipatory and phasic cortical responses that affect subsequent interpersonal space regulation

Alessandro Grecucci 2026-10-27 1 min 59 words FNIRS

Summary: Unpredictable threats can sustain vigilance states that bias perception, cognition, and behaviour beyond immediate stimulus processing. Human screams, as evolutionarily salient alarm signals, are especially effective in eliciting defensive responses, yet their cortical dynamics and consequences for ...

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Human Factors in Brain-Computer Interfaces: Boundaries, Key Issues, and Methodological Approaches

Yunfa Fu 2026-10-27 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: The large-scale, real-world deployment of brain-computer interface (BCI) is confronted with human factors bottlenecks, including reliability and usability. While modern human factors engineering has encompassed domains such as cognitive ergonomics, workload, adaptive automation, and human-in-the-loo...

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EOG Artifact Removal from EEG Using a Gated WGAN with Multi-Scale and Multi-Domain Feature Extraction

👤 Hao Tang 📅 2026-10-27 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: Background: Electroencephalogram (EEG) signals provide important information for brain-computer interfaces (BCIs) and the diagnosis of neurological disorders but are susceptible to electrooculographic (EOG) artifacts. Traditional methods often rely on multichannel data or relatively long windows, li...

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EEG Features of Cortical Intent for Urination and Defecation in Healthy Aging and Spinal Cord Injury: A Descriptive Feasibility Study for BCI Neurorehabilitation

👤 Jie Jia 📅 2026-10-27 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: Background/Objectives: Systematically elucidating the neural mechanisms of bowel and bladder control is crucial for developing closed-loop brain-computer interface (BCI) neurorehabilitation for patients with spinal cord injury (SCI). This study aims to characterize the spatiotemporal Electroencephal...

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The Development and Evolution of Substrate Materials for Brain-Computer Interfaces

👤 Yen Wei 📅 2026-10-27 📖 1 min 📝 58 words BRAIN COMPUTER INTERFACE

Summary: Brain-Computer Interfaces (BCIs) hold tremendous potential for neurological therapeutics and neuroprosthetics; however, their long-term clinical viability is fundamentally limited by chronic interfacial failure. This failure originates from the profound biomechanical mismatch between synthetic micro...

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Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan 📅 2026-10-27 📖 1 min 📝 60 words BRAIN COMPUTER INTERFACE

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

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Agentic EEG Pipeline Synthesis Using Large Language Models and the Model Context Protocol

Katy Tarrit 2026-10-27 1 min 56 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography analysis typically requires multistage processing pipelines for which configuration depends on recording characteristics and the intended analytical objective. Although automated pipelines improve reproducibility, they generally implement predefined processing sequences. We pr...

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Overt and Covert Speech-State Classification from Tripolar Concentric Ring Electrode EEG: A 4 mm and 10 mm Configuration Study

Shoaib Memon 2026-10-27 1 min 76 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) speech-state classification is a noninvasive method to identify the brain's overt and covert behavior. The EEG achieved through the scalp is contaminated by artifacts and volume conduction and suffers from poor spatial resolution. We present the first application of smal...

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The compositional encoding of hand-eye coordinated movements for single neurons in the posterior parietal cortex

Richard A Andersen
2026-10-27 1 min 77 words BRAIN COMPUTER INTERFACE

Summary: Human posterior parietal cortex (PPC) is known to play a role in hand-eye (H-E) coordination, yet the underlying encoding mechanisms remain uncertain. We recorded 412 single neurons across 11 sessions from motor cortex (MC; n = 251) and PPC (n = 161) in a single human participant performing an H-E c...

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NCEL: Convolution-Enhanced Lightweight Neural Architecture for Robust EEG Signal Decoding in Noninvasive Brain-Computer Interfaces

Liangliang Hu
2026-10-27 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) is a cornerstone of brain-computer interfaces (BCIs) and clinical neuro-diagnostics due to its non-invasiveness and high temporal resolution. However, robust EEG decoding remains challenging, particularly for resource-constrained deployments. While lightweight architectu...

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DSGTANet: A semi-supervised dual-stream graph-temporal adaptation network for cross-subject continuous EEG emotion recognition

Wang-Yang Xu 2026-10-27 1 min 61 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: Continuous emotion recognition using electroencephalogram (EEG) is a critical topic in brain-computer inter-faces (BCIs), with applications in mental disorder regulation and human-computer interaction. However, its practical appli-cation is significantly constrained by several key issues,...

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PyPI Removing Reserver

/u/sadrasabouri 2026-10-28 1 min 147 words REDDIT PYTHON

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Code-execution flaw in UEFI firmware threatens most servers on planet Earth

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Expanding our partnership with GSA to bring Claude to all levels of government

👤 matthieu_bl 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.anthropic.com/news/expanding-partnership-gsa Comments URL: https://news.ycombinator.com/item?id=50266701 Points: 31 # Co...

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Moxnox 1.0: NT-Like, POSIX-Compatible OS Running Atop Btrfs

👤 SockThief 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

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https://github.com/maharmstone/moxnox

Git for US Legislation

👤 alansaber 📅 2026-10-28 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://lexifina.com/blog/git-for-the-us-code Comments URL: https://news.ycombinator.com/item?id=50268380 Points: 86 # Comments: 40

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New software package: ERPLAB Studio

👤 Steve Luck 📅 2024-06-12 📖 2 min 📝 449 words ERP BOOT CAMP

> **Summary:** We are excited to announce the release of a new EEG/ERP analysis package, ERPLAB Studio. We think it’s a huge improvement over the classic EEGLAB user interface. See our cheesy <a href="https://www.youtube.com/watch?v=lIaKVQ9DD6E">...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/11/erplab-studio

Brain Age Gap Associations with Externalising Behaviours in Youth: A Longitudinal Investigation in the Adolescent Brain Cognitive Development (ABCD) Cohort

👤 Asikin, A., Visontay, R., Devine, E. K., Squeglia, L. M., Byrne, H., Mewton, L.

📅 2026-10-27 📖 1 min

📝 301 words

BIORXIV NEUROSCIENCE

Summary: Background. Early adolescence is a period of heightened risk for the emergence of externalising psychopathology, making it an important window for identifying neurodevelopmental risk markers. Differences between predicted brain age and chronological age - termed the brain age gap - reflect variation...

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https://www.biorxiv.org/content/10.64898/2026.10.22.761283v1?rss=1

Scalable hardware and software for high-density, multi-regional electrophysiology in non-human primates

👤 Dubey, A., Xu, Q., Khazali, M. F., Choi, J., Wingel, K., Haggerty, J., Stefanowicz, J., Pesaran, B.

📅 2026-10-27 📖 1 min 📝 238 words BIORXIV NEUROSCIENCE

Summary: Understanding how distributed neural circuits generate complex behaviors requires long-duration recordings from populations of individual neurons across multiple brain regions. High-density Neuropixels probes have revolutionized rodent systems neuroscience. Extending these capabilities to non-human ...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.22.761135v1?rss=1

Enhancing working memories using repetitive TMS

👤 Pagnotta, M. F., D'Esposito, M.

📅 2026-10-27 📖 1 min 📝 244 words BIORXIV NEUROSCIENCE

Summary: Working memory, the ability to retain and manipulate information no longer present in the environment, is fundamental for goal-directed behavior (Baddeley, 2012; Baddeley & Hitch, 1974; D'Esposito, 2007; D'Esposito & Postle, 2015). Alpha oscillations (~8-12 Hz) have two apparently contradictory role...

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Manifolds for the neural representation of 3D spatial experience

👤 Doudlah, R., Xiao, M., Postlewaite, S., Kim, B., Cowan, N., Rosenberg, A.

📅 2026-10-27 📖 1 min 📝 182 words

BIORXIV NEUROSCIENCE

Summary: The precision with which animals navigate and interact with objects depends in part on how their brains represent position and orientation in three dimensions. A central challenge is to explain within a common framework how neural systems ranging from insects to primates represent three-dimensional ...

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Dissociable episodic memory performance and subjective experience across the aphantasia imagery spectrum

👤 Duckett, W., Simons, J.

📅 2026-10-27 📖 1 min 📝 263 words

BIORXIV NEUROSCIENCE

Summary: Mental imagery is often assumed to support episodic memory, yet whether this relationship depends on conscious access to imagery, and how reduced imagery differs from total aphantasia, remains unclear. The present study compared objective episodic memory performance, subjective trial-by-trial vividn...

🔗 **Read full article:**
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Daylight exposure improves sleep and subsequent sleepiness

Yang, M., van Bergeyk, T., Hut, R. A., Lok, R.

2026-10-27 1 min 261 words

BIORXIV NEUROSCIENCE

Summary: Study Objectives: The causal effects of daylight exposure on sleep remain poorly understood, particularly under real-world conditions. This study aimed to address this gap by examining the effects of an afternoon daylight intervention on subsequent sleep. Methods: Ten healthy participants (4 males; ...

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A vagal-to-sympathetic reflex controls lung vascular leak

Robinson, E., Ibanez, E. M., Santibanez Ocampo, B., Li, J., Ballal, N., Miller, J. M., Shivdikar, K., Umans, B. D., Prescott, S. L.

2026-10-27 1 min 194 words

BIORXIV NEUROSCIENCE

Summary: Dynamic regulation of the vascular barrier is required for an effective inflammatory response, allowing delivery of plasma proteins and immune cells to sites of insult. Although normally protective, sustained leak becomes costly in organs like the lung, where swelling stiffens tissue and impairs bre...

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https://www.biorxiv.org/content/10.64898/2026.10.22.761052v1?rss=1

Lateral-eyed tree shrews combine panoramic vision with an expanded oculomotor repertoire

Pal, R., Kempkes, E., Aeleti, R., Tubbs, E., Biswas, A., Sedigh-Sarvestani, M.

2026-10-27 1 min

243 words

BIORXIV NEUROSCIENCE

Summary: Mammals differ in how they redirect gaze. Primates shift and hold gaze with rapid, precise eye movements that can be decoupled from the head. In lateral-eyed mammals studied to date, gaze is organized around the head: saccades are coupled to head rotation in freely moving animals and typically accom...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt

2025-01-09 1 min 65 words

OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Surface-Token-Guided Query Refinement for Weakly Supervised RGB Apple Defect Sensing and Visible Surface-Condition Scoring

👤 Yasenjiang Musha 📅 2026-10-27 📖 1 min 📝 69 words LOW VISION

Summary: Low-cost RGB imaging is attractive for postharvest apple grading, yet an image-level defect label does not reveal where damage occurs or how much of the visible surface it affects. This study examines whether folder-labeled images can support defect recognition, weak localization, binary health asse...

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An Energy-Efficient Edge Vision System for Front-End Quality Monitoring in Sustainable Cold Chains: Coffee Fruit Ripeness Detection and Embedded Validation

Shirong Guo 2026-10-27 1 min 64 words LOW VISION

Summary: Reliable quality monitoring at the front end of sustainable cold chains requires direct observations of product condition in addition to environmental measurements. This study developed an energy-efficient edge vision system for six coffee fruit ripeness and visible-condition classes on an RK3576 ne...

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Text-Guided Feature Completion in the CLIP Embedding Space for Occlusion-Robust Facial Expression Recognition

Minsam Ko 2026-10-27 1 min 73 words LOW VISION

Summary: Camera-based affective sensing degrades sharply when the face is partially occluded by hands, masks, eyeglasses or headwear. Existing occlusion-robust facial expression recognition (FER) methods either discard the information carried by occluded regions through attention re-weighting, hallucinate it...

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Intelligent Sensing and Adaptive Control for Low-Loss Soybean Harvesting: A Review of Multi-Scenario Technologies

👤 Zhong Tang 📅 2026-10-27 📖 1 min 📝 60 words LOW VISION

Summary: Soybean harvesting is strongly affected by low pod position, lodging, pod shattering, moisture variation, and changing material flow, making stable low-loss operation difficult across diverse field conditions. This review synthesizes 179 core publications identified through structured literature sea...

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A 3D Vision-Based Localization Method for Automated Loading of Large-Sized Heat Exchangers

👤 Qing Jiang 📅 2026-10-27 📖 1 min 📝 62 words LOW VISION

Summary: Manual loading of large-sized finned-tube heat exchangers has low productivity and imposes substantial physical demands on workers. In addition, industrial 3D vision systems often produce large regions of missing depth data when imaging highly reflective metallic surfaces, which complicates grasp-po...

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Hybrid Vision Perception for Low-Cost Autonomous Vehicle Navigation

Abdelkader El Kamel 2026-10-27 1 min 69 words LOW VISION

Summary: High-performance perception systems for autonomous vehicles (AVs) commonly use LiDAR, RADAR, and multiple cameras. These sensors provide useful information, but their cost and computational requirements can be difficult to justify for small prototypes and low-cost platforms. This work therefore uses...

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Fidelity-oriented pelvic CBCT-to-CT synthesis using an anatomical information-attention diffusion model

Kunyuan Xu 2026-10-27 1 min 63 words LOW VISION

Summary: Cone-beam computed tomography (CBCT) is widely used in image-guided radiotherapy and adaptive radiotherapy, but its direct quantitative use is limited by scatter artifacts, low soft-tissue contrast, and inaccurate Hounsfield unit (HU) values. Diffusion models provide stable image generation capabili...

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Poros: Perception-Enabled Interactive Assistance for Appliance Use by Blind and Low Vision Users

Wei Liang 2026-10-27 1 min 62 words LOW VISION

Summary: Modern home appliances pose significant accessibility barriers for individuals with visual impairments, driven by increasing functional complexity and minimalist design trends. Drawing on formative interviews with blind and low vision (BLV) participants, we propose Poros, a novel system providing pe...

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DELUGE: Decomposed Entropy-coded Live Unstructured Geometry Exchange for Real-time Particle Streaming

Takefumi Hiraki 2026-10-27 1 min 74 words LOW VISION

Summary: Particle-based physics simulations, including fluids, smoke, and granular media, are fundamental to visual realism in immersive VR and AR. With the growing adoption of social VR and digital twins, demand is increasing for shared experiences in which multiple users interact with the same simulation i...

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A multi-attention mechanism-improved ResNet18 approach for lithology identification and classification

Ruifeng Li 2026-10-27 1 min 61 words LOW VISION

Summary: Lithology identification is vital for improving geological exploration accuracy, optimizing petroleum exploration schemes, and ensuring construction engineering safety. Traditional identification methods suffer from low efficiency, high cost, and heavy dependence on domain expertise. This study prop...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

Zhifei Xie
2026-10-24 1 min 66 words TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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In vitro carbapenem activity in Neisseria gonorrhoeae

Béatrice Berçot 2026-09-04 1 min 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan
📅 2026-09-09 📖 1 min 📝 57 words

BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Multimodal premotor stimulation during action observation transiently boosts corticospinal excitability and motor performance

👤 Ishan Mahajan 📅 2026-10-26 📖 1 min 📝 2 words TDCS TACS TRNS

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Combining Cognitive Training and Brain Stimulation: A Novel Approach to Address Disordered Eating and Antipsychotic-Induced Weight Gain in People With Severe Mental Illness

👤 Ulrike Schmidt
📅 2026-10-26 📖 1 min 📝 69 words

TDCS TACS TRNS

Summary: CONCLUSION: The initiation of antipsychotic therapy is a critical clinical window for early intervention to mitigate the long-term metabolic and psychiatric consequences of antipsychotic-induced weight gain. Combining ABMT + tDCS is acceptable and may help reduce binge eating in individuals on antip...

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High-Intensity Transcranial Alternating Current Stimulation for Post-Stroke Cognitive Impairment: A Triple-Blind Randomized Trial

Xiquan Hu 2026-10-26 1 min 60 words TDCS TACS TRNS

Summary: CONCLUSION: Hi-tACS improved global cognition and working memory in patients with PSCI and was generally well tolerated. Exploratory fNIRS findings suggested possible accompanying changes in cortical activation and network organization but did not establish a neurophysiological mechanism. Importantl...

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Comparative analysis of mitochondrial genomes in Zantedeschia parental lines and their hybrid progeny

Hongzhi Wu 2026-10-27 1 min 31 words TDCS TACS TRNS

Summary: CONCLUSIONS: This study expands current knowledge of mt genome diversity in the Araceae and provides valuable genomic resources for future studies of Zantedeschia mitogenome evolution, genetic resource conservation, and molecular breeding.

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Combined Transcranial Direct Current Stimulation and Transcutaneous Cervical Vagus Nerve Stimulation for Anxiety and Depression: An Exploratory Pilot Study

👤 Maria-Manuela Apostol
📅 2026-10-27 📖 1 min 📝 60 words

TDCS TACS TRNS

Summary: Anxiety and depressive disorders are among the leading causes of disability worldwide, yet current treatments remain limited by modest efficacy and restricted accessibility; non-invasive neuromodulation techniques such as transcranial direct current stimulation (tDCS) and transcutaneous cervical vag...

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Non-Invasive Brain Stimulation Techniques and Effects on Neuromodulation and Athletic Performance in Combat Sports: A Systematic Review

Bianca Miarka 2026-10-27 1 min 59 words TDCS TACS TRNS

Summary: CONCLUSIONS: tDCS may influence selected cognitive, neuromuscular, and fatigue-related outcomes in combat sport athletes; however, the direction and magnitude of these effects are inconsistent across studies. Given the substantial methodological heterogeneity and the low-to-very-low certainty of evi...

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Non-Invasive Brain Stimulation for Central Neuropathic Pain: An Exploratory Randomized Sham-Controlled Crossover Trial

Eunhee Sohn 2026-10-27 1 min 64 words TDCS TACS TRNS

Summary: Background/Objectives: Central neuropathic pain (CNP) is difficult to manage, and comparative evidence for non-invasive brain stimulation is limited. We explored the short-term efficacy and safety of repetitive transcranial magnetic stimulation (rTMS) and transcranial direct current stimulation (tDC...

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Noninvasive Neuromodulation for Cognitive Dysfunction in Parkinson's Disease: A Scoping Review

Reginald B O'Hara 2026-10-27 1 min 47 words TDCS TACS TRNS

Summary: CONCLUSIONS: Evidence specific to PD-MCI is limited and mixed. Several studies pairing stimulation with cognitive training reported favorable outcomes, but this pattern was inconsistent and does not establish superiority over stimulation alone. No VNS trial enrolled participants with confirmed PD-MC...

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Comparative Plastid Phylogenomics of 29 Newly Sequenced Bougainvillea Cultivars Resolves the Phylogenetic Placement of Double-Bracted Cultivars

Huiqi Zhao 2026-10-27 1 min 59 words TDCS TACS TRNS

Summary: Bougainvillea (Nyctaginaceae) cultivars are horticulturally prized worldwide for their vivid bracts; however, repeated interspecific hybridization and somatic mutations have obscured their maternal origins, particularly for double-bracted forms. Here, we sequenced and assembled the complete plastome...

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Anodal Transcranial Direct Current Stimulation Reduces Neuroinflammation and Cholinergic Dysregulation After Young Adult Traumatic Brain Injury

👤 Guven Akcay
📅 2026-10-27 📖 1 min 📝 61 words

TDCS TACS TRNS

Summary: Traumatic brain injury (TBI) during childhood and adolescence can result in persistent motor and cognitive dysfunction due to secondary injury cascades involving neuroinflammation and disruption of neurotransmitter homeostasis. Although transcranial Direct Current Stimulation (tDCS) is a promising n...

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Social relationship is associated with distinct patterns of inter-brain synchrony and directed connectivity during interpersonal coordination

Xinhong Jin 2026-10-27 1 min 59 words FNIRS

Summary: How social relationship and effector pairing relate to behavioral coordination and inter-brain interaction during joint timing remains unclear. Using functional near-infrared spectroscopy hyperscanning during a dyadic synchronization-continuation task, we examined 21 romantic couples and 23 stranger...

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Neural-Discursive Dynamics in Collaborative Spatial Problem Solving: Evidence from fNIRS Hyperscanning and Accountable Talk

👤 Xingguang Guo
📅 2026-10-27 📖 1 min 📝 65 words FNIRS

Summary: High-quality collaboration cannot be inferred simply from how much students talk or how strongly their brain activity is synchronized during joint work. This study examined relations among accountable talk, inter-brain synchrony (IBS), and group task performance in fifth-grade students' collaborativ...

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Probing Executive Function in the Toddler Data Desert: A Touchscreen fNIRS Study of Working Memory and ADHD-Related Symptoms

Caitie Busch 2026-10-27 1 min 60 words FNIRS

Summary: Attention-deficit/hyperactivity disorder (ADHD) is associated with early differences in attentional control and working memory (WM), yet relatively few paradigms are developmentally feasible across the toddler-to-preschool period while also distinguishing spatial from featural information use. Addre...

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Mapping the Global Landscape and Thematic Evolution of fNIRS-Gait Research: A Systematic Bibliometric and Network Visualization Analysis

Naime Uluğ 2026-10-27 1 min 20 words FNIRS

Summary: CONCLUSIONS: Establishing standardized preprocessing pipelines and short-separation regressions may prove pivotal for transitioning mobile neuroimaging into objective, closed-loop clinical neurorehabilitation.

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Self-Selected Versus Assigned Emotion Regulation in Pain Expectancy: Behavioral and fNIRS Evidence

👤 Ping Shi 📅 2026-10-27 📖 1 min 📝 63 words FNIRS

Summary: Background: Expectation is a powerful cognitive modulator of pain, but how different types of emotion regulation approaches interact with expectancy to influence pain processing remains unclear. The present study compared self-selected habitual strategies (cognitive reappraisal or deep breathing) wi...

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Comparative Effects of Acute HIIE and MICE on Executive Function, Prefrontal Oxyhaemoglobin Responses, and Related Biomarkers in Sedentary Young Adults: A Randomized Crossover Trial

👤 Xinyue Yang
📅 2026-10-27 📖 1 min 📝 34 words FNIRS

Summary: CONCLUSIONS: Both exercise sessions were followed by faster reaction times on selected tasks. The small sample, multiple analyses, and potential systemic contributions to fNIRS signals preclude firm conclusions about exercise-mode superiority or biological mediation.

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Cerebral Hemodynamic Patterns in People with Subjective Cognitive Decline: A Systematic Review Based on Functional Near-Infrared Spectroscopy

Israel Contador
2026-10-27 1 min 62 words FNIRS

Summary: Background/Objectives: Subjective cognitive decline (SCD) has been proposed as an early symptomatic manifestation of dementia/Alzheimer's disease. However, its associated brain patterns are not yet fully characterized. In recent years, functional near-infrared spectroscopy (fNIRS) has emerged as a c...

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Resting-State Functional Brain Connectivity and Neurodevelopmental Profiles in 3-Year-Olds with Congenital Heart Disease

👤 Anne Gallagher
📅 2026-10-27 📖 1 min 📝 67 words FNIRS

Summary: Background/Objectives: Congenital heart disease (CHD) is associated with altered brain development and an increased risk of neurodevelopmental difficulties, particularly in more severe forms of the condition. Although alterations in functional brain connectivity have been reported before and after t...

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Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan
📅 2026-10-27 📖 1 min 📝 60 words FNIRS

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

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Multimodal Sensing and Digital Biomarkers in Virtual Reality-Based Cognitive Rehabilitation: Clinical Validation, Adaptive Systems, and Real-World Translation

👤 Raffaele Ferri 📅 2026-10-27 📖 1 min 📝 58 words FNIRS

Summary: Virtual reality (VR) is evolving from a platform for cognitive exercises into an instrumented rehabilitation environment capable of simultaneously delivering treatment and quantifying behavior. This critical narrative review examines how behavioral, ocular, kinematic, physiological, and neurophysiol...

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Locomotor-State Dependence of Neuromodulation During Avian Free Flight

👤 Zhigang Shang 📅 2026-10-27 📖 1 min 📝 63 words BRAIN COMPUTER INTERFACE

Summary: Complex avian locomotor behaviors depend on the coordinated operation of neural processes involved in sensory perception, decision-making, and motor execution. Exogenous neural stimulation can alter motor output by intervening in neural processes, but its effects may depend on the current locomotor ...

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STDR-Net: A Spatio-Temporal Dependency Refinement Network for Motor Imagery EEG Decoding

Yongkang Yu 2026-10-27 1 min 56 words BRAIN COMPUTER INTERFACE

Summary: Background/Objectives: Motor imagery electroencephalography (MI-EEG) contains weak, non-stationary patterns that evolve across channels and time. Effective decoding requires local sensorimotor structure, trial-wide context, and ordered temporal evolution to be represented without discarding informat...

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Human Factors in Brain-Computer Interfaces: Boundaries, Key Issues, and Methodological Approaches

Yunfa Fu 2026-10-27 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: The large-scale, real-world deployment of brain-computer interface (BCI) is confronted with human factors bottlenecks, including reliability and usability. While modern human factors engineering has encompassed domains such as cognitive ergonomics, workload, adaptive automation, and human-in-the-loo...

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EOG Artifact Removal from EEG Using a Gated WGAN with Multi-Scale and Multi-Domain Feature Extraction

👤 Hao Tang 📅 2026-10-27 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: Background: Electroencephalogram (EEG) signals provide important information for brain-computer interfaces (BCIs) and the diagnosis of neurological disorders but are susceptible to electrooculographic (EOG) artifacts. Traditional methods often rely on multichannel data or relatively long windows, li...

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EEG Features of Cortical Intent for Urination and Defecation in Healthy Aging and Spinal Cord Injury: A Descriptive Feasibility Study for BCI Neurorehabilitation

👤 Jie Jia 📅 2026-10-27 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: Background/Objectives: Systematically elucidating the neural mechanisms of bowel and bladder control is crucial for developing closed-loop brain-computer interface (BCI) neurorehabilitation for patients with spinal cord injury (SCI). This study aims to characterize the spatiotemporal Electroencephal...

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The Development and Evolution of Substrate Materials for Brain-Computer Interfaces

👤 Yen Wei 📅 2026-10-27 📖 1 min 📝 58 words BRAIN COMPUTER INTERFACE

Summary: Brain-Computer Interfaces (BCIs) hold tremendous potential for neurological therapeutics and neuroprosthetics; however, their long-term clinical viability is fundamentally limited by chronic interfacial failure. This failure originates from the profound biomechanical mismatch between synthetic micro...

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Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan 📅 2026-10-27 📖 1 min 📝 60 words BRAIN COMPUTER INTERFACE

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

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Agentic EEG Pipeline Synthesis Using Large Language Models and the Model Context Protocol

Katy Tarrit 2026-10-27 1 min 56 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography analysis typically requires multistage processing pipelines for which configuration depends on recording characteristics and the intended analytical objective. Although automated pipelines improve reproducibility, they generally implement predefined processing sequences. We pr...

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Overt and Covert Speech-State Classification from Tripolar Concentric Ring Electrode EEG: A 4 mm and 10 mm Configuration Study

Shoaib Memon 2026-10-27 1 min 76 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) speech-state classification is a noninvasive method to identify the brain's overt and covert behavior. The EEG achieved through the scalp is contaminated by artifacts and volume conduction and suffers from poor spatial resolution. We present the first application of smal...

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The compositional encoding of hand-eye coordinated movements for single neurons in the posterior parietal cortex

👤 Richard A Andersen 📅 2026-10-27 📖 1 min 📝 77 words BRAIN COMPUTER INTERFACE

> **Summary:** Human posterior parietal cortex (PPC) is known to play a role in hand-eye (H-E) coordination, yet the underlying encoding mechanisms remain uncertain. We recorded 412 single neurons across 11 sessions from motor cortex (MC; n = 251) and PPC (n = 161) in a single human participant performing an H-E c...

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Survival LLM: AI assistant and emergency library that runs on your own computer

👤 Luc 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

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One year of Dioxus fullstack in production: the good, the painful

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Reviewers rarely talk about reliability. Owners bring it up twice as often

👤 tsxng 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

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A Hierarchical-Dynamic integration model for the formation of interpersonal neural synchrony during spoken communication

📖 1 min 📝 13 words BRAIN RESEARCH

> **Summary:** Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Yating Bai, Qingfang Zhang

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Neurotransmitter receptor architecture links local neurochemistry to semantic memory

Jung, J., Gim, J.-A., Lambon Ralph, M.
2026-10-27 1 min 244 words

BIORXIV NEUROSCIENCE

Summary: A fundamental challenge in neuroscience is understanding how the molecular and neurochemical architecture of the human cortex gives rise to sophisticated higher cognitive functions. The anterior temporal lobe (ATL) is a key hub for semantic memory, and its underlying GABAergic and glutamatergic neur...

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Heterogeneous plasticity rules generate diverse structures in recurrent networks

Cusseddu, C., Festa, D., Miehl, C., Gjorgjieva, J.
2026-10-27 1 min 154 words

BIORXIV NEUROSCIENCE

Summary: How do local changes at individual synapses give rise to structured recurrent connectivity? Experiments have revealed diverse forms of spike-timing-dependent plasticity (STDP) across the brain, making it difficult to predict which circuit structures different rules will generate. We developed an ana...

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Cerebello-basal ganglia networks and behavior across adulthood and aging

Delaney, C., Herrejon, I., Magalhaes, T. N. C., Jackson, T. B., Hicks, T. H., Bernard, J. A.

2026-10-27 1 min 293 words BIORXIV NEUROSCIENCE

Summary: The cerebellum contributes to aging, cognition, and reward related processing, but has been relatively understudied in these domains. The cerebello-basal ganglia (CB-BG) circuit may be key to better understanding how the brain ages. Prior work has implicated this circuit in aging in a small sample, ...

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Ms4a4a loss preferentially remodels microglia with limited, compartment-specific effects across central and peripheral immune compartments in 5xFAD mice

Danhash, E., Verbeck, A., Iyer, A., Karch, C. M.

2026-10-27 1 min 319 words BIORXIV NEUROSCIENCE

Summary: Background: Genome-wide association studies have identified the MS4A locus as a major modifier of Alzheimers disease (AD) risk and resilience. MS4A4A is enriched in microglia, and we previously demonstrated that loss of Ms4a4a reduces amyloid plaque pathology and attenuates amyloid-associated tau sp...

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Information coding strategies in biological reservoir computing for multiclass time-series classification

Nishi, Y., Yamamoto, H., Katori, Y., Sato, S.
2026-10-27 1 min 139 words

BIORXIV NEUROSCIENCE

Summary: Physical reservoir computing (RC) using biological neural networks (BNNs) offers a framework for probing how high-dimensional dynamics arising from multicellular systems support complex information processing in the brain. However, how electrical inputs and extracellular readouts should be designed ...

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Focal neuronal loss and cell replacement in a connectome model of fly steering

Habib, P., Petr, M., Rust, R.
2026-10-27 1 min 203 words

BIORXIV NEUROSCIENCE

Summary: Focal neuronal loss after stroke, traumatic brain injury or spinal cord injury can disrupt circuits and result in persistent motor deficits. Which cells and connections must be restored to recover a specific behavior is difficult to establish experimentally. Here we use a connectome-constrained spik...

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A canonical input to the head direction compass carries predominantly eye movement signals

Stott, J. J., Chang, H. H. V., Bao, C., Zhang, K., Cullen, K. E., van der Meer, M. A.

2026-10-27

1 min

171 words

BIORXIV NEUROSCIENCE

Summary: The mammalian head direction (HD) system is thought to update its internal compass using angular head velocity (AHV) signals relayed from the brainstem. We tested this canonical interpretation by recording from the nucleus prepositus hypoglossi (NPH), a proposed AHV relay known to be important for a...

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High-content imaging of live microglia-containing Alzheimer's disease brain organoids

Quang, D., Summers, R., Dooling, B. R., Esquer, H., Zhou, Q., LaBarbera, D. V., Vergara, M. N., Chial, H. J., Potter, H., Johnson, N. R.

2026-10-27

1 min

148 words

BIORXIV NEUROSCIENCE

Summary: Alzheimer's disease (AD) is the most common progressive neurodegenerative disease for which there is currently no safe and effective treatment. AD is characterized by the pathological aggregation of amyloid-{beta} (A{beta}) in the brain resulting in neuronal dysfunction/death and cognitive decline. ...

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Scalp EEG detects motor sequence replay signatures during wakeful rest

👤 Dash, D., Sugata, H.
📅 2026-10-27 📖 1 min 📝 146 words BIORXIV NEUROSCIENCE

Summary: Brief wakeful rest may support motor-memory processing through replay of recently practiced neural sequences, but human motor replay has been demonstrated mainly with invasive recordings or magnetoencephalography. Here we tested whether ordered motor-sequence replay can be detected from scalp EEG ac...

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Auditory tracking magnitude, temporal alignment, and anatomical distribution during wakefulness and propofol-based anesthesia in human sEEG

👤 Zhu, H., Yao, C., Han, Z., Zhou, W., Wang, Y., Li, S., Wong, P. C. M., Tian, X., Teng, X.
📅 2026-10-27 📖 1 min 📝 155 words BIORXIV NEUROSCIENCE

Summary: General anesthesia attenuates auditory responses, but response magnitude alone does not describe how residual activity aligns with sound over time or varies across the brain. We jointly characterized these aspects of auditory tracking using stereoelectroencephalography (sEEG) in 13 patients with dru...

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Feasibility assessment of novel binocular eye alignment monitor device for intermittent exotropia

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[18F]FDG brain PET-MR versus PET/CT in a clinical cohort: intra-subject quantitative comparisons

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A NEST-based framework unlocks massively parallel simulation of networks of multi-compartment neurons with customizable subcellular dynamics

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Making ‘mind pictures’ takes mental effort

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White matter fiber bundles enable robust conduction and time-critical signaling despite axon caliber variation

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Dynorphin counteracts excitatory co-transmission in a lateral hypothalamus–locus coeruleus circuit underlying stress-induced anxiety

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Ventral CA1 perineuronal nets modulate morphine-associated contextual memory via vCA1–NAcC synaptic remodeling in male mice

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Behavioral genomics identifies Creld1 as a cross-modal regulator of somatosensation

👤 Bailong Xiao 📅 2026-10-22 📖 1 min 📝 42 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 22 October 2026; doi:10.1038/s41593-026-02484-w Jiang et al. developed a robust postnatal knockout platform for somatosensory genomics and behavioral screening, identifying Creld1 as a cross...

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https://www.nature.com/articles/s41593-026-02484-w

Network geometry shapes the transformation of task representations across the human cortex

Michael W. Cole 2026-10-22 1 min 40 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 22 October 2026; doi:10.1038/s41593-026-02486-8 The dimensionality of intrinsic brain connectivity systematically shapes how task representations compress and expand across the cortex, and a...

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https://www.nature.com/articles/s41593-026-02486-8

Different modes of stimulation at the Neiguan acupoint (PC6) regulate different EEG microstates under VR immersion conditions

Guangjun Wang 2026-10-22 1 min 302 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionAcupuncture at the Neiguan acupoint (PC6) alleviates postoperative and chemotherapy-induced nausea and vomiting. Nevertheless, the central regulatory effects of PC6 stimulation on virtual reality motion sickness (VRMS), a form of physiological vertigo triggered by visual-vestibular senso...

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Retraction: Intervention effect of rehabilitation robotic bed under machine learning combined with intensive motor training on stroke patients with hemiplegia

📅 2026-10-14 📖 1 min 📝 0 words FRONTIERS NEUROROBOTICS

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https://www.frontiersin.org/articles/10.3389/fnbot.2026.2023207

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

> **Summary:** Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Safety and Efficacy of a Single Injection of Liposomal Latanoprost in Open-Angle and Angle-Closure Eyes

👤 Tina T Wong 📅 2026-10-27 📖 1 min 📝 67 words LOW VISION

> **Summary:** Background/Objectives: Glaucoma is a leading cause of irreversible global vision loss, with intraocular pressure (IOP) as the sole modifiable risk factor to stall progression. First-line treatment in the form of eye drops suffers from poor bioavailability and low patient adherence, necessitating sus...

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A Lightweight Real-Time Vision Framework for Low-Disturbance Veterinary Management and Behavioral Welfare Assessment in Captive Forest Musk Deer

👤 Ferrante Neri
📅 2026-10-27 📖 1 min 📝 61 words LOW VISION

Summary: Non-contact behavioral monitoring is important for low-disturbance husbandry of captive forest musk deer, but variable illumination, shadows, partial occlusion, and low target-background contrast can reduce recognition reliability. This study developed HGS-YOLO26n, a lightweight real-time detector f...

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Line-Scan Defect Detection for Printed Laminated Films Using Multi-Template Matching and Residual-Driven Column-Band Compensation

Wen Ren 2026-10-27 1 min 58 words LOW VISION

Summary: Machine vision enables efficient, non-contact industrial quality inspection. Online inspection of printed laminated films is challenging because complex graphics and transport-induced stretching cause appearance variations and spatially nonuniform mismatches. Global registration cannot fully correct...

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A Low-Cost Portable Machine Vision System for Non-Destructive Density Estimation of Dunaliella sp. Cultures

Han Wang 2026-10-27 1 min 61 words LOW VISION

Summary: Routine microalgal cell counting requires sampling and microscopy. We developed a low-cost portable machine-vision system (LPMVS) for non-destructive density estimation of visibly green Dunaliella sp. cultures. The compact benchtop LPMVS combines motorized flask loading, a light-isolated dark box wi...

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Surface-Token-Guided Query Refinement for Weakly Supervised RGB Apple Defect Sensing and Visible Surface-Condition Scoring

👤 Yasenjiang Musha
📅 2026-10-27 📖 1 min 📝 69 words

LOW VISION

Summary: Low-cost RGB imaging is attractive for postharvest apple grading, yet an image-level defect label does not reveal where damage occurs or how much of the visible surface it affects. This study examines whether folder-labeled images can support defect recognition, weak localization, binary health asse...

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An Energy-Efficient Edge Vision System for Front-End Quality Monitoring in Sustainable Cold Chains: Coffee Fruit Ripeness Detection and Embedded Validation

Shirong Guo 2026-10-27 1 min 64 words LOW VISION

Summary: Reliable quality monitoring at the front end of sustainable cold chains requires direct observations of product condition in addition to environmental measurements. This study developed an energy-efficient edge vision system for six coffee fruit ripeness and visible-condition classes on an RK3576 ne...

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Text-Guided Feature Completion in the CLIP Embedding Space for Occlusion-Robust Facial Expression Recognition

Minsam Ko 2026-10-27 1 min 73 words LOW VISION

Summary: Camera-based affective sensing degrades sharply when the face is partially occluded by hands, masks, eyeglasses or headwear. Existing occlusion-robust facial expression recognition (FER) methods either discard the information carried by occluded regions through attention re-weighting, hallucinate it...

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Intelligent Sensing and Adaptive Control for Low-Loss Soybean Harvesting: A Review of Multi-Scenario Technologies

👤 Zhong Tang 📅 2026-10-27 📖 1 min 📝 60 words LOW VISION

Summary: Soybean harvesting is strongly affected by low pod position, lodging, pod shattering, moisture variation, and changing material flow, making stable low-loss operation difficult across diverse field conditions. This review synthesizes 179 core publications identified through structured literature sea...

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A 3D Vision-Based Localization Method for Automated Loading of Large-Sized Heat Exchangers

👤 Qing Jiang 📅 2026-10-27 📖 1 min 📝 62 words LOW VISION

Summary: Manual loading of large-sized finned-tube heat exchangers has low productivity and imposes substantial physical demands on workers. In addition, industrial 3D vision systems often produce large regions of missing depth data when imaging highly reflective metallic surfaces, which complicates grasp-po...

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Hybrid Vision Perception for Low-Cost Autonomous Vehicle Navigation

Abdelkader El Kamel 2026-10-27 1 min 69 words LOW VISION

Summary: High-performance perception systems for autonomous vehicles (AVs) commonly use LiDAR, RADAR, and multiple cameras. These sensors provide useful information, but their cost and computational requirements can be difficult to justify for small prototypes and low-cost platforms. This work therefore uses...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

Zhifei Xie
2026-10-24 1 min 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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In vitro carbapenem activity in Neisseria gonorrhoeae

Béatrice Berçot 2026-09-04 1 min 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan
📅 2026-09-09 📖 1 min 📝 57 words

BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Multimodal premotor stimulation during action observation transiently boosts corticospinal excitability and motor performance

👤 Ishan Mahajan 📅 2026-10-26 📖 1 min 📝 2 words TDCS TACS TRNS

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Combining Cognitive Training and Brain Stimulation: A Novel Approach to Address Disordered Eating and Antipsychotic-Induced Weight Gain in People With Severe Mental Illness

👤 Ulrike Schmidt
📅 2026-10-26 📖 1 min 📝 69 words

TDCS TACS TRNS

Summary: CONCLUSION: The initiation of antipsychotic therapy is a critical clinical window for early intervention to mitigate the long-term metabolic and psychiatric consequences of antipsychotic-induced weight gain. Combining ABMT + tDCS is acceptable and may help reduce binge eating in individuals on antip...

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High-Intensity Transcranial Alternating Current Stimulation for Post-Stroke Cognitive Impairment: A Triple-Blind Randomized Trial

Xiquan Hu 2026-10-26 1 min 60 words TDCS TACS TRNS

Summary: CONCLUSION: Hi-tACS improved global cognition and working memory in patients with PSCI and was generally well tolerated. Exploratory fNIRS findings suggested possible accompanying changes in cortical activation and network organization but did not establish a neurophysiological mechanism. Importantl...

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Comparative analysis of mitochondrial genomes in Zantedeschia parental lines and their hybrid progeny

Hongzhi Wu 2026-10-27 1 min 31 words TDCS TACS TRNS

Summary: CONCLUSIONS: This study expands current knowledge of mt genome diversity in the Araceae and provides valuable genomic resources for future studies of Zantedeschia mitogenome evolution, genetic resource conservation, and molecular breeding.

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Combined Transcranial Direct Current Stimulation and Transcutaneous Cervical Vagus Nerve Stimulation for Anxiety and Depression: An Exploratory Pilot Study

👤 Maria-Manuela Apostol
📅 2026-10-27 📖 1 min 📝 60 words

TDCS TACS TRNS

Summary: Anxiety and depressive disorders are among the leading causes of disability worldwide, yet current treatments remain limited by modest efficacy and restricted accessibility; non-invasive neuromodulation techniques such as transcranial direct current stimulation (tDCS) and transcutaneous cervical vag...

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Non-Invasive Brain Stimulation Techniques and Effects on Neuromodulation and Athletic Performance in Combat Sports: A Systematic Review

👤 Bianca Miarka 📅 2026-10-27 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: CONCLUSIONS: tDCS may influence selected cognitive, neuromuscular, and fatigue-related outcomes in combat sport athletes; however, the direction and magnitude of these effects are inconsistent across studies. Given the substantial methodological heterogeneity and the low-to-very-low certainty of evi...

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Non-Invasive Brain Stimulation for Central Neuropathic Pain: An Exploratory Randomized Sham-Controlled Crossover Trial

👤 Eunhee Sohn 📅 2026-10-27 📖 1 min 📝 64 words TDCS TACS TRNS

Summary: Background/Objectives: Central neuropathic pain (CNP) is difficult to manage, and comparative evidence for non-invasive brain stimulation is limited. We explored the short-term efficacy and safety of repetitive transcranial magnetic stimulation (rTMS) and transcranial direct current stimulation (tDC...

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Noninvasive Neuromodulation for Cognitive Dysfunction in Parkinson's Disease: A Scoping Review

Reginald B O'Hara 2026-10-27 1 min 47 words TDCS TACS TRNS

Summary: CONCLUSIONS: Evidence specific to PD-MCI is limited and mixed. Several studies pairing stimulation with cognitive training reported favorable outcomes, but this pattern was inconsistent and does not establish superiority over stimulation alone. No VNS trial enrolled participants with confirmed PD-MC...

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Comparative Plastid Phylogenomics of 29 Newly Sequenced Bougainvillea Cultivars Resolves the Phylogenetic Placement of Double-Bracted Cultivars

Huiqi Zhao 2026-10-27 1 min 59 words TDCS TACS TRNS

Summary: Bougainvillea (Nyctaginaceae) cultivars are horticulturally prized worldwide for their vivid bracts; however, repeated interspecific hybridization and somatic mutations have obscured their maternal origins, particularly for double-bracted forms. Here, we sequenced and assembled the complete plastome...

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Anodal Transcranial Direct Current Stimulation Reduces Neuroinflammation and Cholinergic Dysregulation After Young Adult Traumatic Brain Injury

👤 Guven Akcay
📅 2026-10-27 📖 1 min 📝 61 words

TDCS TACS TRNS

Summary: Traumatic brain injury (TBI) during childhood and adolescence can result in persistent motor and cognitive dysfunction due to secondary injury cascades involving neuroinflammation and disruption of neurotransmitter homeostasis. Although transcranial Direct Current Stimulation (tDCS) is a promising n...

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Social relationship is associated with distinct patterns of inter-brain synchrony and directed connectivity during interpersonal coordination

👤 Xinhong Jin 📅 2026-10-27 📖 1 min 📝 59 words FNIRS

Summary: How social relationship and effector pairing relate to behavioral coordination and inter-brain interaction during joint timing remains unclear. Using functional near-infrared spectroscopy hyperscanning during a dyadic synchronization-continuation task, we examined 21 romantic couples and 23 stranger...

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Neural-Discursive Dynamics in Collaborative Spatial Problem Solving: Evidence from fNIRS Hyperscanning and Accountable Talk

👤 Xingguang Guo
📅 2026-10-27 📖 1 min 📝 65 words FNIRS

Summary: High-quality collaboration cannot be inferred simply from how much students talk or how strongly their brain activity is synchronized during joint work. This study examined relations among accountable talk, inter-brain synchrony (IBS), and group task performance in fifth-grade students' collaborativ...

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Probing Executive Function in the Toddler Data Desert: A Touchscreen fNIRS Study of Working Memory and ADHD-Related Symptoms

👤 Caitie Busch 📅 2026-10-27 📖 1 min 📝 60 words FNIRS

Summary: Attention-deficit/hyperactivity disorder (ADHD) is associated with early differences in attentional control and working memory (WM), yet relatively few paradigms are developmentally feasible across the toddler-to-preschool period while also distinguishing spatial from featural information use. Addre...

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Mapping the Global Landscape and Thematic Evolution of fNIRS-Gait Research: A Systematic Bibliometric and Network Visualization Analysis

👤 Naime Uluğ 📅 2026-10-27 📖 1 min 📝 20 words FNIRS

Summary: CONCLUSIONS: Establishing standardized preprocessing pipelines and short-separation regressions may prove pivotal for transitioning mobile neuroimaging into objective, closed-loop clinical neurorehabilitation.

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Self-Selected Versus Assigned Emotion Regulation in Pain Expectancy: Behavioral and fNIRS Evidence

👤 Ping Shi 📅 2026-10-27 📖 1 min 📝 63 words FNIRS

Summary: Background: Expectation is a powerful cognitive modulator of pain, but how different types of emotion regulation approaches interact with expectancy to influence pain processing remains unclear. The present study compared self-selected habitual strategies (cognitive reappraisal or deep breathing) wi...

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Comparative Effects of Acute HIIE and MICE on Executive Function, Prefrontal Oxyhaemoglobin Responses, and Related Biomarkers in Sedentary Young Adults: A Randomized Crossover Trial

👤 Xinyue Yang
📅 2026-10-27 📖 1 min 📝 34 words FNIRS

Summary: CONCLUSIONS: Both exercise sessions were followed by faster reaction times on selected tasks. The small sample, multiple analyses, and potential systemic contributions to fNIRS signals preclude firm conclusions about exercise-mode superiority or biological mediation.

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Cerebral Hemodynamic Patterns in People with Subjective Cognitive Decline: A Systematic Review Based on Functional Near-Infrared Spectroscopy

👤 Israel Contador
📅 2026-10-27 📖 1 min 📝 62 words FNIRS

Summary: Background/Objectives: Subjective cognitive decline (SCD) has been proposed as an early symptomatic manifestation of dementia/Alzheimer's disease. However, its associated brain patterns are not yet fully characterized. In recent years, functional near-infrared spectroscopy (fNIRS) has emerged as a c...

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Resting-State Functional Brain Connectivity and Neurodevelopmental Profiles in 3-Year-Olds with Congenital Heart Disease

👤 Anne Gallagher
📅 2026-10-27 📖 1 min 📝 67 words FNIRS

Summary: Background/Objectives: Congenital heart disease (CHD) is associated with altered brain development and an increased risk of neurodevelopmental difficulties, particularly in more severe forms of the condition. Although alterations in functional brain connectivity have been reported before and after t...

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Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan
📅 2026-10-27 📖 1 min 📝 60 words FNIRS

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

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Multimodal Sensing and Digital Biomarkers in Virtual Reality-Based Cognitive Rehabilitation: Clinical Validation, Adaptive Systems, and Real-World Translation

Raffaele Ferri 2026-10-27 1 min 58 words FNIRS

Summary: Virtual reality (VR) is evolving from a platform for cognitive exercises into an instrumented rehabilitation environment capable of simultaneously delivering treatment and quantifying behavior. This critical narrative review examines how behavioral, ocular, kinematic, physiological, and neurophysiol...

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Brain-Computer Interfaces for Cognitive Rehabilitation: A Narrative Review

Ruben I Carino-Escobar 2026-10-27 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Cognitive diseases/disorders are a major cause of disability affecting millions of people worldwide due to pathologies or conditions such as Alzheimer's disease, Parkinson's disease, vascular cognitive impairment, and attention deficit/hyperactivity disorder. The effects of these diseases/disorders ...

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Locomotor-State Dependence of Neuromodulation During Avian Free Flight

Zhigang Shang 2026-10-27 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Complex avian locomotor behaviors depend on the coordinated operation of neural processes involved in sensory perception, decision-making, and motor execution. Exogenous neural stimulation can alter motor output by intervening in neural processes, but its effects may depend on the current locomotor ...

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STDR-Net: A Spatio-Temporal Dependency Refinement Network for Motor Imagery EEG Decoding

Yongkang Yu 2026-10-27 1 min 56 words BRAIN COMPUTER INTERFACE

Summary: Background/Objectives: Motor imagery electroencephalography (MI-EEG) contains weak, non-stationary patterns that evolve across channels and time. Effective decoding requires local sensorimotor structure, trial-wide context, and ordered temporal evolution to be represented without discarding informat...

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Human Factors in Brain-Computer Interfaces: Boundaries, Key Issues, and Methodological Approaches

👤 Yunfa Fu 📅 2026-10-27 📖 1 min 📝 62 words BRAIN COMPUTER INTERFACE

Summary: The large-scale, real-world deployment of brain-computer interface (BCI) is confronted with human factors bottlenecks, including reliability and usability. While modern human factors engineering has encompassed domains such as cognitive ergonomics, workload, adaptive automation, and human-in-the-loo...

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EOG Artifact Removal from EEG Using a Gated WGAN with Multi-Scale and Multi-Domain Feature Extraction

👤 Hao Tang 📅 2026-10-27 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: Background: Electroencephalogram (EEG) signals provide important information for brain-computer interfaces (BCIs) and the diagnosis of neurological disorders but are susceptible to electrooculographic (EOG) artifacts. Traditional methods often rely on multichannel data or relatively long windows, li...

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EEG Features of Cortical Intent for Urination and Defecation in Healthy Aging and Spinal Cord Injury: A Descriptive Feasibility Study for BCI Neurorehabilitation

👤 Jie Jia 📅 2026-10-27 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: Background/Objectives: Systematically elucidating the neural mechanisms of bowel and bladder control is crucial for developing closed-loop brain-computer interface (BCI) neurorehabilitation for patients with spinal cord injury (SCI). This study aims to characterize the spatiotemporal Electroencephal...

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The Development and Evolution of Substrate Materials for Brain-Computer Interfaces

👤 Yen Wei 📅 2026-10-27 📖 1 min 📝 58 words BRAIN COMPUTER INTERFACE

Summary: Brain-Computer Interfaces (BCIs) hold tremendous potential for neurological therapeutics and neuroprosthetics; however, their long-term clinical viability is fundamentally limited by chronic interfacial failure. This failure originates from the profound biomechanical mismatch between synthetic micro...

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Bridging Temporal and Cepstral Hemodynamic Intelligence: An Adaptive Multi-Domain Feature Fusion Framework for Multi-Class fNIRS Brain-Computer Interfaces

👤 Sajjad Ahmad Khan 📅 2026-10-27 📖 1 min 📝 60 words BRAIN COMPUTER INTERFACE

Summary: Reliable four-class decoding remains challenging for functional near-infrared spectroscopy (fNIRS) brain-computer interfaces (BCIs), because a single feature domain captures only part of the hemodynamic response. This study proposes an Adaptive Multi-Domain Feature Fusion (AMFF) framework that combi...

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Agentic EEG Pipeline Synthesis Using Large Language Models and the Model Context Protocol

👤 Katy Tarrit 📅 2026-10-27 📖 1 min 📝 56 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography analysis typically requires multistage processing pipelines for which configuration depends on recording characteristics and the intended analytical objective. Although automated pipelines improve reproducibility, they generally implement predefined processing sequences. We pr...

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Overt and Covert Speech-State Classification from Tripolar Concentric Ring Electrode EEG: A 4 mm and 10 mm Configuration Study

Shoaib Memon 2026-10-27 1 min 76 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) speech-state classification is a noninvasive method to identify the brain's overt and covert behavior. The EEG achieved through the scalp is contaminated by artifacts and volume conduction and suffers from poor spatial resolution. We present the first application of smal...

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A research on the typical images of animals in the dreams of Naxi.

2026-09-10 1 min 157 words DREAMING

Summary: Animal images in dreams have long been studied, yet existing research is not systematic or in-depth. The Naxi, a small Chinese ethnic group (about 320,000), possess a rich cultural heritage. In their belief, animals mediate between humans, deities, and nature, while dreams are key communication chan...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000353

Philosophical and medical analysis of dreams in avicenna's perspective: Nature, mechanism, and functions.

📅 2026-06-15 📖 1 min 📝 218 words DREAMING

Summary: Avicenna was an eminent Islamic philosopher and physician. Drawing primarily on The Book of Healing and The Canon of Medicine, this article presents a systematic account of his theory of dreaming within the framework of his psychology and natural philosophy. Avicenna conceives drea...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000341

Threatening events and other themes in COVID-19-related dreams: Examining pandemic dreams from the perspective of current dream simulation theories.

📅 2026-06-25 📖 1 min 📝 218 words DREAMING

Summary: In this study on pandemic dream experiences in rural Nepal, 54 interviews were collected 1 year into the COVID-19 pandemic and compared with 62 interviews collected some years earlier. As in comparable studies, the dream narratives were categorized and analyzed according to their content, employing ...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000348

Correction to “Big-five personality traits and dream experiences: A systematic review and meta-analysis” by Marin et al. (2026).

📅 2026-09-10 📖 1 min 📝 399 words DREAMING

Summary: Reports an error in "Big-Five personality traits and dream experiences: A systematic review and meta-analysis" by Jai Marin, Noah Thomas, Lan Nguyen and Jared Walters (Dreaming, Advanced Online Publication, Mar 26, 2026, np; see record 2027-47158-001). The article is being made available op...

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http://doi.org/10.1037/drm0000349

Monthly Updates [March]

📅 2026-03-01 📖 2 min 📝 571 words FMHY

Summary: INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

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Monthly Updates [June]

📅 2026-06-01 📖 4 min 📝 954 words FMHY

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Monthly Updates [August]

📅 2026-08-01 📖 3 min 📝 700 words FMHY

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Monthly Updates [Sept]

📅 2026-08-31 📖 5 min 📝 1079 words FMHY

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Towards closing the gap between declaring and verifying Python dependencies

👤 /u/s5s11 📅 2026-10-27 📖 1 min 📝 129 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> ⁉️ Why does <code>pydantic>=2.11</code> sit in your <code>pyproject.toml</code>? Is it the lowest version that actually works, or just a tool default? And what evidence backs that claim? 💡 Dependency declarations are claims about package versions, yet they...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xgwbul/towards_closing_the_gap_between_declaring_and/

Pulling numbers from 10-K and 10-Q filings: XBRL or parse the document?

👤 /u/Early-Sir-932 📅 2026-10-27 📖 1 min 📝 260 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> for income statement, balance sheet and cash flow numbers from 10-k and 10-q filings there are two routes and theyre good at different things. xbrl uses the tagged data that comes with the filing. the secs companyfacts api returns every tagged fact for a comp...

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PHP2Shell, LiteLLM, GitLab: Find What You Need to Patch

👤 bigfish2026 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

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Amazon Is Gutting Its AI Division After Sustained Failure

👤 dpflan 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

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Vicious Cycles: Theses on a philosophy of news

👤 eustoria 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

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Classification of Finite Simple Groups Formalized in Lean

👤 pfdietz 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

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Police are using AI to pull over drivers for 'suspicious' behavior

👤 gumby 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

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Ghostty 1.4.0

👤 deaux 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

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https://ghostty.org/docs/install/release-notes/1-4-0

Introducing LFM3: Every Modality for Specialized Intelligence

👤 Philpax 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.liquid.ai/blog/lfm3 Comments URL: https://news.ycombinator.com/item?id=50262229 Points: 54 # Comments: 15

🔗 **Read full article:**
https://www.liquid.ai/blog/lfm3

Three TLS Bugs in GoAhead

👤 owenshen24 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://foxio.io/blog/choose-your-backend-three-tls-bugs-in-goahead Comments URL: https://news.ycombinator.com/item?id=50263701 ...

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Discord will soon let anyone find public server messages through Google

👤 DeepLogin 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://discord.com/blog/making-content-from-discoverable-servers-easier-to-discover Comments URL: <a href="https://news.ycombinator.com/item?id=50263854">https://news.ycombinato...

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https://discord.com/blog/making-content-from-discoverable-servers-easier-to-discover

US President says he will ban the term 'artificial intelligence'

👤 anigbrowl 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://thehill.com/homenews/administration/6175633-trump-bans-artificial-intelligence-term/ Comments URL: <a href="https://news.ycombinator.com/item?id=50264228">https:/...

🔗 **Read full article:**
https://thehill.com/homenews/administration/6175633-trump-bans-artificial-intelligence-term/

Recording and slides now available for ERPLAB Studio webinar

👤 Steve Luck 📅 2024-06-28 📖 1 min 📝 30 words ERP BOOT CAMP

> **Summary:** We held a webinar to demonstration ERPLAB Studio on 28 June 2024. Click here to access a recording. Click here to access a PDF of the slides.<...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/28/recording-and-slides-now-available-for-erplab-studio-webinar

New Paper: Does the P3b component reflect working memory updating?

👤 Steve Luck 📅 2025-03-21 📖 7 min 📝 1540 words ERP BOOT CAMP

> **Summary:** Carrasco, C. D., Simmons, A. M., Kiat, J. E., & Luck, S. J. (in press). Enhanced working memory representations for rare events. Psychophysiology. https://doi.org/10.1111/psyp.70038 [<a href="https://doi.org/10.1101/2024.03.20...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/3/20/new-paper-oddball

10-Day ERP Boot Camp to be held June 15-24, 2026 in Davis, California

👤 Steve Luck 📅 2025-08-20 📖 1 min 📝 125 words ERP BOOT CAMP

> **Summary:** We have received another 5 years of funding from the National Institute of Mental Health, so we plan to hold ERP Boot Camps in each of the next 5 summers. The next one will be June 15-24, 2026 in Davis, California. The application portal will open around January 1, 2026. A...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/8/20/boot-camp-summer-2026

Call for Papers: IEEE Brain Special Issue

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 36 words BRAIN

Summary: In a unique interdisciplinary collaboration with the IEEE's Society on Social Implications of Technology (SSIT) and IEEE Brain, J-FLEX is joining forces to explore both the technology of the Internet-of-Medical-Things (IoMT) solutions and medical wearables/ implantables.

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-journal-on-flexible-electronics/

IEEE Brain Joins the American Brain Coalition

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 61 words BRAIN

Summary: IEEE Brain is pleased to announce its acceptance as a nonprofit member of the American Brain Coalition (ABC), a prestigious alliance of over 150 organizations dedicated to advancing brain research, advocacy, and improving treatments for individuals affected by brain conditions. The ABC Board has ent...

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-brain-joins-the-american-brain-coalition-as-a-nonprofit-member/

Call for Papers: IEEE Transactions on Human-Machine Systems

👤 Adriel Carridice 📅 2025-06-18 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/call-for-papers-ieee-transactions-on-human-machine-systems/

Q&A with Dr. Alain Dagher, Neurologist and Professor at McGill University and the Montreal Neurological Institute

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dr-alain-dagher-neurologist-and-professor-at-mcgill-university-and-the-montreal-neurological-institute/

IEEE Advances Responsible Neurotechnology Standard

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/news-and-insights/ieee-brain-news/ieee-advances-responsible-neurotechnology-standard/

Q&A with Dhruv Vaish, PhD student at UC Berkeley and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dhruv-vaish-phd-student-at-uc-berkeley-and-best-poster-award-winner-at-the-2025-ieee-brain-discovery-and-neurotechnology-workshop/

Technical Parameters and Feedback Control for Blood-Brain Barrier Permeability Enhancement by Focused Ultrasound

📅 2025-12-02 📖 1 min 📝 166 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Focused ultrasound combined with intravenously infused microbubbles has been shown to effectively enhance the permeability of the blood-brain barrier, facilitating drug delivery to the brain. A wide range of technical parameters has been evaluated through preclinical studies and clinical trials. Gen...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11272135

Decoding Spikes From Multiunit Data

📅 2026-01-22 📖 1 min 📝 250 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Communication and control in biological systems is mediated by the timing of discharges –spikes– from excitable cells such as neurons and muscle fibers. Each spike is associated to a characteristic waveform that can be captured by sensors. The waveform's characteristics depend on the cell's biophysi...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11361153

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📅 2026-01-29 📖 1 min 📝 1 words REVIEWS BIOMEDICAL ENGINEERING

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IEEE Engineering in Medicine and Biology Society

📅 2026-01-29 📖 1 min 📝 1 words REVIEWS BIOMEDICAL ENGINEERING

🔗 **Read full article:**
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Front Cover

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Welcoming the Newly Elected EMBS AdCom Members for 2026–2028

👤 dziura 📅 2025-12-10 📖 1 min 📝 18 words EMBS

Summary: The post Welcoming the Newly Elected EMBS AdCom Members for 2026–2028 appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/uncategorized/welcoming-the-newly-elected-embs-adcom-members-for-2026-2028/

Welcoming & Congratulating Our Newly Elected EMBS ExCom Members

👤 Nancy Zimmerman 📅 2025-12-14 📖 1 min 📝 18 words EMBS

Summary: The post Welcoming & Congratulating Our Newly Elected EMBS ExCom Members appeared first on IEEE EMBS.

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EMBS AdCom Nominations Open

👤 Nancy Zimmerman 📅 2026-05-04 📖 1 min 📝 51 words EMBS

Summary: Nominate volunteers for terms starting Jan 1, 2027: Asia Pacific Rep (3 yr), Technical Rep (3 yr), Student Rep (2 yr), Young Professional Rep (2 yr). Help shape the future… <a class="continue" href="https://www.embs.org/uncategorized/adcom-nominations-2027/">Continue Reading EMBS AdCo...

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IEEE Pulse Editor-in-Chief Search – Submit an Application!

👤 Deidre Artis 📅 2026-08-28 📖 1 min 📝 17 words EMBS

> **Summary:** The post IEEE Pulse Editor-in-Chief Search – Submit an Application! appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/ieee-pulse-editor-in-chief-search-submit-and-application/

Metformin alleviates neuronal injury in MK-801-Induced Schizophrenia-Like mice by inhibiting neuronal ferroptosis

📖 1 min 📝 25 words BRAIN RESEARCH

> **Summary:** Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Yanwen Lin, Gaoxia Wen, Xiaoyue Hu, Ran Hu, Yahong Duan, Na Wu, Peng Xiong, Xianlai Jin

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Distinctive patterns of right-hemisphere vulnerability in cerebral small vessel disease: a gray matter asymmetry study

📖 1 min 📝 15 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Jingdian Lu, Yanli Wu, Lei Gao

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https://www.sciencedirect.com/science/article/pii/S0006899326004646?dgcid=rss_sd_all

N-methyl-D-aspartate receptors in the prelimbic cortex modulate chronic neuropathic pain and comorbid generalised anxiety in a rainbow boa snake threat paradigm

📖 1 min 📝 38 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Priscila Medeiros, Ana Carolina Medeiros, Ohana Turcato Macacare, Weverton Castro Coelho-Silva, Renata Moreira Acunha, Fernando René Bendaña-Córdoba, Farhad Ullah, Sérgio Henrique Ferreira, Renato L...

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Exploratory bone–brain transcriptomic associations and an ODE-based digital-twin framework for fracture healing

📖 1 min 📝 15 words BRAIN RESEARCH

> **Summary:** Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Chao Ma, Wei Lu, Ruipeng Guo

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Zona incerta circuits in cognition and the regulation of sleep and wakefulness: potential relevance to Alzheimer’s disease

📖 1 min 📝 14 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Jun Kang, Yang Li, Wenjun Yan

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State transitions in amyotrophic lateral sclerosis: mechanistic convergence and biomarker-guided stratification

📖 1 min 📝 23 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Mo Nasb, Dan Yang, Tayyab Hamid Malik, Dandan Xu, Lamis Dayoub, Jing Zhou, Yan Zhao

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Dynamic multi-scale topology learning and cross-modal graph fusion for autism spectrum disorder diagnosis

📖 1 min 📝 22 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Huadong Zhang, Junyong Zhao, Shaoqi Wu, Krisztián Szigeti, Kellermayer Miklós, Jiande Sun, Jianping Qiao

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Diagnostic classification of ASD improves with positively sparsified DTI inter-regional correlation and graph-theoretical artificial intelligence

📖 1 min 📝 22 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Ravi Ratnaik, Sriram Kumar P., Lavanya Selvaganesh, Sandeep Singh Sengar, Jac Fredo Agastinose Ronickom

🔗 **Read full article:**
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A shape deformation-based framework for constructing individualized subcortical morphological networks

📖 1 min 📝 16 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Shutong Lu, Jiashun Zhang, Junle Li, Jinhui Wang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006208?dgcid=rss_sd_all

Digital twin control framework uncovers individualized cross-scale links between functional connectivity deviations, communication impairment, and molecular mechanisms in autism

📖 1 min 📝 12 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Guanlu Liu, Jianfeng Qiu

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006269?dgcid=rss_sd_all

Decoding individual differences in mental information from human brain responses predicted by deep neural networks

📖 1 min 📝 20 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Kiichi Kawahata, Jiaxin Wang, Antoine Blanc, Naoya Maeda, Shinji Nishimoto, Satoshi Nishida

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006270?dgcid=rss_sd_all

Insula functional connectivity development follows a canonical antero-posterior axis of organization

👤 Flook, E. A., Shafiei, G., Cieslak, M., Horien, C., Luo, A. C., Meisler, S. L., Salo, T., Sydnor, V. J., Tong, T. T., Barch, D. M., Goldsmith, J., Gur, R. E., Somerville, L. H., Uddin, L. Q., Alexander-Bloch, A., Shinohara, R. T., Satterthwaite, T. D.

📅 2026-10-27 📖 1 min 📝 150 words BIORXIV NEUROSCIENCE

Summary: The insula is a heterogeneous region associated with a range of functional domains including sensory, cognitive, and emotional processing. Cytoarchitectural and functional studies have described an antero-posterior axis of insula organization. However, the insula's functional development remains spa...

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Loss of PEA-15 Alters ERK/MAPK Signaling and Circadian Function in the Suprachiasmatic Nucleus

👤 Wheaton, K., Halloy, N., Formanowicz, M., Ramos, J. W., Hoyt, K. R., Obrietan, K.

📅 2026-10-27

📖 1 min 📝 261 words BIORXIV NEUROSCIENCE

Summary: Circadian rhythms are a fundamental biological process that coordinates physiological functions, behavior, and metabolism with an approximately 24-hour periodicity. At the center of the mammalian circadian timing system is the suprachiasmatic nucleus (SCN) of the hypothalamus, which generates robust...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.25.761398v1?rss=1

Predictable Timing Enhances the Use of Predictable Content

Magnusson, O., Hobbelen, H., Berkhof, W., Sack, A. T., Leunissen, I., ten Oever, S.

2026-10-27

1 min

224 words

BIORXIV NEUROSCIENCE

Summary: As events unfold, they carry structure in both their timing and content, with these dimensions often interrelated. Most research has treated these dimensions separately, although processing them jointly may provide important computational benefits. It is an open question whether the brain processes ...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.23.760967v1?rss=1

Linking cross-species hemodynamics with neuronal activity in prefrontal cortex during abstract sequence viewing

Conen, K. E., Stewart, B., Rivera Nunez, M. V., Hyde, H., Yusif Rodriguez, N., Ahuja, A., Basu, D., McKim, T., Desrochers, T. M.

2026-10-27

1 min

199 words

BIORXIV NEUROSCIENCE

Summary: The ability to identify and predict abstract sequential patterns enables a wide range of complex cognition and behavior, but the mechanisms mediating it remain poorly understood. Lateral prefrontal cortex (LPFC) has been implicated in abstract sequence tracking in both humans and macaques, but it is...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.23.761183v1?rss=1

Habitual sleep duration is associated with the cortical organisation of brain network controllability

Liu, X., Lu, R., Wang, S., Yu, Q., Zhang, M., Cui, R., Xiao, J., Liu, X., Fan, T.

2026-10-27 1 min

148 words

BIORXIV NEUROSCIENCE

Summary: Sleep is fundamental to adaptive brain function, yet how habitual sleep relates to the structural network properties that support flexible transitions between brain states remains poorly understood. Combining wrist-accelerometer sleep measures with diffusion MRI connectomes from 15,799 UK Biobank ad...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.25.761633v1?rss=1

Familiarity synchronizes brain rhythms during natural viewing

Noad, K. N., Kaiser, D.

2026-10-27 1 min 200 words

BIORXIV NEUROSCIENCE

Summary: Our perception of the world is shaped not only by sensory information, but also by prior experience. However, how familiarity alters the coordination of neural representations across distributed brain networks remains unclear. Rhythmic neural activity provides a potential mechanism for coordinating ...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.22.760672v1?rss=1

The human-specific lncRNA HD-LINC drives lipid droplet-associated neurodegeneration in Huntington's disease

Kwon, J.-S., Wang, L., Ganti, N. R., Li, J., Xia, S., Al-Dalahmah, O., Yoo, A. S.

2026-10-27

1 min

230 words

BIORXIV NEUROSCIENCE

Summary: Huntington's disease (HD) is characterized by degeneration of striatal medium spiny neurons (MSNs), yet the human-specific regulatory mechanisms that contribute to neuronal vulnerability remain poorly understood. Here, we identify HD-LINC (LINC01139), a human-specific long intergenic noncoding RNA s...

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https://www.biorxiv.org/content/10.64898/2026.10.23.761349v1?rss=1

Neuroscience in Conversation: Tracking Scientific Change Through an Open Seminar Archive

Bellin, I., Tulner, L. K., Beyh, A., Wearn, A., Forkel, S. J.

2026-10-27

1 min

262 words

BIORXIV NEUROSCIENCE

Summary: Scientific talks capture how researchers frame questions, debate ideas, and communicate emerging methods, providing a complementary record of scientific change to the published literature. We investigated whether an open neuroscience seminar archive could reveal shifting scientific attention and con...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.22.761073v1?rss=1

Early life adversity enhances prosocial helping towards a cagemate and reshapes brain-wide neural activity and network organization

Breton, J. M., Lynch, A., Chiang, J., Wijethunga, S., Abdallah, I., Hickman-Maynard, J., Poon, Z., Ben-Ami Bartal, I., Brenhouse, H.

2026-10-27 1 min 238 words BIORXIV NEUROSCIENCE

Summary: Early life adversity (ELA) is linked to changes in social behavior and emotional regulation and to increased risk for developing mood and conduct disorders. However, the effects of ELA on empathy and its associated behaviors, including prosocial helping, remain poorly understood. Animal models of EL...

Read full article:
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Collateral-defined prefrontal-striatal circuits partition behavioral vigor during appetitive goal pursuit

Ma, X., Shao, M., Ren, B., Lei, J., Liu, Y., Hong, F., Huang, D., He, M., Zhang, S., Tai, Y., Lu, J.

2026-10-27 1 min 215 words BIORXIV NEUROSCIENCE

Summary: Motivated behavior unfolds through a sequence of appetitive processes that transform an intention to act into pursuit of a goal. Yet how behavioral vigor is regulated across distinct components of appetitive goal pursuit, and the neural mechanisms underlying this regulation, remain unclear. Here we ...

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Model selection in linear mixed-effects analysis of craving outcomes in a randomized controlled trial of medications for opioid use disorders

2026-10-27 1 min 0 words NATURE NEUROSCIENCE SUBJECTS

Read full article:
https://www.nature.com/articles/s41598-026-77858-y

Circuit mechanisms for disentangled representations and generalization in the hippocampus

Antonio Fernandez-Ruiz 2026-10-27 1 min 39 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 27 October 2026; doi:10.1038/s41593-026-02491-x The hippocampus encodes different aspects of experience as disentangled variables, enabling flexible recombination and generalization to new s...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02491-x

Stimulus-specific recruitment of human amygdala neurons predicts episodic memory encoding success

Ueli Rutishauser 2026-10-27 1 min 40 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 27 October 2026; doi:10.1038/s41593-026-02493-9 The study shows that stimulus-specific neurons in the human amygdala determine which experiences become lasting memory, revealing a mechanism ...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02493-9

Multiple sources contribute to predictive tactile attenuation in the mouse whisker system

Kalpana GuptaShubhodeep ChakrabartiCornelius Schwarzahttps://ror.org/03a1kwz48Systems Neurophysiology, Werner Reichardt Center for Integrative Neuroscience, Eberhard Karls University Tübingen, Tübingen 72076, Germanybhttps://ror.org/04zzwzx41Systems Neurophysiology, Hertie Institute for Clinical Brain Research, Eberhard Karls University Tübingen, Tübingen 72076, Germanychttps://ror.org/03a1kwz48Graduate Training Center for Neuroscience, Eberhard Karls University Tübingen, Tübingen 72076, Germany

2026-10-20 1 min 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceThe brain has evolved multiple interacting neuronal circuits that predict varying aspects of the consequences of one’s own actions. The predicted consequences range from sensory consequences of mov...

Read full article:
https://www.pnas.org/doi/abs/10.1073/pnas.2617276123?af=R

Connexins are essential for the contribution of latent progenitors to healing after spinal cord injury

María Victoria FalcoGabriela FabbianiDaniel PrietoMateo VidalMilagros BenítezMaría Inés RehermannMaría Constanza SilveraRenata SimeoneFederico F. TrigoRaúl E. Russoahttps://ror.org/05b50ej63Departamento de Neurofisiología Celular y Molecular, División Neurociencias, Instituto de Investigaciones Biológicas Clemente Estable, Montevideo CP11600, Uruguaybhttps://ror.org/05b50ej63Departamento de Señalización Neuronal y Glial, División Neurociencias, Instituto de Investigaciones Biológicas Clemente Estable, Montevideo CP11600, Uruguay

2026-10-20 1 min 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceEpendymal cells of the spinal cord are latent progenitors whose manipulation is regarded as a potential strategy to improve healing after injury. Therefore, a better understanding of the mechanisms...

Read full article:
https://www.pnas.org/doi/abs/10.1073/pnas.2605647123?af=R

CX3CL1 (fractalkine)–CX3CR1 interface drives CD8⁺ T cell–induced neurodegeneration in human brain organoids

👤 Woo Sub YangMin Sun ShinMu Seog ChoeFerdi Ridvan KiralJumee KimHong-Jai ParkMario FelixLais OsmaniIn-Hyun ParkInsoo KangaDepartment of Genetics, Interdepartmental Neuroscience Program, Yale Stem Cell Center, Yale Child Study Center, Wu Tsai Institute, Yale School of Medicine, New Haven, CT 06520bDepartment of Internal Medicine, Section of Rheumatology, Allergy & Immunology, Yale School of Medicine, New Haven, CT 06520chttps://ror.org/00vvvt117College of Pharmacy, Sookmyung Women's University, Seoul 04310, Republic of KoreadDepartment of Pathology, Yale School of Medicine, New Haven, CT 06520

📅 2026-10-20 📖 1 min 📝 45 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceNeurodegenerative diseases involve not only intrinsic neuronal pathology but also dysregulated immune responses. However, how peripheral adaptive immune cells directly influence human neurons under...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2605209123?af=R

FMRI-guided focal transcranial magnetic stimulation of anterior cingulate cortex reduces opioid seeking and restores connectivity

👤 Zilu MaYing DuanHieu NguyenSabrina LinMd Mohaiminul HaqueDanni WangSamantha HoffmanAidan CarneyTaylor ScottEleni VarlasElliot SteinZheng-Xiong XiYavin ShahamHanbing LuYihong YangaNeuroimaging Research Branch, Intramural Research Program, National Institute on Drug Abuse, National Institutes of Health, Baltimore, MD 21224bSchool of Electrical Engineering, International University, VNU-HCM, Ho Chi Minh City 71309, Vietnamchttps://ror.org/00waaqh38Viet Nam National University Ho Chi Minh City, Ho Chi Minh City 71309, Vietnamdhttps://ror.org/043mer456Department of Electrical and Electronic Engineering, University of Nebraska-Lincoln, Lincoln, NE 68588eMolecular Targets and Medication Discovery Branch, Intramural Research Program, National Institute on Drug Abuse, National Institutes of Health, Baltimore, MD 21224fBehavioral Neuroscience Branch, Intramural Research Program, National Institute on Drug Abuse, National Institutes of Health, Baltimore, MD 21224

📅 2026-10-26 📖 1 min 📝 48 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceA major challenge in developing brain stimulation therapies for opioid addiction is identifying which neural circuits to target and tracking how stimulation reshapes relapse-related circuit functi...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2619013123?af=R

Prefrontal cortical morphometric covariance and cognitive task performance in healthy young adults: an exploratory analysis of Human Connectome Project data

Roman Chivers 2026-10-22 1 min 343 words FRONTIERS HUMAN NEUROSCIENCE

Summary: The prefrontal cortex (PFC) is implicated in many cognitive tasks, including working memory, reward-based decision-making, executive control, and metacognition. This exploratory analysis investigated whether individual expression of PFC cortical thickness and surface area covariance components is as...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1976371

CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-10-22 1 min 231 words FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundMotor imagery-based brain-computer interface (MI-BCI) holds great potential in neurorehabilitation and motor assistance. Data quality and motor intention decoding effectiveness are closely associated with participants' engagement. However, previous MI tasks ignored active engagement regula...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1908608

Correction: The impact of sleep deprivation on cognitive function in healthy adults: insights from auditory P300 and reaction time analysis

Wei Wang 2026-10-27 1 min 0 words FRONTIERS NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnins.2026.2023176

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Clinical significance of "complex" multimorbidity in hospitalised adult patients

👤 Gino Roberto Corazza 📅 2026-10-24 📖 1 min 📝 65 words LOW VISION

Summary: The term "complex multimorbidity" lacks universal consensus and its relevance remains unclear. We aimed to assess the prevalence of complex multimorbidity and its association with major healthcare outcomes. This was a prospective, single-centre study (internal medicine ward). We collected sociodemog...

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Outcomes of subthreshold diode micropulse laser and anti-VEGF vs anti-VEGF monotherapy for diabetic macular edema: a meta-analysis

Marko M Popovic 2026-10-26 1 min 41 words LOW VISION

Summary: CONCLUSIONS: In patients with DME, SDMLP and anti-VEGF combination therapy appears comparable, and possibly modestly superior to anti-VEGF monotherapy, with a reduced injection burden over 12 months. BCVA and CMT differences were below clinically important thresholds and supported by low-certainty e...

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Tropospheric Ozone Assessment Report II: Ozone distribution and variability in the oceanic and polar regions

Alfonso Saiz-Lopez 2026-10-26 1 min 76 words LOW VISION

Summary: Oceanic and polar regions are critical for global atmospheric processes, but an evaluation of ozone distributions and trends in these regions has been missing. In this assessment, part of the TOAR-II project, we report lower atmosphere ozone concentrations over open oceans and polar regions: with da...

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Telemedicine-Enabled Medication Optimization for Adults with Polypharmacy: A Systematic Review and Implications for Saudi Vision 2030 Healthcare Transformation

👤 Mark Sim 📅 2026-10-26 📖 1 min 📝 32 words LOW VISION

Summary: CONCLUSIONS: Telemedicine-enabled medication optimization appears feasible, but evidence of clinical effectiveness and cost savings remains uncertain. Saudi implementation should begin with monitored pilots emphasizing safety, equity, Arabic-language accessibility, governance, and prospective evalua...

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Accurate Extraction of Phenotypic Parameters for DUS Testing of Oudemansiella raphanipes Based on Deep Learning

Linyi Li 2026-10-26 1 min 65 words LOW VISION

Summary: To enable the rapid and accurate acquisition of phenotypic parameters during distinctness, uniformity, and stability (DUS) testing of edible mushroom varieties, and to address the low efficiency, strong subjectivity, and insufficient standardization of conventional manual measurement methods, this s...

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Understanding decision making factors influencing referrals to vision rehabilitation services at a tertiary medical center

Christina L Thomas-Virnig 2026-10-26 1 min 74 words LOW VISION

Summary: Functional vision loss at any stage affects quality of life even before clinically quantified to be moderate or severe. Eye care providers as the gatekeepers to vision rehabilitation services often underestimate how small variations in functional vision loss can have a significant impact in a patien...

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Comparative longitudinal evaluation of DIMS, progressive addition and single-vision lenses on myopia progression, binocular and visual performance in Spanish children

José R Jiménez 2026-10-26 1 min 58 words LOW VISION

Summary: An 18-month prospective, non-randomized comparative observational study evaluated the effects of different optical corrections on myopia progression, binocular vision, and visual performance in Spanish children. Forty-five myopic participants aged 9-14 years were allocated to single-vision lenses (S...

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Will Artificial Intelligence Augment Global Health Inequalities?

Matthew J Burton 2026-10-26 1 min 66 words LOW VISION

Summary: AbstractArtificial intelligence (AI) has been promoted as a tool to reduce global health inequities by addressing workforce shortages and expanding access to specialist care. However, if AI development and validation are concentrated in high-income countries, innovation may disproportionately benefi...

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A secure and explainable hybrid CNN-transformer framework for trustworthy skin lesion classification

Annavarapu Chandra Sekhara Rao 2026-10-26 1 min 67 words LOW VISION

Summary: High diagnostic accountability and data integrity in digital health systems are a prerequisite for accurate early-stage classification of malignant skin lesions. Although deep learning models are remarkably successful in laboratory settings, their clinical deployment is limited due to their opaque "...

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Highly Linear Perovskite Synapses for Neuromorphic Computing

Yue Zhang 2026-10-27 1 min 65 words LOW VISION

Summary: Neuromorphic computing offers a compelling route to data-intensive applications with boosted computing power and energy efficiency. However, its accuracy for edge vision tasks is critically bottlenecked by the inherent non-linearity and asymmetry of conductance updates in analog memristive synapses....

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

Zhifei Xie
2026-10-24 1 min 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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In vitro carbapenem activity in Neisseria gonorrhoeae

Béatrice Berçot 2026-09-04 1 min 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan
📅 2026-09-09 📖 1 min 📝 57 words

BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Efficacy and Safety of Perioperative Transcranial Electrical Stimulation for Postoperative Pain and Related Outcomes: A Systematic Review and Meta-Analysis With Trial Sequential and Subgroup Analyses

👤 Qing Zhong 📅 2026-10-24 📖 1 min 📝 64 words TDCS TACS TRNS

Summary: CONCLUSIONS: Current evidence suggests that tES, as an adjuvant perioperative intervention, may alleviate postoperative pain, reduce opioid consumption, and decrease the incidence of postoperative delirium. However, the effect of tES on postoperative quality of recovery remains unclear. Considering ...

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Accelerated high-capacity 6 mA transcranial direct current stimulation for major depression

Mark S George 2026-10-25 1 min 14 words TDCS TACS TRNS

Summary: CONCLUSIONS: Accelerated 6 mA HC-tDCS was feasible, well tolerated, and warrants a sham-controlled trial.

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Rapid and robust effects of high-dose intermittent Transcranial Brain Stimulation (iTBS) on depression and suicidal ideation in adolescents with acute major depression: a 10-day randomised, controlled, single-blind trial

Yutao Lan 2026-10-26 1 min 56 words TDCS TACS TRNS

Summary: CONCLUSION: Current research indicates that a 10-day course of high-dose iTBS is effective in alleviating depressive symptoms and suicidal ideation in adolescents with depression, and that the therapeutic effects persist for up to one month after the end of treatment. We also found that high-dose iT...

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Comparative Efficacy of Noninvasive Neuromodulation in Patients with Disorders of Consciousness: A Systematic Review and Network Meta-Analysis of Randomized Controlled Trials

👤 Quan Wei
📅 2026-10-26
📖 1 min
📝 64 words
TDCS TACS TRNS

Summary: CONCLUSIONS: HF-rTMS-M1 and RMNS plus HF-rTMS-DLPFC, RMNS, HF-rTMS-DLPFC, and tDCS have shown positive signals in patients with DoC. The duration of the disorder and underlying etiology may influence the effectiveness of different noninvasive neuromodulation techniques in DoC. Given the marked heter...

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Associations of Combined Transcranial Magnetic Stimulation and Transcranial Direct Current Stimulation With Psychological and Functional Outcomes After Stroke: A Retrospective Cohort Study

👤 Meng Huang
📅 2026-10-26 📖 1 min 📝 46 words

TDCS TACS TRNS

Summary: CONCLUSIONS: In this retrospective cohort, combined TMS and tDCS with standard rehabilitation was associated with lower depression and anxiety symptom scores and more favorable neurological and motor outcome trajectories than standard rehabilitation. Non-random treatment allocation and residual conf...

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A tool to evaluate the impact of lived experience involvement in research: the Brain and Genomics Hub Impact Log protocol

Sophie E Legge 2026-10-26 1 min 74 words TDCS TACS TRNS

Summary: CONCLUSION: The potential value of this research is to implement a novel tool and framework for facilitating, recording and evaluating LEI in complex mental health research, which can be adapted for wider use. Strengths in design are LE leadership and cross-cutting LE research integration, incorpora...

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Multimodal premotor stimulation during action observation transiently boosts corticospinal excitability and motor performance

Ishan Mahajan 2026-10-26 1 min 2 words TDCS TACS TRNS

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Combining Cognitive Training and Brain Stimulation: A Novel Approach to Address Disordered Eating and Antipsychotic-Induced Weight Gain in People With Severe Mental Illness

👤 Ulrike Schmidt
📅 2026-10-26 📖 1 min 📝 69 words

TDCS TACS TRNS

Summary: CONCLUSION: The initiation of antipsychotic therapy is a critical clinical window for early intervention to mitigate the long-term metabolic and psychiatric consequences of antipsychotic-induced weight gain. Combining ABMT + tDCS is acceptable and may help reduce binge eating in individuals on antip...

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High-Intensity Transcranial Alternating Current Stimulation for Post-Stroke Cognitive Impairment: A Triple-Blind Randomized Trial

Xiquan Hu 2026-10-26 1 min 60 words TDCS TACS TRNS

Summary: CONCLUSION: Hi-tACS improved global cognition and working memory in patients with PSCI and was generally well tolerated. Exploratory fNIRS findings suggested possible accompanying changes in cortical activation and network organization but did not establish a neurophysiological mechanism. Importantl...

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Comparative analysis of mitochondrial genomes in Zantedeschia parental lines and their hybrid progeny

Hongzhi Wu 2026-10-27 1 min 31 words TDCS TACS TRNS

Summary: CONCLUSIONS: This study expands current knowledge of mt genome diversity in the Araceae and provides valuable genomic resources for future studies of Zantedeschia mitogenome evolution, genetic resource conservation, and molecular breeding.

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The test-retest reliability of cortical activation and behavioral metrics in adolescent basketball players during dual-task postural control: an fNIRS-based assessment

👤 Lulu Song 📅 2026-10-24 📖 1 min 📝 59 words FNIRS

Summary: CONCLUSION: These findings provide preliminary test-retest reliability estimates for selected multimodal neurobehavioral measures in male adolescent basketball players although reliability alone does not establish clinical utility and estimates with wide confidence intervals require confirmation in ...

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How School-Based Interventions Shape the Developing Brain during 3-12 Years: A Systematic Review and Exploratory Meta-analysis

👤 Hui Li 📅 2026-10-24 📖 1 min 📝 56 words FNIRS

Summary: The effects of school-based interventions (SBIs) on children's neurodevelopment are poorly synthesized. This systematic review integrates neuroscientific evidence on how structured educational activities influence brain development in children aged 3-12 years. Following PRISMA guidelines, we identif...

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The Walking Brain: Mapping feature-specific cortical responses to naturally unfolding sensory experience in the real world

👤 Meryem A Yücel
📅 2026-10-25 📖 1 min 📝 58 words FNIRS

Summary: Understanding how the brain processes the everyday world is fundamental, yet laboratory paradigms exclude self-generated movement, unscripted events, and a continuously evolving multisensory environment. Wearable neuroimaging enables real-world measurement, but linking neural activity to uncontrolle...

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Functional near infrared spectroscopy-based dual-stream adaptive language mapping in adults with and without post-stroke aphasia

👤 Meryem Yücel
📅 2026-10-25 📖 1 min 📝 67 words FNIRS

Summary: In aphasia, variability in lesion location and post-stroke network reorganization complicates efforts to identify the regions that support semantic versus phonological processing. In prior work, researchers have often examined these domains in separate patient cohorts, further limiting direct compar...

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How Task Load Modulates the Effect of State Anxiety on Working Memory and Its Neural Mechanisms: Evidence from Verbal and Spatial Modalities

👤 Senqing Qi 📅 2026-10-25 📖 1 min 📝 73 words FNIRS

Summary: The impact of state anxiety on working memory (WM) and its neural mechanisms, especially the role of task load, remain uncertain. To address this issue, we conducted two experiments to examine how state anxiety affects WM under different task loads and whether these effects are consistent across ver...

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Prefrontal dysfunction and impulsive decision-making in female methamphetamine use disorder: a fNIRS study

👤 Peng Lei 📅 2026-10-26 📖 1 min 📝 67 words FNIRS

Summary: CONCLUSIONS: Women with MUD showed altered intertemporal choice and reduced prefrontal hemodynamic responses in regions implicated in value evaluation and cognitive control. These findings support the relevance of prefrontal dysfunction to impulsive decision-making in female MUD while also indicatin...

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Melanin Bias in Cerebral fNIRS: A Monte Carlo Study Toward Equitable Optical Brain Sensing

👤 Murad Althobaiti
📅 2026-10-26 📖 1 min 📝 63 words FNIRS

Summary: Functional near-infrared spectroscopy (fNIRS) is an optical biosensing modality that transduces cortical hemodynamics into a measurable near-infrared signal, underpinning wearable brain monitors and brain-computer interfaces (BCIs). Because epidermal melanin is a strong near-infrared absorber, devic...

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Professional Experience Moderates Behavioral Performance and Sensorimotor-Related Cortical Activity During Scoring Decision-Making in Aerobic Gymnastics Judges: An fNIRS Study

👤 Bin Shao 📅 2026-10-26 📖 1 min 📝 58 words FNIRS

Summary: Understanding how professional experience shapes cognitive and neural resource allocation across different task demands is important for elucidating expert decision-making. This study examines whether professional experience influences behavioral performance and sensorimotor-related cortical activit...

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High-Intensity Transcranial Alternating Current Stimulation for Post-Stroke Cognitive Impairment: A Triple-Blind Randomized Trial

👤 Xiquan Hu
📅 2026-10-26 📖 1 min 📝 60 words

FNIRS

Summary: CONCLUSION: Hi-tACS improved global cognition and working memory in patients with PSCI and was generally well tolerated. Exploratory fNIRS findings suggested possible accompanying changes in cortical activation and network organization but did not establish a neurophysiological mechanism. Importantl...

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fNIRS-VFT is a complementary tool to (18)F-FDG PET in presurgical evaluation of drug-resistant epilepsy: an exploratory study

Yuanxiang Lin 2026-10-27 1 min 59 words FNIRS

Summary: CONCLUSIONS: In this exploratory, descriptive analysis, interictal ^(18)F-FDG PET showed higher concordance with lateralization of surgical areas compared with fNIRS-VFT among patients with DRE. Nevertheless, fNIRS may provide complementary information for cases with inconclusive PET findings. Addit...

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Fabrication Strategies and Performance Optimization Strategies of Flexible Brain-Computer Interface Electrodes

Jianguo Tang 2026-10-25 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) provides a direct communication pathway between the brain and external devices and has been applied in neurological disease treatment, motor rehabilitation, and human-computer interaction. Electrodes play essential roles in recording and modulating neural signals, and ...

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Melanin Bias in Cerebral fNIRS: A Monte Carlo Study Toward Equitable Optical Brain Sensing

Murad Althobaiti 2026-10-26 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Functional near-infrared spectroscopy (fNIRS) is an optical biosensing modality that transduces cortical hemodynamics into a measurable near-infrared signal, underpinning wearable brain monitors and brain-computer interfaces (BCIs). Because epidermal melanin is a strong near-infrared absorber, devic...

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A Complementary Dual-Branch Receptive Field Network for Cross-Channel-Density Motor Imagery EEG Decoding

Zuguang Rao 2026-10-26 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) enable direct communication between the brain and external devices, providing a bio-inspired link between humans and artificial systems. However, electroencephalography (EEG)-based motor imagery (MI) decoding continues to pose challenges, due to limited temporal expl...

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A Biodegradable Fish-Derived Gelatin-Chitosan Sponge as a Novel Hemostatic Agent in a Rat Tooth Extraction Model

Dwi Ariawan 2026-10-26 1 min 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: These findings demonstrate the potential of gelatin-rich fish-derived gelatin-chitosan composite sponges as biodegradable hemostatic biomaterials for oral surgical applications.

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Beyond Overlap: Development and Validation of the Inclusion of AI in Self (IAIS) Scale and Its Moderating Role in the Relationship Between AI Identity and AI Threat

Xiang Yao 2026-10-26 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Artificial intelligence (AI) is becoming increasingly integrated into people's self-concepts, yet a psychometrically robust multi-item pictorial instrument for assessing the inclusion of AI in self (IAIS) is lacking. Drawing on self-expansion theory and the AI-mediated communication (AI-MC) model, w...

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REFS: Robust EEG feature selection with incomplete labels for multi-dimensional emotion recognition

Li Zhuo 2026-10-26 1 min 61 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: Compared to single-type features, multi-type electroencephalography (EEG) features provide a multi-level representation for analyzing multi-dimensional emotions. However, the high dimensionality of multi-type EEG features, combined with the relatively small number of high-quality EEG samp...

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Theoretical system of brain-computer interface for basic research

Rongzhang Luo 2026-10-26 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) has achieved remarkable progress in brain signal acquisition, decoding algorithms, and related fields. Nevertheless, its underlying theoretical system remains largely fragmented, lacking a systematic characterization of system decodability, performance boundaries, and ...

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Multi-scale motor imagery electroencephalogram classification framework with dynamic weighted multi-source domain alignment

Jingxia Chen 2026-10-26 1 min 55 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) technology facilitates information exchange between the brain and external devices, offering new approaches to human-computer interaction and medical rehabilitation. However, distribution shifts caused by inter-subject variability in electroencephalogram (EEG) signals ...

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Mechanisms and outcome evaluation of neurofeedback in post-traumatic stress disorder intervention

Yunfa Fu 2026-10-26 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Patients with post-traumatic stress disorder (PTSD) still face challenges such as limited treatment response in some cases and substantial inter-individual variability when receiving existing interventions. Neurofeedback (NFB), as a closed-loop neuromodulation approach based on brain-computer interf...

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Auditory Rhythm Biases Visual Speed Perception and Shows Tempo-Dependent Signatures During Motion Extrapolation

Lihan Chen 2026-10-27 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: Auditory signals can bias visual motion judgements, but how dynamic auditory structure affects variable-speed motion during perception and subsequent extrapolation remains unclear. Across three experiments, we paired changing visual motion with changes in auditory tempo or pitch. In Experiment 1, sl...

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When the body suffers, the mind dreams: The lived experience of cancer.

2026-09-10 1 min 194 words DREAMING

Summary: Dreams have intrigued humanity across centuries, serving as portals to the unconscious, reflections of lived experiences, and sources of spiritual guidance. The study of dreaming spans neuroscience, psychology, anthropology, sociology, and religious thoughts, with each discipline offering unique ins...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000355

Big-Five personality traits and dream experiences: A systematic review and meta-analysis.

📅 2026-03-26 📖 1 min 📝 369 words DREAMING

Summary: [Correction Notice: An Erratum for this article was reported in Vol 36(3) of Dreaming (see record 2028-23196-002). The article is being made available open access under the CC-BY license following the Griffith University opt-in to the CAUL/ APA Read and Publish agreement. The correct copyrig...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000336

The dream insight framework: Harnessing the power of dreams in leadership.

📅 2026-08-20 📖 1 min 📝 80 words DREAMING

Summary: This article introduces the Dream Insight Framework, an approach for integrating dreamwork into leadership development. It explores how dreams, often dismissed in existing leadership approaches, can be powerful tools for enhancing creativity, decision-making, and problem-solving. By reviewing existi...

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http://doi.org/10.1037/drm0000354

Interpretability of dreams elicited by progressive-interval rapid-eye-movement and fixed-schedule awakening methods: Trajectories of dream affect, themes, and insight across the night.

📅 2026-07-30 📖 1 min 📝 201 words DREAMING

Summary: This study examined whether interpreting REM dream reports elicited using a REM-cycle-accommodating awakening protocol yielded more associations and greater insight than interpreting REM and non-REM dream reports obtained through a fixed-schedule awakening method. It also explored how interpretabili...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000350

Coping strategies and dreams—The impact of waking-life distress on negative dream emotions.

📅 2026-05-18 📖 1 min 📝 183 words DREAMING

Summary: The relationship between waking life and dreaming seems complex, and various studies show that dreams are influenced by waking-life experiences. However, only a few studies deal with the interactions between coping strategies in waking life and dream emotions. The main goal of this study is to inves...

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Software may be getting worse, but it used to be horrible

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Here Be Monsters: What Do You Still Need to Know When Agents Do the Work?

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PHP2Shell: Exploitation Attempts and Remediation Trends

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Giving chat feedback gives Anthropic your data

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The US Government Is Hiding Secret Court Ruling About How It Spies on Americans

👤 helterskelter2 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

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Toward Neuroadaptive Communication Interfaces: A Framework for Personalized Speech and Language Support

Carlos Bruno, Luis Ricardo Santos

2026-10-27 1 min 136 words ARXIV CS HC

Summary: arXiv:2610.34457v1 Announce Type: new Abstract: Speech and language technologies have expanded access to communication, but the support they provide is often fixed. The same linguistic or visual aid may help one user and hinder another, or become less useful as task demands change. This project pro...

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https://arxiv.org/abs/2610.34457

Perceived Acceptability of Sensory Augmentation

Michel Witter, Alwin de Rooij, Michel van Dartel, Emiel Krahmer

2026-10-27 1 min 153 words ARXIV CS HC

Summary: arXiv:2610.34451v1 Announce Type: new Abstract: Sensory augmentation (SA) can extend how people perceive and engage with public environments through the body, offering opportunities to support shared multimodal experiences between deaf and hearing visitors. Designing such systems requires understan...

🔗 **Read full article:**
https://arxiv.org/abs/2610.34451

Detecting and Understanding Ambiguity in Visualizations with Role-Playing Vision Language Model Interpretations

Garrick Chan, Yun-Hsin Kuo, Sam Yu-Te Lee, Yinuo Tang, Kwan-Liu Ma

2026-10-27 1 min

219 words

ARXIV CS HC

Summary: arXiv:2610.33997v1 Announce Type: new Abstract: Communicative visualizations can support multiple chart-grounded takeaways, including divergent or conflicting ones; we refer to this variation as interpretive ambiguity. Such ambiguity can undermine a visualization's communicative goal when the suppo...

🔗 **Read full article:**
https://arxiv.org/abs/2610.33997

A spinal-inspired spiking neural network for continuous finger-joint kinematics estimation from high-density surface EMG

Cristian David Guerrero-Mendez, Leonardo Abdala Elias

2026-10-27 1 min 239 words

ARXIV CS HC

Summary: arXiv:2610.33979v1 Announce Type: new Abstract: Decoding simultaneous and proportional finger kinematics from high-density surface electromyography (HD sEMG) remains challenging. Although data-driven methods have shown promising performance for finger movement decoding, their limited physiological ...

🔗 **Read full article:**
https://arxiv.org/abs/2610.33979

InteractiveWorld: A Verifiable Human-Centric Environment for Building Better Interactive MLLM Assistants

Qinsi Wang, Wei Wen, Chong Zhou, Saksham Suri, Zechun Liu, Lemeng Wu, Hai Helen Li, Yiran Chen, Raghuraman Krishnamoorthi, Qi Qian

2026-10-27 1 min 271 words ARXIV CS HC

Summary: arXiv:2610.33966v1 Announce Type: new Abstract: People have long dreamed of an assistant that is always there to guide them and remind what they forget, from new dishes to lost keys. Powered by multimodal large language models (MLLMs), egocentric AI assistants on wearable devices are bringing this ...

Read full article:
https://arxiv.org/abs/2610.33966

Human-Aware Agentic Driving: Rethinking Driver State, Trust, and Intervention

Lu Feng

2026-10-27 1 min 169 words ARXIV CS HC

Summary: arXiv:2610.33915v1 Announce Type: new Abstract: Agentic driving brings goal-directed reasoning, memory, tools, and adaptive interaction into automated vehicles. A vehicle can clarify a request, explain a maneuver, or prepare a driver to take control, then adapt its next action to the response. Thes...

Read full article:
https://arxiv.org/abs/2610.33915

FlashScript: Compiling Recurring Agent Workflows into Reusable Neuro-Symbolic Programs

👤 Sadra Sabouri, Hosein Hasanbeig, Adam Sage, Shreya Sinha, Cale Darling, Ashish Tiwari, Katie Garms, Souti Chattopadhyay, Sumit Gulwani

📅 2026-10-27 📖 1 min 📝 247 words ARXIV CS HC

Summary: arXiv:2610.33876v1 Announce Type: new Abstract: Many agent tasks recur with new data or small changes, yet agents regenerate their reasoning and actions on each run, producing different results. Distilling a prior run into a skill helps the agent avoid earlier mistakes, but it still re-runs the ful...

🔗 **Read full article:**
https://arxiv.org/abs/2610.33876

The As-If Machine: Personalized counterfactuals make distant policy risks feel close

👤 Joshua Ashkinaze, Yubo Zhou, Pengyu Chen, Avi Bhattacharya, Hyebin Park, Yanman Zhang, Stephanie Preston, Eric Gilbert, Ceren Budak

📅 2026-10-27 📖 1 min 📝 244 words ARXIV CS HC

Summary: arXiv:2610.33301v1 Announce Type: new Abstract: Life experiences can shape policy attitudes, but we only live one life. Citizens may therefore fail to support policies addressing risks they have never personally experienced. Narratives can make such risks immediate and personal, but personalized st...

🔗 **Read full article:**
https://arxiv.org/abs/2610.33301

Whale You Go Green? Immediate and Longitudinal Effects of Embodied Conversational AI Avatars on Sustainable Behavioral Tendencies and Pro-Environmental Attitudes

👤 Rachel Poonsiriwong, Pat Pataranutaporn
📅 2026-10-27 📖 1 min 📝 160 words ARXIV CS HC

Summary: arXiv:2610.33191v1 Announce Type: new Abstract: Embodied conversational AI can be persuasive, but can it motivate people to develop sustainable behavioral tendencies and increase pro-environmental attitudes? We conducted a randomized experiment (N = 576) comparing four embodied avatars--a beluga wh...

🔗 **Read full article:**
https://arxiv.org/abs/2610.33191

GlanceAR: Repurposing Phone Sensors via Low-Cost Reflectors for Always-On Obstacle Awareness

👤 Boyang Zhou, Xizhe Hao, Minbeom Kim, Jaewook Lee, Jon E. Froehlich
📅 2026-10-27 📖 1 min 📝 145 words ARXIV CS HC

Summary: arXiv:2610.33142v1 Announce Type: new Abstract: Smartphone use while walking reduces environmental awareness and can lead to accidents and injury, yet existing obstacle-awareness systems remain unadopted because they require users to hold the phone upright and aimed forward, conflicting with how pe...

🔗 **Read full article:**
https://arxiv.org/abs/2610.33142

Neural-DISCO: Source-Conditioned Counterfactual Editing of Neural Population Activity

👤 Nico Policzer, James Ho, Joseph Soo, Xin Tang

📅 2026-10-27 📖 1 min 📝 237 words

ARXIV QBIO NC

Summary: arXiv:2610.33145v1 Announce Type: cross Abstract: Understanding how neural activity encodes behavioral and sensory variables is a key challenge in systems neuroscience. This requires models that go beyond predicting activity from labels and instead support source-conditioned counterfactual editing:...

🔗 **Read full article:**
https://arxiv.org/abs/2610.33145

Simulated transcranial ultrasound of human language cortex: verifying heard speech needs far less data and signal than decoding it

👤 Omer Shalev

📅 2026-10-27 📖 1 min 📝 168 words

ARXIV QBIO NC

Summary: arXiv:2610.33093v1 Announce Type: cross Abstract: Functional ultrasound could image cortical blood volume through the temporal bone, but the skull leaves a weak signal, and task-evoked cortical imaging through the intact adult skull has not yet been shown. We built a voxel-specific Monte Carlo link...

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https://arxiv.org/abs/2610.33093

On the stability of learning

John J. Vastola
2026-10-27 1 min 158 words ARXIV QBIO NC

Summary: arXiv:2610.35856v1 Announce Type: new Abstract: Learning algorithms are usually analyzed under the assumption that informative gradients are continuously available. But this is not always true---learning signals in biological neural networks, for example, may be sparse, local, and only intermittent...

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https://arxiv.org/abs/2610.35856

Phase response approach to networked neural activity models with delays

Martin R Nelson, Jonathan J Crofts
2026-10-27 1 min 245 words ARXIV QBIO NC

Summary: arXiv:2610.35638v1 Announce Type: new Abstract: Networks of delay-coupled neural mass models alternate between synchrony and anti-phase locking as the delay between brain regions increases. The origin of these delay-dependent bands has remained unclear. Here, we show that a single function, the pha...

🔗 **Read full article:**
https://arxiv.org/abs/2610.35638

Retrieval-Augmented Parkinson's Disease Classification from Resting-State Functional Connectivity

Tara Niknam, Sana Hassan Imam, Daniel Kristanto

2026-10-27 1 min 249 words

ARXIV QBIO NC

Summary: arXiv:2610.35107v1 Announce Type: new Abstract: Altered resting-state functional connectivity (FC) is a candidate marker for the diagnosis of Parkinson's disease (PD). Publicly available PD datasets, however, are small and fragmented across sites, scanners and clinical criteria. Transfer learning m...

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https://arxiv.org/abs/2610.35107

Intrinsic Goals for Autonomous Agents: Hierarchical Exploration Captures Future Volitional Action in Drosophila

Declan Kutscher, Reece Keller, Alyn Kirsch, Felix Pei, Aran Nayebi

2026-10-27 1 min 248 words

ARXIV QBIO NC

Summary: arXiv:2610.34638v1 Announce Type: new Abstract: Animals have a remarkable capacity for autonomy---driven by intrinsic motivation---without external sensory cues or extrinsic objectives, yet the computational mechanisms behind this ability largely remain a mystery. We combine whole-brain recordings ...

🔗 **Read full article:**
https://arxiv.org/abs/2610.34638

Analysis of Biomarkers from Visual Explorations in Individuals Suspected to Be in the Prodromal Phase of Parkinson's Disease

👤 Maria F. Alcala-Durand, J. Camilo Puerta-Acevedo, Valle P\'erez del Olmo, Elisa Luque-Buzo, Juli\'an D. Arias-Londo\~no, Francisco J. Grandas-P\'erez, Juan I. Godino-Llorente

📅 2026-10-27 📖 1 min 📝 294 words ARXIV QBIO NC

Summary: arXiv:2610.34524v1 Announce Type: new Abstract: Identifying prodromal Parkinson's Disease is a precondition for any future treatment that modifies the course of the disease, since such therapies would ideally act on the initial pathogenic events, early enough to delay the clinical onset of the dise...

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https://arxiv.org/abs/2610.34524

The dominance of feedback mechanisms for perception in the central visual field: the flip tilt illusion by backward masking and the Feedforward-Feedback-Verify-reWeight algorithm

👤 Li Zhaoping 📅 2026-10-27 📖 1 min 📝 295 words ARXIV QBIO NC

Summary: arXiv:2610.34455v1 Announce Type: new Abstract: The flip tilt illusion is the misperception of the orientation of the alignment between a pair of dots by 90 degrees. It occurs when the dot-pair contains one black dot and one white dot, is typically visible only in the peripheral visual field. It is...

🔗 **Read full article:**
https://arxiv.org/abs/2610.34455

New developments in the geometry and topology of 3D head-direction coding

👤 Jean Laurens 📅 2026-10-27 📖 1 min 📝 242 words ARXIV QBIO NC

Summary: arXiv:2610.34106v1 Announce Type: new Abstract: The head-direction (HD) system is a neuronal compass that represents head orientation in 3D. Its encoding of 3D head orientation can be decomposed into a multiplexed representation of 2D head tilt relative to gravity and 1D head orientation in the hor...

🔗 **Read full article:**
https://arxiv.org/abs/2610.34106

Sensorimotor behaviour from an untrained brain-and-nerve-cord connectome

👤 John R. Minnick, Jesus Gonzalez-Ferrer, Jinghui Geng, Jason K. Eshraghian, Mohammed A. Mostajo-Radji, David Haussler, Mircea Teodorescu

📅 2026-10-27 📖 1 min 📝 204 words ARXIV QBIO NC

Summary: arXiv:2610.33881v1 Announce Type: new Abstract: Connectomes now map the neurons and synapses of an adult fly's brain and nerve cord, but how much of behaviour follows from the wiring is unknown. Here we turn one such connectome into a spiking network of 172,433 neurons with a published brain model'...

🔗 **Read full article:**
https://arxiv.org/abs/2610.33881

Peripheral pathway rerouting shapes whole-brain functional network

📖 1 min 📝 25 words NEUROSCIENCE JOURNAL

Summary: Publication date: 7 December 2026 Source: Neuroscience, Volume 618 Author(s): Jin-Jing Hu, Yun-Ting Xiang, Bei-Bei Huo, Yi-Fei Zuo, Chen-Yi Peng, He-Wang, Jia-Jia Wu, Mou-Xiong Zheng, Jian-Guang Xu

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Corrigendum to “Acupuncture combined with repetitive transcranial magnetic stimulation vs repetitive transcranial magnetic stimulation alone for recovery of hemiparesis after ischemic stroke: A systematic review and meta-analysis” [Neuroscience 616 (2026) 73–91]

📖 1 min 📝 28 words NEUROSCIENCE JOURNAL

Summary: Publication date: 7 December 2026 Source: Neuroscience, Volume 618 Author(s): Geng Li, Mengqi Yue, Shuochen Fan, Qichen Bai, Shuai Li, Xiaolin Yang, Xu Chen, Yihao Zhou, Jing Zhong, Jing Shi

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Polysaccharide-based drug conjugates and related delivery systems for Alzheimer’s disease: from molecular design to translational challenges

📖 1 min 📝 24 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Ngoc Hong Nguyen, Huong Thuy Le, Nhat Thang Thi Nguyen, Thuy Trang Nguyen, Giau Van Vo

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Partial convergence of alternative polyadenylation changes across ALS-Associated RNA-Binding proteins

📖 1 min 📝 10 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Changhe Ji

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Genetic variants in NEF2L2 rs35652124 and GABRB3 rs2081648 genes: association with autism spectrum disorder in Iraqi children

📖 1 min 📝 14 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Zainab Mohammed Burhan, Wisam Hindawi Hoidy

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Propofol suppresses intrinsic excitability and alters excitatory synaptic plasticity in ventral tegmental area dopamine neurons

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Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Min Hee Lee, Dong Kwan Kim

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Heavy-tailed connections support robust neural communication

👤 Betancourt, J. M., Varanasi, V., Clark, D. A., Lynn, C. W. 📅 2026-10-26 📖 1 min 📝 172 words

BIORXIV NEUROSCIENCE

Summary: In the synaptic wiring between neurons, some rare connections are orders of magnitude stronger than most. This heavy-tailed connectivity is ubiquitous across different species, brain regions, and cell types, suggesting that it serves a general purpose for neural function. Here, we show, both analyti...

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https://www.biorxiv.org/content/10.64898/2026.10.21.760834v1?rss=1

Somatotopic Organization of the Ventroposterior Nucleus of the Thalamus in Macaque Monkeys with Dorsal Spinal Cord Injuries Sustained During Early Postnatal Development

Jain, N., Qi, H., Raman, A., Lyon, D. C., Kaas, J. H.

2026-10-26 1 min 235 words

BIORXIV NEUROSCIENCE

Summary: Injuries to peripheral nerves or the spinal cord lead to changes in the topographic organization in the brain at multiple levels along the sensory pathways. In adult monkeys large-scale deafferentations due to transection of the dorsal columns of the spinal cord at cervical levels result in large-sc...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.21.759785v1?rss=1

Neuromuscular junction pathology in amyotrophic lateral sclerosis and emerging molecular therapeutic strategies

2026-10-27 1 min 0 words

NATURE NEUROSCIENCE SUBJECTS

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Lycium barbarum polysaccharides protect ischemic brain injury through regulating RPLP1 expression on ischemia-reperfusion nerve cells

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https://www.nature.com/articles/s41598-026-63504-0

Gender differences in leisure activity engagement and cognitive resilience among older Chinese adults: evidence from CHARLS, 2011–2020

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Communicative strategy selection in language anxiety: eye-tracking evidence of EFL learners' multimodal pragmatic regulation

📅 2026-10-27 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41539-026-00461-3

SIK3 controls synaptic localization of NMDA receptors by phosphorylating GluN1

Thomas C. Südhof 2026-10-21 1 min 41 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 21 October 2026; doi:10.1038/s41593-026-02469-9 SIK3 kinase controls synaptic NMDA receptor recruitment by phosphorylating GluN1 at a single site. Loss or inhibition of SIK3 markedly reduces...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02469-9

A corticohypothalamic circuit that mediates maternal awakening to pup cries

Feixue Liang 2026-10-21 1 min 45 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 21 October 2026; doi:10.1038/s41593-026-02458-y Tang, Wu et al. identify a brain circuit that enables mother mice to wake selectively to pup cries. Motherhood strengthens this pathway throug...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02458-y

Computational approaches to deep learning–driven EEG analysis for neurological disease diagnosis: a systematic review

Jisu Elsa Jacob 2026-10-13 1 min 229 words FRONTIERS NEUROINFORMATICS

Summary: Electroencephalography (EEG) is a widely used non-invasive neurophysiological technique for analyzing brain dynamics and diagnosing neurological diseases. However, the non-stationary nature, low signal-to-noise ratio, and inter-subject variability of EEG signals cause significant challenges to tradi...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1778980

Wave-function formalism for evolving neural dynamics: a mathematical framework informed by EEG

Madelyn Klebba 2026-10-12 1 min 268 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: The organization of distributed neural activity remains one of the central challenges in computational neuroscience. Contemporary frameworks, including Integrated Information Theory, Global Workspace Theory, entropy-based approaches, and electromagnetic field theories, have advanced our understandin...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1812909

Implementing the ILAE 2025 seizure taxonomy in medical software

Jukka Peltola 2026-10-13 1 min 241 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: The International League Against Epilepsy (ILAE), working via a multidisciplinary task force, has issued a position paper with updated seizure classification guidelines (2025) replacing its previous guidance (2017), establishing a revised de-facto standard for clinical assessment and classification ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1877135

Correction: Arts engagement, active inference, and allostatic regulation: a neurobiological framework for understanding the health effects of aesthetic experience

Pier Luigi Sacco 2026-10-13 1 min 0 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.2004139

Cross-modal alignment with language tokens for unified multilingual image-to-text generation

Guanjun Lin 2026-10-14 1 min 274 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Current multimodal translation systems typically employ a cascaded pipeline where image captioning and machine translation are performed sequentially, leading to error propagation and semantic drift. This paper introduces a unified neural architecture that integrates visual encoding, English caption...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1826455

From a surgeon's stroke to a third hemisphere—The auxilium architecture for brain restoration and extension

Mijail Demian Serruya 2026-10-21 1 min 344 words FRONTIERS HUMAN NEUROSCIENCE

Summary: A surgeon was exercising on a stationary bicycle when he had a large ischemic left middle cerebral artery territory stroke, complicated by hemorrhagic transformation requiring hemicraniectomy and cranioplasty. He developed right-sided spastic hemiplegia and a mixed aphasia. This paper takes that cas...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1893045

Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI

Christoffer Olsson 2026-10-21 1 min 245 words FRONTIERS HUMAN NEUROSCIENCE

Summary: IntroductionPredictions of the electric field (E-field) induced by transcranial direct current stimulation (tDCS) depend on how brain conductivity is modeled. White matter anisotropy is conventionally estimated from single-shell diffusion tensor imaging (DTI). Multidimensional diffusion MRI (MD-dMRI...

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Embodied multimodal human-computer interaction for English language learning: integrating speech, gestures, emotion, and adaptive feedback

Yao Tang 2026-10-14 1 min 238 words FRONTIERS NEUROROBOTICS

Summary: Current English language learning systems are often limited by single-dimensional learning-state perception, insufficient modeling of affective-cognitive characteristics, and delayed or inflexible feedback regulation. To address these limitations, this study proposes an affective-cognitive gated cro...

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Temporal signatures of prediction error reveal external forces on an embodied robot

Christian Balkenius
2026-10-14 1 min 218 words FRONTIERS NEUROROBOTICS

Summary: Embodied robots must distinguish self-generated sensory consequences from externally caused perturbations. In predictive processing, such deviation is accounted for as prediction error, but it is commonly treated primarily as a signal to be minimised. Here, we test whether the temporal profile of lo...

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Effectiveness of bimanual kinesthetic teaching through different controller settings

Andrej Gams
2026-10-21 1 min 94 words FRONTIERS NEUROROBOTICS

Summary: Physical human-robot interaction (pHRI) through kinesthetic teaching enables intuitive robot programming, but extending this approach to bimanual systems introduces challenges in synchronized control. Thus, the best way to control bimanual robot systems for kinesthetic teaching remains unclear. In t...

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Case Report: Representational geometry of multi-digit coordination in the macaque grasping network for intracortical BCIs

Hansjörg Scherberger 2026-10-27 1 min 351 words FRONTIERS NEUROSCIENCE

Summary: IntroductionIntracortical brain-computer interfaces (BCIs) hold considerable promise for restoring dexterous hand function in individuals with severe paralysis or limb loss. However, reliable control of coordinated multi-digit movements remains challenging because the cortical representation of fing...

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Stress-induced dorsal striatal dopamine release is associated with changes in functional connectivity to the prefrontal cortex in nicotine-dependent individuals—a preliminary report

Evan D. Morris 2026-10-27 1 min 322 words FRONTIERS NEUROSCIENCE

Summary: IntroductionThe neural mechanism linking stress to addiction pathophysiology remains unclear. A proposed pathway involves stress-induced dopamine (DA) transmission impairing inhibitory control, leading to relapse. Lipopolysaccharide (LPS), a biological stressor inducing systemic inflammation, amplif...

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Warfarin anticoagulation control and risks of cerebral small vessel disease and cognitive impairment among elderly patients with ischemic stroke and atrial fibrillation: a retrospective cohort study

👤 Man Zhang 📅 2026-10-27 📖 1 min 📝 318 words FRONTIERS NEUROSCIENCE

Summary: ObjectiveThis study applied individual exposure metrics derived from population pharmacokinetic-pharmacodynamic (PopPK/PD) modelling to investigate the associations between warfarin anticoagulation quality, cerebral small vessel disease (CSVD) burden, and cognitive outcomes among older adults with i...

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3D confocal imaging reveals quantitative bias in cerebral microhemorrhage counts from standard histology

👤 Bernard Choi 📅 2026-10-27 📖 1 min 📝 192 words FRONTIERS NEUROSCIENCE

Summary: Standard histological quantification of cerebral microhemorrhages (CMH), the pathologic substrate of cerebral microbleeds, may systematically undercount lesion burden due to sparse tissue sampling. CMH are visualized using Prussian blue-stained 2D tissue sections. This method provides a snapshot of ...

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Effects of brain–computer interface output modalities on motor recovery after stroke: a systematic review and network meta-analysis

👤 Lili Fang
📅 2026-10-27 📖 1 min 📝 240 words FRONTIERS NEUROSCIENCE

Summary: BackgroundThe relative effects of different brain-computer interface output modalities for post-stroke motor rehabilitation remain unclear.ObjectivesTo synthesize randomized evidence on brain–computer interface-assisted motor rehabilitation after stroke, with emphasis on functional electrical stimul...

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Ultra high-density, 4096-channel intraoperative neurophysiological brain mapping for functional localization of the human central sulcus

👤 Alexandra Fink Skular, Gelana Tostaeva, Elton Ho, Jacqueline Dister, Daniel D Cummins, Katrina Barth, Lizbeth Nunez, Adam Poole, Craig H Mermel, Benjamin Rapoport, Joshua B Bederson and Ignacio Saez

📅 2026-10-20 📖 1 min 📝 378 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Intraoperative localization of the central sulcus (CS) using somatosensory evoked-potential (SSEP) phase reversal (PR) is routinely performed with sparse subdural strip electrodes. We evaluated whether a multi-array, 4096-channel surface micro-electrocorticography (μECoG) configuration co...

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Cybertissue neuroprosthetics: a new implantable neuroprosthetics paradigm to achieve seamless lifetime integration

Liang Guo
2026-10-27 1 min 234 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Despite decades of advances in neural interface technology, the long-term instability of implantable neuroprostheses remains a major barrier to achieving reliable lifetime performance. Chronic foreign-body reactions and fibrotic encapsulation progressively degrade the electrode-tissue int...

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Combined voltammetry and optical brain imaging: toward high resolution neurochemical measurements

Vladislav Zavialov, Vassiliy Tsytsarev, Anna B Volnova, Raul R Gainetdinov and Mikhail Inyushin

2026-10-27 1 min 316 words JOURNAL NEURAL ENGINEERING

Summary: The question of the mechanisms by which cellular interactions in the brain are converted into integrative behavioral phenomena is closely related to functional localization, and also to the relationship between spatial and temporal resolution when interpreting experimental data. Classical electrophy...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 2025-06-26 1 min 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 2025-08-13 1 min 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

Robert Oostenveld 2025-08-13 1 min 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

Suzy J Styles 2025-12-08 1 min 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

Julia-Katharina Pfarr 2026-02-12 1 min 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

Robert Oostenveld 2026-02-12 1 min 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Diagnosis-Related Group Payment Systems and Hospital Efficiency, Quality, and Coding Behavior: A Systematic Review and Narrative Synthesis

👤 Wael Alsebaei 📅 2026-10-24 📖 1 min 📝 66 words LOW VISION

Summary: Diagnosis-related group (DRG)-based payment systems are used globally to incentivize hospital efficiency, yet their impact on clinical quality and documentation behavior remains variable. As Saudi Arabia implements the Australian Refined DRGs (AR-DRG) framework under Vision 2030, a synthesis of inte...

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Clinical significance of "complex" multimorbidity in hospitalised adult patients

👤 Gino Roberto Corazza 📅 2026-10-24 📖 1 min 📝 65 words LOW VISION

Summary: The term "complex multimorbidity" lacks universal consensus and its relevance remains unclear. We aimed to assess the prevalence of complex multimorbidity and its association with major healthcare outcomes. This was a prospective, single-centre study (internal medicine ward). We collected sociodemog...

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Outcomes of subthreshold diode micropulse laser and anti-VEGF vs anti-VEGF monotherapy for diabetic macular edema: a meta-analysis

Marko M Popovic 2026-10-26 1 min 41 words LOW VISION

Summary: CONCLUSIONS: In patients with DME, SDMLP and anti-VEGF combination therapy appears comparable, and possibly modestly superior to anti-VEGF monotherapy, with a reduced injection burden over 12 months. BCVA and CMT differences were below clinically important thresholds and supported by low-certainty e...

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Tropospheric Ozone Assessment Report II: Ozone distribution and variability in the oceanic and polar regions

Alfonso Saiz-Lopez 2026-10-26 1 min 76 words LOW VISION

Summary: Oceanic and polar regions are critical for global atmospheric processes, but an evaluation of ozone distributions and trends in these regions has been missing. In this assessment, part of the TOAR-II project, we report lower atmosphere ozone concentrations over open oceans and polar regions: with da...

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Telemedicine-Enabled Medication Optimization for Adults with Polypharmacy: A Systematic Review and Implications for Saudi Vision 2030 Healthcare Transformation

👤 Mark Sim
📅 2026-10-26 📖 1 min 📝 32 words LOW VISION

Summary: CONCLUSIONS: Telemedicine-enabled medication optimization appears feasible, but evidence of clinical effectiveness and cost savings remains uncertain. Saudi implementation should begin with monitored pilots emphasizing safety, equity, Arabic-language accessibility, governance, and prospective evalua...

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Accurate Extraction of Phenotypic Parameters for DUS Testing of Oudemansiella raphanipes Based on Deep Learning

Linyi Li 2026-10-26 1 min 65 words LOW VISION

Summary: To enable the rapid and accurate acquisition of phenotypic parameters during distinctness, uniformity, and stability (DUS) testing of edible mushroom varieties, and to address the low efficiency, strong subjectivity, and insufficient standardization of conventional manual measurement methods, this s...

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Understanding decision making factors influencing referrals to vision rehabilitation services at a tertiary medical center

Christina L Thomas-Virnig 2026-10-26 1 min 74 words LOW VISION

Summary: Functional vision loss at any stage affects quality of life even before clinically quantified to be moderate or severe. Eye care providers as the gatekeepers to vision rehabilitation services often underestimate how small variations in functional vision loss can have a significant impact in a patien...

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Comparative longitudinal evaluation of DIMS, progressive addition and single-vision lenses on myopia progression, binocular and visual performance in Spanish children

José R Jiménez 2026-10-26 1 min 58 words LOW VISION

Summary: An 18-month prospective, non-randomized comparative observational study evaluated the effects of different optical corrections on myopia progression, binocular vision, and visual performance in Spanish children. Forty-five myopic participants aged 9-14 years were allocated to single-vision lenses (S...

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Will Artificial Intelligence Augment Global Health Inequalities?

Matthew J Burton 2026-10-26 1 min 66 words LOW VISION

Summary: AbstractArtificial intelligence (AI) has been promoted as a tool to reduce global health inequities by addressing workforce shortages and expanding access to specialist care. However, if AI development and validation are concentrated in high-income countries, innovation may disproportionately benefi...

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A secure and explainable hybrid CNN-transformer framework for trustworthy skin lesion classification

Annavarapu Chandra Sekhara Rao 2026-10-26 1 min 67 words LOW VISION

Summary: High diagnostic accountability and data integrity in digital health systems are a prerequisite for accurate early-stage classification of malignant skin lesions. Although deep learning models are remarkably successful in laboratory settings, their clinical deployment is limited due to their opaque "...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

Zhifei Xie
2026-10-24 1 min 66 words TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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In vitro carbapenem activity in Neisseria gonorrhoeae

Béatrice Berçot 2026-09-04 1 min 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan
📅 2026-09-09 📖 1 min 📝 57 words

BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Disrupting objectification: transcranial Direct Current Stimulation (tDCS) enhances the inversion effect toward sexualized women

👤 Arwen Stephens 📅 2026-10-24 📖 1 min 📝 68 words TDCS TACS TRNS

Summary: Sexual objectification is associated with viewing women in an object-like manner, often indexed perceptually by a reduced inversion effect for sexualized women relative to men. Prior research has shown that this index can be attenuated through social manipulations; here we investigated whether it ca...

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Efficacy and Safety of Perioperative Transcranial Electrical Stimulation for Postoperative Pain and Related Outcomes: A Systematic Review and Meta-Analysis With Trial Sequential and Subgroup Analyses

Qing Zhong 2026-10-24 1 min 64 words TDCS TACS TRNS

Summary: CONCLUSIONS: Current evidence suggests that tES, as an adjuvant perioperative intervention, may alleviate postoperative pain, reduce opioid consumption, and decrease the incidence of postoperative delirium. However, the effect of tES on postoperative quality of recovery remains unclear. Considering ...

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Accelerated high-capacity 6 mA transcranial direct current stimulation for major depression

Mark S George 2026-10-25 1 min 14 words TDCS TACS TRNS

Summary: CONCLUSIONS: Accelerated 6 mA HC-tDCS was feasible, well tolerated, and warrants a sham-controlled trial.

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Rapid and robust effects of high-dose intermittent Transcranial Brain Stimulation (iTBS) on depression and suicidal ideation in adolescents with acute major depression: a 10-day randomised, controlled, single-blind trial

👤 Yutao Lan 📅 2026-10-26 📖 1 min 📝 56 words TDCS TACS TRNS

Summary: CONCLUSION: Current research indicates that a 10-day course of high-dose iTBS is effective in alleviating depressive symptoms and suicidal ideation in adolescents with depression, and that the therapeutic effects persist for up to one month after the end of treatment. We also found that high-dose iT...

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Comparative Efficacy of Noninvasive Neuromodulation in Patients with Disorders of Consciousness: A Systematic Review and Network Meta-Analysis of Randomized Controlled Trials

👤 Quan Wei
📅 2026-10-26
📖 1 min
📝 64 words
TDCS TACS TRNS

Summary: CONCLUSIONS: HF-rTMS-M1 and RMNS plus HF-rTMS-DLPFC, RMNS, HF-rTMS-DLPFC, and tDCS have shown positive signals in patients with DoC. The duration of the disorder and underlying etiology may influence the effectiveness of different noninvasive neuromodulation techniques in DoC. Given the marked heter...

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Associations of Combined Transcranial Magnetic Stimulation and Transcranial Direct Current Stimulation With Psychological and Functional Outcomes After Stroke: A Retrospective Cohort Study

👤 Meng Huang
📅 2026-10-26 📖 1 min 📝 46 words

TDCS TACS TRNS

Summary: CONCLUSIONS: In this retrospective cohort, combined TMS and tDCS with standard rehabilitation was associated with lower depression and anxiety symptom scores and more favorable neurological and motor outcome trajectories than standard rehabilitation. Non-random treatment allocation and residual conf...

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A tool to evaluate the impact of lived experience involvement in research: the Brain and Genomics Hub Impact Log protocol

Sophie E Legge 2026-10-26 1 min 74 words TDCS TACS TRNS

Summary: CONCLUSION: The potential value of this research is to implement a novel tool and framework for facilitating, recording and evaluating LEI in complex mental health research, which can be adapted for wider use. Strengths in design are LE leadership and cross-cutting LE research integration, incorpora...

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Multimodal premotor stimulation during action observation transiently boosts corticospinal excitability and motor performance

Ishan Mahajan 2026-10-26 1 min 2 words TDCS TACS TRNS

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Combining Cognitive Training and Brain Stimulation: A Novel Approach to Address Disordered Eating and Antipsychotic-Induced Weight Gain in People With Severe Mental Illness

👤 Ulrike Schmidt
📅 2026-10-26 📖 1 min 📝 69 words

TDCS TACS TRNS

Summary: CONCLUSION: The initiation of antipsychotic therapy is a critical clinical window for early intervention to mitigate the long-term metabolic and psychiatric consequences of antipsychotic-induced weight gain. Combining ABMT + tDCS is acceptable and may help reduce binge eating in individuals on antip...

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High-Intensity Transcranial Alternating Current Stimulation for Post-Stroke Cognitive Impairment: A Triple-Blind Randomized Trial

Xiquan Hu 2026-10-26 1 min 60 words TDCS TACS TRNS

Summary: CONCLUSION: Hi-tACS improved global cognition and working memory in patients with PSCI and was generally well tolerated. Exploratory fNIRS findings suggested possible accompanying changes in cortical activation and network organization but did not establish a neurophysiological mechanism. Importantl...

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Correction-stage information leakage and target-input asymmetry can distort comparisons of fNIRS short-separation channel correction strategies

Esra Suzen 2026-10-23 1 min 16 words FNIRS

Summary: OBJECTIVE: We examined how evaluation design affects comparisons of functional near-infrared spectroscopy (fNIRS) short-separation correction strategies.

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The test-retest reliability of cortical activation and behavioral metrics in adolescent basketball players during dual-task postural control: an fNIRS-based assessment

👤 Lulu Song 📅 2026-10-24 📖 1 min 📝 59 words FNIRS

Summary: CONCLUSION: These findings provide preliminary test-retest reliability estimates for selected multimodal neurobehavioral measures in male adolescent basketball players although reliability alone does not establish clinical utility and estimates with wide confidence intervals require confirmation in ...

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How School-Based Interventions Shape the Developing Brain during 3-12 Years: A Systematic Review and Exploratory Meta-analysis

👤 Hui Li 📅 2026-10-24 📖 1 min 📝 56 words FNIRS

Summary: The effects of school-based interventions (SBIs) on children's neurodevelopment are poorly synthesized. This systematic review integrates neuroscientific evidence on how structured educational activities influence brain development in children aged 3-12 years. Following PRISMA guidelines, we identif...

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The Walking Brain: Mapping feature-specific cortical responses to naturally unfolding sensory experience in the real world

Meryem A Yücel 2026-10-25 1 min 58 words FNIRS

Summary: Understanding how the brain processes the everyday world is fundamental, yet laboratory paradigms exclude self-generated movement, unscripted events, and a continuously evolving multisensory environment. Wearable neuroimaging enables real-world measurement, but linking neural activity to uncontrolle...

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Functional near infrared spectroscopy-based dual-stream adaptive language mapping in adults with and without post-stroke aphasia

👤 Meryem Yücel
📅 2026-10-25 📖 1 min 📝 67 words FNIRS

Summary: In aphasia, variability in lesion location and post-stroke network reorganization complicates efforts to identify the regions that support semantic versus phonological processing. In prior work, researchers have often examined these domains in separate patient cohorts, further limiting direct compar...

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How Task Load Modulates the Effect of State Anxiety on Working Memory and Its Neural Mechanisms: Evidence from Verbal and Spatial Modalities

Senqing Qi 2026-10-25 1 min 73 words FNIRS

Summary: The impact of state anxiety on working memory (WM) and its neural mechanisms, especially the role of task load, remain uncertain. To address this issue, we conducted two experiments to examine how state anxiety affects WM under different task loads and whether these effects are consistent across ver...

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Prefrontal dysfunction and impulsive decision-making in female methamphetamine use disorder: a fNIRS study

Peng Lei 2026-10-26 1 min 67 words FNIRS

Summary: CONCLUSIONS: Women with MUD showed altered intertemporal choice and reduced prefrontal hemodynamic responses in regions implicated in value evaluation and cognitive control. These findings support the relevance of prefrontal dysfunction to impulsive decision-making in female MUD while also indicatin...

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Melanin Bias in Cerebral fNIRS: A Monte Carlo Study Toward Equitable Optical Brain Sensing

Murad Althobaiti
2026-10-26 1 min 63 words FNIRS

Summary: Functional near-infrared spectroscopy (fNIRS) is an optical biosensing modality that transduces cortical hemodynamics into a measurable near-infrared signal, underpinning wearable brain monitors and brain-computer interfaces (BCIs). Because epidermal melanin is a strong near-infrared absorber, devic...

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Professional Experience Moderates Behavioral Performance and Sensorimotor-Related Cortical Activity During Scoring Decision-Making in Aerobic Gymnastics Judges: An fNIRS Study

👤 Bin Shao 📅 2026-10-26 📖 1 min 📝 58 words FNIRS

Summary: Understanding how professional experience shapes cognitive and neural resource allocation across different task demands is important for elucidating expert decision-making. This study examines whether professional experience influences behavioral performance and sensorimotor-related cortical activit...

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High-Intensity Transcranial Alternating Current Stimulation for Post-Stroke Cognitive Impairment: A Triple-Blind Randomized Trial

Xiquan Hu 2026-10-26 1 min 60 words FNIRS

Summary: CONCLUSION: Hi-tACS improved global cognition and working memory in patients with PSCI and was generally well tolerated. Exploratory fNIRS findings suggested possible accompanying changes in cortical activation and network organization but did not establish a neurophysiological mechanism. Importantl...

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Correction: Perioperative management for invasive brain-computer interface implantation under surgical navigation in a patient with traumatic spinal cord injury: a case report

Wenyao Cui 2026-10-24 1 min 6 words BRAIN COMPUTER INTERFACE

Summary: [This corrects the article DOI: 10.3389/fnhum.2026.1892331.].

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Limits of Trial-Adaptive Neural-Language Fusion Across Large Language Models in P300 Brain-Computer Interfaces

Yiftach Barash 2026-10-24 1 min 21 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: A tuned fixed weight is a difficult-to-beat default across the tested scale range; reliability estimation gave no deployable adaptive advantage.

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Creating online cognitive assessment platform for children's literacy and math skills: a case study illustrated with the ACE kids platform

U Maurer 2026-10-24 1 min 70 words BRAIN COMPUTER INTERFACE

Summary: Computerized versions of paper-and-pencil tests made behavioral assessments of children more automatic and precise. The COVID-19 pandemic further demonstrated the need for assessments to be offered online to allow for remote data collection. However, the design and implementation of computerized tes...

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Combined neuromodulation techniques via a central-peripheral-central strategy facilitate functional recovery from AIS A to AIS C in a patient with complete spinal cord injury: A case report

Liansheng Zhong
2026-10-24 1 min 48 words BRAIN COMPUTER INTERFACE

Summary: CONTEXT: Complete spinal cord injury (SCI) has traditionally been considered irreversible. However, accumulating evidence suggests that subclinical neural pathways may persist. Multimodal combined neuromodulation targeting brain-spinal-peripheral circuits holds promise to unlock latent neural plasti...

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Fabrication Strategies and Performance Optimization Strategies of Flexible Brain-Computer Interface Electrodes

Jianguo Tang 2026-10-25 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) provides a direct communication pathway between the brain and external devices and has been applied in neurological disease treatment, motor rehabilitation, and human-computer interaction. Electrodes play essential roles in recording and modulating neural signals, and ...

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Melanin Bias in Cerebral fNIRS: A Monte Carlo Study Toward Equitable Optical Brain Sensing

Murad Althobaiti 2026-10-26 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Functional near-infrared spectroscopy (fNIRS) is an optical biosensing modality that transduces cortical hemodynamics into a measurable near-infrared signal, underpinning wearable brain monitors and brain-computer interfaces (BCIs). Because epidermal melanin is a strong near-infrared absorber, devic...

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A Complementary Dual-Branch Receptive Field Network for Cross-Channel-Density Motor Imagery EEG Decoding

Zuguang Rao 2026-10-26 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) enable direct communication between the brain and external devices, providing a bio-inspired link between humans and artificial systems. However, electroencephalography (EEG)-based motor imagery (MI) decoding continues to pose challenges, due to limited temporal expl...

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A Biodegradable Fish-Derived Gelatin-Chitosan Sponge as a Novel Hemostatic Agent in a Rat Tooth Extraction Model

Dwi Ariawan 2026-10-26 1 min 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: These findings demonstrate the potential of gelatin-rich fish-derived gelatin-chitosan composite sponges as biodegradable hemostatic biomaterials for oral surgical applications.

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Beyond Overlap: Development and Validation of the Inclusion of AI in Self (IAIS) Scale and Its Moderating Role in the Relationship Between AI Identity and AI Threat

👤 Xiang Yao 📅 2026-10-26 📖 1 min 📝 68 words BRAIN COMPUTER INTERFACE

Summary: Artificial intelligence (AI) is becoming increasingly integrated into people's self-concepts, yet a psychometrically robust multi-item pictorial instrument for assessing the inclusion of AI in self (IAIS) is lacking. Drawing on self-expansion theory and the AI-mediated communication (AI-MC) model, w...

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REFS: Robust EEG feature selection with incomplete labels for multi-dimensional emotion recognition

👤 Li Zhuo 📅 2026-10-26 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: Compared to single-type features, multi-type electroencephalography (EEG) features provide a multi-level representation for analyzing multi-dimensional emotions. However, the high dimensionality of multi-type EEG features, combined with the relatively small number of high-quality EEG samp...

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Implicit learning of melodic structure: A role for pitch?

📅 2024-01-22 📖 1 min 📝 180 words PSYCHOMUSICOLOGY

Summary: Growing evidence suggests that pitch influences musical processing, with melodic processing being enhanced in higher pitch ranges (e.g., Fujioka et al., 2005) and rhythmic processing being enhanced in lower pitches, and these effects may have a basis in elementary properties of the auditory system (...

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http://doi.org/10.1037/pmu0000303

To those who feel unloved, misunderstood, or afraid

📅 2026-09-22 📖 3 min 📝 687 words FMHY

Summary: I understand the world can feel devoid of love, but I swear, there is a spark within humanity that has the capability to care deeply. To love unconditionally. It doesn't always shine through us all, but it exists, it's real, it's tangible, and no one is exempt from it. We often see it shin...

🔗 **Read full article:**
https://fmhy.net/posts/love

I still teach OpenGL ES 3.0

👤 majorchord 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://eliasfarhan.ch/graphics/cpp/2026/01/27/why-i-teach-opengles.html Comments URL: <a href="https://news.ycombinator.com/item?id=50253939">https://news.ycombinator.com/item?id=50253939</...

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Writing about pratt parsing so I remember it

👤 dverlaeckt80 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.unix.dog/~yosh/blog/2026-10-25_pratt-parsing.html Comments URL: https://news.ycombinator.com/item?id=50255177 Points: 15</...

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The Mythical Man-Month in the Age of AI

👤 ilreb 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

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A simplified proof of ETH is False [pdf]

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Easily download files from the Open Science Framework with Papercheck

noreply@blogger.com (Daniel Lakens)
2025-07-22 3 min 765 words

TWENTY PERCENT STATISTICIAN

Summary: <p>Researchers increasingly use the Open Science Framework (OSF) to share files, such as data and code underlying scientific publications, or presentations and materials for scientific workshops. The OSF is an amazing service that has contributed immensely to a changed ...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/07/easily-download-files-from-open-science.html

How I use AI in my work

noreply@blogger.com (Daniel Lakens)
2026-10-11 21 min 4292 words

TWENTY PERCENT STATISTICIAN

Summary: <p>Tomorrow we will have a discussion in our lab about how to use AI as a scientist. I have increasingly started to use AI, having moved from Copilot Chat that our university provides last year, to some free credit on GitHub Copilot under their educational program, to Claude Pro integrated in Visual...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/10/how-i-use-ai-in-my-work.html

Active Lubrication Brakes in Tabletop and Hand-Held Haptic Surgical Drilling Simulators

📅 2026-06-29 📖 1 min 📝 222 words TRANSACTIONS HAPTICS

Summary: Haptic user interfaces that provide both kinesthetic and vibrotactile feedback have strong potential to enhance a wide range of medical training simulations. Yet, designing such devices remains challenging due to constraints related to integration, workspace, cost, and the need for diverse feedback ...

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http://ieeexplore.ieee.org/document/11585967

Reconciling Reduced the Sense of Ownership and Agency Toward a Virtual Keypad With Enhanced Haptic Experience Using Visually Adjusted Self-Haptics for Key-Input Interaction

📅 2026-07-06 📖 1 min 📝 250 words TRANSACTIONS HAPTICS

Summary: Self-Haptics is a tactile feedback technique where users utilize their own body to generate tactile sensations. This method allows the body to act purely as a prop, making it useful for virtual objects and user interfaces not directly related to the body. However, any method for maintaining the sens...

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http://ieeexplore.ieee.org/document/11595768

VibGrasp: Spatiotemporal Vibration Based Multimodal Haptic Rendering With a Lightweight Exo-Glove for 3D Shape Perception

📅 2026-04-20 📖 1 min 📝 173 words TRANSACTIONS HAPTICS

Summary: Haptic feedback has been employed to enhance shape perception during grasp interactions, improving immersion and task performance in extended reality (XR). Although wearable devices offer precise multi-phalangeal haptic feedback, their bulkiness and complexity often hinder practicality. In addition,...

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http://ieeexplore.ieee.org/document/11488477

Propeller-Based Handheld Force-Feedback Device for Navigation

📅 2026-04-22 📖 1 min 📝 168 words TRANSACTIONS HAPTICS

Summary: This article presents an evaluation of CrazyJoystick, a propeller-based handheld force-feedback device, for spatial navigation tasks in real-world and virtual reality environments. Building upon prior work validating the device's on-demand aerial deployment and directional cue discriminability, we i...

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A Wearable Shoulder Skin-Deformation Device Generating Illusory Forces for Posture Correction

📅 2026-04-22 📖 1 min 📝 208 words TRANSACTIONS HAPTICS

Summary: The hanger reflex is a haptic illusion in which skin-shear deformation elicits a directional force sensation. This study presents a wearable haptic device that uses four pneumatic airbags to apply controlled skin shear to both shoulders for posture guidance. Although previous work has shown that sho...

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http://ieeexplore.ieee.org/document/11489488

Sensitivity at the Edges of the Finger Enables Mixed Reality Force Feedback

📅 2026-04-06 📖 1 min 📝 155 words TRANSACTIONS HAPTICS

Summary: We introduce a new paradigm for rendering high-force sensations by stimulating the edges of the fingerpad rather than the central contact surface. This non-blocking approach preserves natural finger use, enabling simultaneous physical and virtual interactions. To investigate its feasibility, we prop...

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http://ieeexplore.ieee.org/document/11475177

ThermalTexture: Temperature-Integrated Surface Rendering Using Thermal Referral Induced by Texture Slip-Feedback

📅 2026-03-23 📖 1 min 📝 204 words TRANSACTIONS HAPTICS

Summary: Thermal–vibration feedback has been reported to elicit the thermal referral phenomenon, offering new possibilities for enhancing user experience. Yet, little research has explored whether thermal referral occurs in the presence of texture-slip feedback. To address this, we developed a custom haptic ...

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Guest Editorial: World Haptics Conference 2025

📅 2026-09-30 📖 1 min 📝 1 words TRANSACTIONS HAPTICS

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http://ieeexplore.ieee.org/document/11716332

Table of Contents

📅 2026-09-30 📖 1 min 📝 1 words TRANSACTIONS HAPTICS

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Multi-Channel Electromagnetic Interference Elimination for Shielding-Free MRI Using Null Operations

📅 2026-01-23 📖 1 min 📝 184 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Emerging technologies for electromagnetic interference (EMI) elimination have enabled radio frequency (RF) shielding-free magnetic resonance imaging (MRI), significantly reduced costs and increased accessibility. Existing methods often rely on multiple external sensors for EMI elimination...

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Double Banking on Knowledge: A Unified All-in-One Framework for Unpaired Multi-Modality Semi-Supervised Medical Image Segmentation

📅 2026-01-21 📖 1 min 📝 231 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Multi-modality (MM) semi-supervised learning (SSL) based medical image segmentation has recently gained increasing attention for its ability to utilize MM data and reduce reliance on labeled images. However, current methods face several challenges: (1) Multiple networks assignment hinder scalability...

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Virtual Imaging-Guided Harmonization of Chest X-Rays Using a Generative Adversarial Neural Network

📅 2026-01-26 📖 1 min 📝 247 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: To reduce variability in chest radiography (CXR) from acquisition and post-processing, and assess whether harmonization improves image quality and downstream diagnostic performance. Methods: A generative adversarial network (GAN) was trained exclusively on virtual images produced by a Vir...

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Self-Supervised Speech Representations for Sleep Apnea Severity Prediction

📅 2026-01-20 📖 1 min 📝 253 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Obstructive sleep apnea (OSA) is a prevalent yet underdiagnosed condition characterized by repetitive upper airway obstruction during sleep. Current gold standard diagnostic standards rely on polysomnography (PSG), which is resource-intensive. Since upper airway characteristics impact bot...

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Rectification of Cornea Induced Distortions in Microscopic Images for Assisted Ophthalmic Surgery

📅 2026-01-20 📖 1 min 📝 240 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Image distortions induced by the high refractive power of the eye's optical components challenge the accuracy of geometric information derived from intra-operative sensor data in ophthalmic surgery. Correcting these distortions is vital for advancing surgical assistance systems that rely ...

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Distinct Patterns of Syntactic Errors Doubly Dissociate in Chronic Poststroke Aphasia

📅 2026-10-20 📖 1 min 📝 195 words COGNITIVE NEUROSCIENCE

Summary: The lesion correlates of syntactic deficits in aphasia remain poorly understood. Previous studies have suggested that distinct error types in expressive syntax may be associated with damage to different brain regions, but the specific lesion correlates of these errors have not been fully delineated....

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Neural Distinctions between Social and Nonsocial Reward Anticipation: An Examination of Activation Magnitude, Representational Patterns, Individual Differences, and Functional Connectivity

📅 2026-10-19 📖 1 min 📝 235 words COGNITIVE NEUROSCIENCE

Summary: Prevailing theories emphasize a “common neural currency” for social and nonsocial rewards, particularly during reward anticipation. However, direct comparisons of social and nonsocial reward anticipation remain limited, and clinical observations suggest that social reward processing can be selective...

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Exploring Neural Mechanisms Underlying Error-related Impairments in Active Working Memory Suggests an Adaptive Shielding of Contents during Cognitive Control

📅 2026-10-19 📖 1 min 📝 302 words COGNITIVE NEUROSCIENCE

Summary: Goal-directed behavior relies on cognitive flexibility—the ability to rapidly adapt ongoing thoughts and behaviors while preserving task-relevant information. The performance monitoring system optimizes such behavior by detecting and evaluating errors, whereas the working memory (WM) system maintain...

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Slow Oscillations Modulate Overnight Brain Changes in Working Memory Function

📅 2026-10-19 📖 1 min 📝 227 words COGNITIVE NEUROSCIENCE

Summary: Working memory (WM), the transient storage and manipulation of information, is a cognitive function that can improve with training. Accumulating evidence suggests that sleep, particularly slow oscillations (SOs) that occur during nonrapid eye movement sleep, supports this improvement. Yet, how sleep...

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http://ieeexplore.ieee.org/document/11741002

Transformer Mechanisms Mimic Frontostriatal Gating Operations When Trained on Human Working Memory Tasks

📅 2026-10-20 📖 1 min 📝 237 words COGNITIVE NEUROSCIENCE

Summary: Working memory (WM) is thought to rely on sophisticated frontostriatal mechanisms for selective gating, supporting updating and readout of information to and from distinct memory “addresses” (neural populations). According to computational models, capacity limitations in WM arise due to challenges i...

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http://ieeexplore.ieee.org/document/11743154

Visual Perspective Shapes Subjective Experience: Dissociable Parietal Contributions to the Constructive Nature of Memory

📅 2026-10-19 📖 1 min 📝 223 words COGNITIVE NEUROSCIENCE

Summary: Subjective features of memory are often treated as secondary to the objective content of remembered events. However, growing evidence suggests that these features actively shape how memories are constructed, experienced, and used. Rather than treating visual perspective as a peripheral correlate of ...

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Memory Traces Reflect How They Were Last Accessed

📅 2026-10-19 📖 1 min 📝 216 words COGNITIVE NEUROSCIENCE

Summary: Episodic memories are known to change with each act of retrieval. We hypothesize that accessing semantic knowledge in different ways during episodic retrieval—from unique perceptual features to taxonomic and thematic context—shapes how those memories are subsequently represented during recognition. ...

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http://ieeexplore.ieee.org/document/11740999

The Temporal Dynamics of Attentional Allocation during Counterfactual Learning

📅 2026-10-19 📖 1 min 📝 194 words COGNITIVE NEUROSCIENCE

Summary: Counterfactual learning, the ability to learn from what could have happened under different circumstances, is a key cognitive mechanism supporting behavioral adaptation. While its neural and computational underpinnings are increasingly understood, the temporal dynamics of attention toward factual an...

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http://ieeexplore.ieee.org/document/11741005

Theta and High-alpha Phase-locking Dynamics Unveil Face Perception in Ambiguous Stimuli

📅 2026-10-19 📖 1 min 📝 227 words COGNITIVE NEUROSCIENCE

Summary: Previous studies emphasize phase synchronization as a fundamental mechanism for integrating local features into coherent percepts. We employed a novel paradigm and dynamic graph analysis based on EEG to track neural dynamics associated with perceiving a face (or not) while keeping the stimuli identi...

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Capturing the Attentional Trade-off between Speech Planning and Comprehension

📅 2026-10-19 📖 1 min 📝 255 words COGNITIVE NEUROSCIENCE

Summary: In conversation, future speakers often plan speech simultaneously with comprehension, which means that they must divide attentional resources between these processes. In this EEG study, we used responses to linguistic attention probes (i.e., syllable “BA” presented during spoken sentences) to track ...

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http://ieeexplore.ieee.org/document/11740942

Decoding spatial attention in the cocktail party problem using whole-head high-density fNIRS

📖 1 min 📝 47 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Sudan Duwadi, De’Ja Rogers, Alex D. Boyd, Laura B. Carlton, Yiwen Zhang, Anna Kawai Gaona, Aneesa Diya Pathiyaparambath, Ravin Chaudhury, Grace Magee, Bernhard B Zimmermann, Walker J O'Brien, Alexander...

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Subjective emotional ratings dissociate brain activation and network patterns during counterfactual outcome processing

📖 1 min 📝 24 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Tingting Zhang, Denghui Bai, Xiaotong Wei, Lei Zhuo, Qiuzhu Zhang, Junjun Zhang, Zhenlan Jin, Ling Li

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S105381192600618X?dgcid=rss_sd_all

Whole-brain normative modeling across multiple functional connectivity metrics in MEG

👤 Kobbersmed, J. R. L., Gohil, C., Woolrich, M. W., Vidaurre, D. 📅 2026-10-26 📖 1 min 📝 200 words

BIORXIV NEUROSCIENCE

Summary: Normative modeling has recently emerged in structural and functional neuroimaging as a promising approach to characterize neuropathology at the individual level. Trained on large healthy cohorts, these models predict expected measurements from covariates such as age and sex while quantifying normal ...

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https://www.biorxiv.org/content/10.64898/2026.10.21.760411v1?rss=1

Practical Identifiability of a Minimal Adaptive Learning-Gain Model for Binary Feedback: A Simulation Study

Li, W., Zhong, J.
2026-10-26 3 min 688 words BIORXIV NEUROSCIENCE

Summary: Social learning is jointly determined by several computationally distinct quantities, namely baseline belief (prior mean), sensitivity to feedback (learning gain), and the temporal uncertainty of belief (drift), which are rarely separated in empirical designs. Here we systematically characterise the...

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https://www.biorxiv.org/content/10.64898/2026.10.21.760725v1?rss=1

Encoding activity predicts distinct recognition signals during working and long-term memory in the human hippocampus

Tallman, C. W., Wixted, J. T., Steinmetz, P. N.
2026-10-26 1 min 150 words BIORXIV NEUROSCIENCE

Summary: Neurocomputational models predict that episodic memories are represented by sparse, pattern-separated hippocampal codes. Recognition within working memory capacity may instead rely on overlapping cortical codes. We analyzed human intracranial microwire recordings during a continuous recognition task...

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https://www.biorxiv.org/content/10.64898/2026.10.21.759719v1?rss=1

Covert attentional tracking with invisible phase tagging

Arora, K., Zinkovic, K., Kenemans, J. L., Van der Stigchel, S., Gayet, S., Chota, S.

2026-10-26

1 min | 213 words

BIORXIV NEUROSCIENCE

Summary: Attention forms a fundamental component of visual processing. The locus and spread of covert attention reveals how specific locations or features in the environment are preferentially processed, however, tracking this process in a time-resolved fashion is not easy. Rapid Invisible Frequency Tagging ...

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https://www.biorxiv.org/content/10.64898/2026.10.21.760697v1?rss=1

Self-enhancement and devaluation of others during belief formation are linked to narcissistic traits

Matthiesen, M., Mayer, A. V., Schröder, A., Rosenbusch, L., Klein, J. P., Wilhelm-Groch, I., Stolz, D. S., Paulus, F. M., Müller-Pinzler, L., Krach, S., Czekalla, N.

2026-10-26 | 1 min | 253 words

BIORXIV NEUROSCIENCE

Summary: Background: Narcissistic personality traits are associated with the tendency to maintain a grandiose self-image through self-enhancement and devaluation of others. Using computational modeling, we investigated how narcissistic traits and the experience of pride are linked to how individuals form bel...

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https://www.biorxiv.org/content/10.64898/2026.10.21.760328v1?rss=1

Dissociating Resting-State Networks of Pedohebephilic Preference and Child Sexual Offending Using Network-Based Prediction

Serin, E., Khan, K. R., Krueger, T., Sinke, C., Beier, K. M., Walter, M., Schiffer, B., Amelung, T., Walter, H.

2026-10-26 1 min 237 words BIORXIV NEUROSCIENCE

Summary: Background: Pedohebephilia (P+), the sexual preference for prepubescent and early pubertal minors, regularly causes distress in those affected. While associated with child sexual offending (CSO), P+ is not a prerequisite for CSO. Limited neuroimaging studies have identified alterations associated wi...

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https://www.biorxiv.org/content/10.64898/2026.10.21.760530v1?rss=1

Sustained antidepressant-like effects of psychedelics through 5-HT_{2A}R–β-arrestin signaling

Sheng Wang

2026-10-23 1 min 40 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 23 October 2026; doi:10.1038/s41593-026-02497-5 This study shows that the sustained antidepressant-like effect of psychedelics is driven by 5-HT2A receptor–β-arrestin signaling. Repeated low...

Read full article:
https://www.nature.com/articles/s41593-026-02497-5

Identification of candidate markers for the isolation of neuron-specific extracellular vesicles

David R. Walt
2026-10-23 1 min 41 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 23 October 2026; doi:10.1038/s41593-026-02466-y The authors developed methods to identify extracellular vesicles (EVs) in the blood that come from the brain by capturing EV surface proteins....

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02466-y

Neuropharmacology of language: ERP evidence for catecholaminergic modulation of syntactic processing

Yingying TanAshley G. LewisIris C. SchmitsRobert-Jan VerkesRoshan CoolsPeter Hagoortahttps://ror.org/01bn89z48Shanghai International Studies University, Institute of Language Sciences, Shanghai 201620, Chinabhttps://ror.org/00671me87Max Planck Institute for Psycholinguistics, Neurobiology of Language Department, Nijmegen 6525 XD, the Netherlandschttps://ror.org/03q8dnn23City University of Hong Kong, Department of Linguistics and Translation, Kowloon, Hong Kong Special Administrative Region, Chinadhttps://ror.org/016xsfp80Radboud University, Donders Institute for Brain, Cognition and Behaviour, Center for Cognitive Neuroimaging, Nijmegen 6525 EN, the Netherlandsehttps://ror.org/05wg1m734Radboud University Medical Center, Department of Psychiatry, Nijmegen 6525 GA, the Netherlands

📅 2026-10-22 📖 1 min 📝 42 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceHow the brain’s neuromodulatory systems influence language, a defining human capacity, remains poorly understood. Here we show that catecholaminergic neurotransmission—primarily involving dopamine ...

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https://www.pnas.org/doi/abs/10.1073/pnas.2606806123?af=R

Safety of lecanemab in real-world patients with Alzheimer’s disease: a prospective, observational case series supported by FAERS signal analysis

Erhao Bao 2026-10-02 1 min 354 words FRONTIERS NEUROINFORMATICS

Summary: ObjectiveLecanemab, a humanized immunoglobulin monoclonal antibody, has been approved for the treatment of Alzheimer’s disease (AD). Although clinical trials have demonstrated its favorable safety profile, real-world studies are crucial for identifying rare adverse events (AEs) and evaluating post-m...

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EEG emotion-recognition datasets for neuroarchitecture: an appraisal of openness, AI-readiness and ontology maturity

Han Jong Jun 2026-10-20 1 min 341 words FRONTIERS NEUROINFORMATICS

Summary: IntroductionNeuroarchitecture studies how designed space shapes emotion, often with electroencephalography (EEG), and computational models that decode emotion from EEG must be trained on labelled recordings. Yet none of the widely shared EEG emotion datasets was recorded while people moved through d...

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A systematic review of machine learning approaches for MRI-based brain tumor analysis

Almira Melsovna Kustubayeva
2026-10-12 1 min 213 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Machine learning is now widely applied to magnetic resonance imaging-based brain tumor analysis, with task-specific performance metrics: the Dice similarity coefficient (DSC) quantifies segmentation overlap, while the area under the curve (AUC) and accuracy measure classification performance. A PRIS...

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Are cortical microcircuits optimized for information flux?—A simulation-based reverse engineering study

Patrick Krauss
2026-10-20 1 min 271 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionA sufficiently large information flux in recurrent neural networks, quantified by the mutual information between successive network states, has been proposed as a marker of rich information-processing capabilities. This raises the question of whether biological neural networks, such as c...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1886601

Artificial intelligence-generated and algorithmically personalized music interventions and their effects on brain function, behavior, and health: a critical review

Jingting Jin 2026-10-23 1 min 310 words FRONTIERS HUMAN NEUROSCIENCE

Summary: This critical review synthesizes empirical studies of algorithmically personalized and artificial-intelligence-generated music interventions reporting neurophysiological, neuroimaging, or objective behavioral outcomes. Literature was identified through structured searches of PubMed, Scopus, Google S...

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Effect of robotic assistance on upper-limb sensorimotor recovery: a systematic review and meta-analysis

Sivakumar Balasubramanian 2026-10-20 1 min 359 words FRONTIERS NEUROROBOTICS

Summary: IntroductionRobot-assisted upper-limb neurorehabilitation enables several beneficial therapy components, including repetitive task-specific practice, high training intensity, real-time feedback, gamification, and continuous physical interaction with the patient’s limb. However, conventional comparis...

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Research progress on the mechanism of microglia–astrocyte signal crosstalk in central sensitization of vestibular migraine

Huailiang Zhang 2026-10-23 1 min 127 words FRONTIERS NEUROSCIENCE

Summary: Vestibular migraine (VM) is a vestibular disorder characterized by recurrent episodes of vertigo, frequently accompanied by headache, nausea, and vomiting, which substantially reduce patients' quality of life. The pathogenesis of VM remains incompletely understood. Current studies mainly focus on io...

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From confidence to caution: conformal gating for cross-session motor imagery brain–computer interfaces

Sixue Xing and Aarthy Nagarajan 2026-10-22 1 min 278 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Erroneous motor imagery brain–computer interface (MI-BCI) commands can cause unintended device actions, yet standard classifiers provide no mechanism to withhold uncertain predictions. We evaluate whether conformal prediction (CP) can serve as a command gate under cross-session EEG shift,...

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https://iopscience.iop.org/article/10.1088/1741-2552/aead26

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 2025-06-26 1 min 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 2025-08-13 1 min 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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From task-specific AI to multimodal foundation models in PET/CT: a readiness framework for clinical translation

👤 Jia Wu 📅 2026-10-24 📖 1 min 📝 59 words LOW VISION

Summary: Artificial intelligence in PET/CT is evolving from narrowly optimized systems toward reusable pretrained representations and multimodal models. PET/CT is a demanding test case because clinically meaningful interpretation must preserve quantitative molecular information while integrating anatomic CT,...

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Diagnosis-Related Group Payment Systems and Hospital Efficiency, Quality, and Coding Behavior: A Systematic Review and Narrative Synthesis

👤 Wael Alsebaei 📅 2026-10-24 📖 1 min 📝 66 words LOW VISION

Summary: Diagnosis-related group (DRG)-based payment systems are used globally to incentivize hospital efficiency, yet their impact on clinical quality and documentation behavior remains variable. As Saudi Arabia implements the Australian Refined DRGs (AR-DRG) framework under Vision 2030, a synthesis of inte...

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Clinical significance of "complex" multimorbidity in hospitalised adult patients

Gino Roberto Corazza 2026-10-24 1 min 65 words LOW VISION

Summary: The term "complex multimorbidity" lacks universal consensus and its relevance remains unclear. We aimed to assess the prevalence of complex multimorbidity and its association with major healthcare outcomes. This was a prospective, single-centre study (internal medicine ward). We collected sociodemog...

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Outcomes of subthreshold diode micropulse laser and anti-VEGF vs anti-VEGF monotherapy for diabetic macular edema: a meta-analysis

Marko M Popovic 2026-10-26 1 min 41 words LOW VISION

Summary: CONCLUSIONS: In patients with DME, SDMLP and anti-VEGF combination therapy appears comparable, and possibly modestly superior to anti-VEGF monotherapy, with a reduced injection burden over 12 months. BCVA and CMT differences were below clinically important thresholds and supported by low-certainty e...

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Tropospheric Ozone Assessment Report II: Ozone distribution and variability in the oceanic and polar regions

👤 Alfonso Saiz-Lopez 📅 2026-10-26 📖 1 min 📝 76 words LOW VISION

Summary: Oceanic and polar regions are critical for global atmospheric processes, but an evaluation of ozone distributions and trends in these regions has been missing. In this assessment, part of the TOAR-II project, we report lower atmosphere ozone concentrations over open oceans and polar regions: with da...

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Telemedicine-Enabled Medication Optimization for Adults with Polypharmacy: A Systematic Review and Implications for Saudi Vision 2030 Healthcare Transformation

👤 Mark Sim 📅 2026-10-26 📖 1 min 📝 32 words LOW VISION

Summary: CONCLUSIONS: Telemedicine-enabled medication optimization appears feasible, but evidence of clinical effectiveness and cost savings remains uncertain. Saudi implementation should begin with monitored pilots emphasizing safety, equity, Arabic-language accessibility, governance, and prospective evalua...

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Accurate Extraction of Phenotypic Parameters for DUS Testing of Oudemansiella raphanipes Based on Deep Learning

Linyi Li 2026-10-26 1 min 65 words LOW VISION

Summary: To enable the rapid and accurate acquisition of phenotypic parameters during distinctness, uniformity, and stability (DUS) testing of edible mushroom varieties, and to address the low efficiency, strong subjectivity, and insufficient standardization of conventional manual measurement methods, this s...

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Understanding decision making factors influencing referrals to vision rehabilitation services at a tertiary medical center

Christina L Thomas-Virnig 2026-10-26 1 min 74 words LOW VISION

Summary: Functional vision loss at any stage affects quality of life even before clinically quantified to be moderate or severe. Eye care providers as the gatekeepers to vision rehabilitation services often underestimate how small variations in functional vision loss can have a significant impact in a patien...

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Comparative longitudinal evaluation of DIMS, progressive addition and single-vision lenses on myopia progression, binocular and visual performance in Spanish children

José R Jiménez 2026-10-26 1 min 58 words LOW VISION

Summary: An 18-month prospective, non-randomized comparative observational study evaluated the effects of different optical corrections on myopia progression, binocular vision, and visual performance in Spanish children. Forty-five myopic participants aged 9-14 years were allocated to single-vision lenses (S...

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Will Artificial Intelligence Augment Global Health Inequalities?

Matthew J Burton 2026-10-26 1 min 66 words LOW VISION

Summary: AbstractArtificial intelligence (AI) has been promoted as a tool to reduce global health inequities by addressing workforce shortages and expanding access to specialist care. However, if AI development and validation are concentrated in high-income countries, innovation may disproportionately benefi...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux 📅 2026-07-18 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

Zhifei Xie
2026-10-24 1 min 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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In vitro carbapenem activity in Neisseria gonorrhoeae

Béatrice Berçot 2026-09-04 1 min 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan
📅 2026-09-09 📖 1 min 📝 57 words

BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Neuromodulatory interventions for Alzheimer's disease

Wandong Zhang 2026-10-24 1 min 60 words TDCS TACS TRNS

Summary: Alzheimer's disease (AD) and mild cognitive impairment (MCI) represent an escalating global public health challenge with major socioeconomic consequences. Although anti-amyloid monoclonal antibodies have introduced disease-modifying treatment options for early symptomatic AD, their benefits remain m...

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Disrupting objectification: transcranial Direct Current Stimulation (tDCS) enhances the inversion effect toward sexualized women

Arwen Stephens
2026-10-24 1 min 68 words TDCS TACS TRNS

Summary: Sexual objectification is associated with viewing women in an object-like manner, often indexed perceptually by a reduced inversion effect for sexualized women relative to men. Prior research has shown that this index can be attenuated through social manipulations; here we investigated whether it ca...

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Efficacy and Safety of Perioperative Transcranial Electrical Stimulation for Postoperative Pain and Related Outcomes: A Systematic Review and Meta-Analysis With Trial Sequential and Subgroup Analyses

Qing Zhong 2026-10-24 1 min 64 words TDCS TACS TRNS

Summary: CONCLUSIONS: Current evidence suggests that tES, as an adjuvant perioperative intervention, may alleviate postoperative pain, reduce opioid consumption, and decrease the incidence of postoperative delirium. However, the effect of tES on postoperative quality of recovery remains unclear. Considering ...

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Accelerated high-capacity 6 mA transcranial direct current stimulation for major depression

Mark S George 2026-10-25 1 min 14 words TDCS TACS TRNS

Summary: CONCLUSIONS: Accelerated 6 mA HC-tDCS was feasible, well tolerated, and warrants a sham-controlled trial.

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Rapid and robust effects of high-dose intermittent Transcranial Brain Stimulation (iTBS) on depression and suicidal ideation in adolescents with acute major depression: a 10-day randomised, controlled, single-blind trial

👤 Yutao Lan 📅 2026-10-26 📖 1 min 📝 56 words TDCS TACS TRNS

Summary: CONCLUSION: Current research indicates that a 10-day course of high-dose iTBS is effective in alleviating depressive symptoms and suicidal ideation in adolescents with depression, and that the therapeutic effects persist for up to one month after the end of treatment. We also found that high-dose iT...

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Comparative Efficacy of Noninvasive Neuromodulation in Patients with Disorders of Consciousness: A Systematic Review and Network Meta-Analysis of Randomized Controlled Trials

Quan Wei
2026-10-26
1 min
64 words
TDCS TACS TRNS

Summary: CONCLUSIONS: HF-rTMS-M1 and RMNS plus HF-rTMS-DLPFC, RMNS, HF-rTMS-DLPFC, and tDCS have shown positive signals in patients with DoC. The duration of the disorder and underlying etiology may influence the effectiveness of different noninvasive neuromodulation techniques in DoC. Given the marked heter...

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Associations of Combined Transcranial Magnetic Stimulation and Transcranial Direct Current Stimulation With Psychological and Functional Outcomes After Stroke: A Retrospective Cohort Study

👤 Meng Huang
📅 2026-10-26 📖 1 min 📝 46 words

TDCS TACS TRNS

Summary: CONCLUSIONS: In this retrospective cohort, combined TMS and tDCS with standard rehabilitation was associated with lower depression and anxiety symptom scores and more favorable neurological and motor outcome trajectories than standard rehabilitation. Non-random treatment allocation and residual conf...

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A tool to evaluate the impact of lived experience involvement in research: the Brain and Genomics Hub Impact Log protocol

Sophie E Legge 2026-10-26 1 min 74 words TDCS TACS TRNS

Summary: CONCLUSION: The potential value of this research is to implement a novel tool and framework for facilitating, recording and evaluating LEI in complex mental health research, which can be adapted for wider use. Strengths in design are LE leadership and cross-cutting LE research integration, incorpora...

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Multimodal premotor stimulation during action observation transiently boosts corticospinal excitability and motor performance

Ishan Mahajan 2026-10-26 1 min 2 words TDCS TACS TRNS

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Combining Cognitive Training and Brain Stimulation: A Novel Approach to Address Disordered Eating and Antipsychotic-Induced Weight Gain in People With Severe Mental Illness

👤 Ulrike Schmidt 📅 2026-10-26 📖 1 min 📝 69 words TDCS TACS TRNS

Summary: CONCLUSION: The initiation of antipsychotic therapy is a critical clinical window for early intervention to mitigate the long-term metabolic and psychiatric consequences of antipsychotic-induced weight gain. Combining ABMT + tDCS is acceptable and may help reduce binge eating in individuals on antip...

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Local and network-level prefrontal reactivity to intermittent theta burst stimulation: a randomized, sham-controlled iTBS/fNIRS study

👤 Georg S Kranz 📅 2026-10-23 📖 1 min 📝 2 words FNIRS

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Correction-stage information leakage and target-input asymmetry can distort comparisons of fNIRS short-separation channel correction strategies

Esra Suzen 2026-10-23 1 min 16 words FNIRS

Summary: OBJECTIVE: We examined how evaluation design affects comparisons of functional near-infrared spectroscopy (fNIRS) short-separation correction strategies.

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The test-retest reliability of cortical activation and behavioral metrics in adolescent basketball players during dual-task postural control: an fNIRS-based assessment

Lulu Song 2026-10-24 1 min 59 words FNIRS

Summary: CONCLUSION: These findings provide preliminary test-retest reliability estimates for selected multimodal neurobehavioral measures in male adolescent basketball players although reliability alone does not establish clinical utility and estimates with wide confidence intervals require confirmation in ...

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How School-Based Interventions Shape the Developing Brain during 3-12 Years: A Systematic Review and Exploratory Meta-analysis

👤 Hui Li 📅 2026-10-24 📖 1 min 📝 56 words FNIRS

Summary: The effects of school-based interventions (SBIs) on children's neurodevelopment are poorly synthesized. This systematic review integrates neuroscientific evidence on how structured educational activities influence brain development in children aged 3-12 years. Following PRISMA guidelines, we identif...

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The Walking Brain: Mapping feature-specific cortical responses to naturally unfolding sensory experience in the real world

Meryem A Yücel 2026-10-25 1 min 58 words FNIRS

Summary: Understanding how the brain processes the everyday world is fundamental, yet laboratory paradigms exclude self-generated movement, unscripted events, and a continuously evolving multisensory environment. Wearable neuroimaging enables real-world measurement, but linking neural activity to uncontrolle...

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Functional near infrared spectroscopy-based dual-stream adaptive language mapping in adults with and without post-stroke aphasia

👤 Meryem Yücel
📅 2026-10-25 📖 1 min 📝 67 words FNIRS

Summary: In aphasia, variability in lesion location and post-stroke network reorganization complicates efforts to identify the regions that support semantic versus phonological processing. In prior work, researchers have often examined these domains in separate patient cohorts, further limiting direct compar...

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How Task Load Modulates the Effect of State Anxiety on Working Memory and Its Neural Mechanisms: Evidence from Verbal and Spatial Modalities

👤 Senqing Qi 📅 2026-10-25 📖 1 min 📝 73 words FNIRS

Summary: The impact of state anxiety on working memory (WM) and its neural mechanisms, especially the role of task load, remain uncertain. To address this issue, we conducted two experiments to examine how state anxiety affects WM under different task loads and whether these effects are consistent across ver...

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Prefrontal dysfunction and impulsive decision-making in female methamphetamine use disorder: a fNIRS study

👤 Peng Lei 📅 2026-10-26 📖 1 min 📝 67 words FNIRS

Summary: CONCLUSIONS: Women with MUD showed altered intertemporal choice and reduced prefrontal hemodynamic responses in regions implicated in value evaluation and cognitive control. These findings support the relevance of prefrontal dysfunction to impulsive decision-making in female MUD while also indicatin...

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Melanin Bias in Cerebral fNIRS: A Monte Carlo Study Toward Equitable Optical Brain Sensing

👤 Murad Althobaiti
📅 2026-10-26 📖 1 min 📝 63 words FNIRS

Summary: Functional near-infrared spectroscopy (fNIRS) is an optical biosensing modality that transduces cortical hemodynamics into a measurable near-infrared signal, underpinning wearable brain monitors and brain-computer interfaces (BCIs). Because epidermal melanin is a strong near-infrared absorber, devic...

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Professional Experience Moderates Behavioral Performance and Sensorimotor-Related Cortical Activity During Scoring Decision-Making in Aerobic Gymnastics Judges: An fNIRS Study

👤 Bin Shao 📅 2026-10-26 📖 1 min 📝 58 words FNIRS

Summary: Understanding how professional experience shapes cognitive and neural resource allocation across different task demands is important for elucidating expert decision-making. This study examines whether professional experience influences behavioral performance and sensorimotor-related cortical activit...

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Case Report: An MDT-guided multimodal protocol for large-area stroke plateau - integrating neuromodulation with multimodal assessment

Xiao-Xia Du 2026-10-24 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In this patient with large-area stroke who showed limited response to conventional rehabilitation, an MDT-guided, dynamically adjusted multimodal intervention was temporally associated with concurrent recovery across cognitive, language, motor, and daily living domains. The concomitant n...

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Correction: Perioperative management for invasive brain-computer interface implantation under surgical navigation in a patient with traumatic spinal cord injury: a case report

Wenyao Cui 2026-10-24 1 min 6 words BRAIN COMPUTER INTERFACE

Summary: [This corrects the article DOI: 10.3389/fnhum.2026.1892331.].

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Limits of Trial-Adaptive Neural-Language Fusion Across Large Language Models in P300 Brain-Computer Interfaces

Yiftach Barash 2026-10-24 1 min 21 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: A tuned fixed weight is a difficult-to-beat default across the tested scale range; reliability estimation gave no deployable adaptive advantage.

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Creating online cognitive assessment platform for children's literacy and math skills: a case study illustrated with the ACE kids platform

U Maurer 2026-10-24 1 min 70 words BRAIN COMPUTER INTERFACE

Summary: Computerized versions of paper-and-pencil tests made behavioral assessments of children more automatic and precise. The COVID-19 pandemic further demonstrated the need for assessments to be offered online to allow for remote data collection. However, the design and implementation of computerized tes...

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Combined neuromodulation techniques via a central-peripheral-central strategy facilitate functional recovery from AIS A to AIS C in a patient with complete spinal cord injury: A case report

Liansheng Zhong
2026-10-24 1 min 48 words BRAIN COMPUTER INTERFACE

Summary: CONTEXT: Complete spinal cord injury (SCI) has traditionally been considered irreversible. However, accumulating evidence suggests that subclinical neural pathways may persist. Multimodal combined neuromodulation targeting brain-spinal-peripheral circuits holds promise to unlock latent neural plasti...

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Fabrication Strategies and Performance Optimization Strategies of Flexible Brain-Computer Interface Electrodes

Jianguo Tang 2026-10-25 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) provides a direct communication pathway between the brain and external devices and has been applied in neurological disease treatment, motor rehabilitation, and human-computer interaction. Electrodes play essential roles in recording and modulating neural signals, and ...

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Melanin Bias in Cerebral fNIRS: A Monte Carlo Study Toward Equitable Optical Brain Sensing

Murad Althobaiti 2026-10-26 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Functional near-infrared spectroscopy (fNIRS) is an optical biosensing modality that transduces cortical hemodynamics into a measurable near-infrared signal, underpinning wearable brain monitors and brain-computer interfaces (BCIs). Because epidermal melanin is a strong near-infrared absorber, devic...

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A Complementary Dual-Branch Receptive Field Network for Cross-Channel-Density Motor Imagery EEG Decoding

Zuguang Rao 2026-10-26 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) enable direct communication between the brain and external devices, providing a bio-inspired link between humans and artificial systems. However, electroencephalography (EEG)-based motor imagery (MI) decoding continues to pose challenges, due to limited temporal expl...

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A Biodegradable Fish-Derived Gelatin-Chitosan Sponge as a Novel Hemostatic Agent in a Rat Tooth Extraction Model

Dwi Ariawan 2026-10-26 1 min 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: These findings demonstrate the potential of gelatin-rich fish-derived gelatin-chitosan composite sponges as biodegradable hemostatic biomaterials for oral surgical applications.

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Beyond Overlap: Development and Validation of the Inclusion of AI in Self (IAIS) Scale and Its Moderating Role in the Relationship Between AI Identity and AI Threat

👤 Xiang Yao 📅 2026-10-26 📖 1 min 📝 68 words BRAIN COMPUTER INTERFACE

> **Summary:** Artificial intelligence (AI) is becoming increasingly integrated into people's self-concepts, yet a psychometrically robust multi-item pictorial instrument for assessing the inclusion of AI in self (IAIS) is lacking. Drawing on self-expansion theory and the AI-mediated communication (AI-MC) model, w...

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Tuesday Daily Thread: Advanced questions

👤 /u/ AutoModerator 📅 2026-10-27 📖 1 min 📝 288 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> <h1>Weekly Wednesday Thread: Advanced Questions 🐍</h1> Dive deep into Python with our Advanced Questions thread! This space is reserved for questions about more advanced Python topics, frameworks, and best practices. <h2>How it Works:</h2> Ask A...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xgdey8/tuesday_daily_thread_advanced_questions/

Python 3.15.2, 3.14.10, 3.13.18, 3.12.16 and 3.11.18 are out

👤 /u/japaget 📅 2026-10-26 📖 1 min 📝 40 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> See https://blog.python.org/2026/10/python-3152-31410-31318-31216-31118/ . This is another security update that addresses the <a href="https://discuss.python.org/t/information-for-pyt...

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https://www.reddit.com/r/Python/comments/1xgcu6d/python_3152_31410_31318_31216_and_31118_are_out/

The Exponential Time Hypothesis Is False [pdf]

👤 vok 📅 2026-10-26 📖 1 min 📝 26 words HACKER NEWS

> **Summary:** See also: Announcing three new discoveries in mathematics, computer science, and physics - https://news.ycombinator.com/item?id=50251344 <hr /> Comments URL: <a href="https://news.ycombinator.com/item?id=50251661">https://n...

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https://github.com/openai/math/blob/main/preprints/The-Exponential-Time-Hypothesis-is-False-October-25-2026/paper.pdf

Is PHP Running on Your ColdFusion Server? How to Check, and Remove It

👤 rmason 📅 2026-10-26 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.gregoryalexander.com/blog/2026/10/23/is-php-quietly-running-on-your-coldfusion-server-how-to-check-and-remove-it Comments URL:...

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16-bit Windows (Windows 1.x, 2.x, 3.0, 3.1, etc.) on 64-bit Windows

👤 doener 📅 2026-10-27 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/otya128/winevdm Comments URL: https://news.ycombinator.com/item?id=50253529 Points: 14 # Comments: 1

🔗 **Read full article:**
https://github.com/otya128/winevdm

Dogmatic Bayesianism Disorder

👤 noreply@blogger.com (Daniel Lakens)
📅 2025-12-06 📖 4 min 📝 910 words

TWENTY PERCENT STATISTICIAN

Summary: <p>Note: This humorously intended sarcastic blog post directly mimics classifications of psychological disorders in the Diagnostic and Statistical Manual of the American Psychological Association, but Dogmatic Bayesianism is not in the DSM-5 as an actual psychologic...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/12/dogmatic-bayesianism-disorder.html

On the reliability and reproducibility of qualitative research

👤 noreply@blogger.com (Daniel Lakens)
📅 2026-02-08 📖 14 min 📝 2928 words

TWENTY PERCENT STATISTICIAN

Summary: <p class="MsoNormal">With my collaborators, I am increasingly performing qualitative research. I find qualitative research projects a useful way to improve my understanding of behaviors that I want to explore with other methods in the future. For example, some years ago I performe...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/02/on-reliability-and-reproducibility-of.html

New Papers: Optimal Filter Settings for ERP Research

👤 Steve Luck 📅 2024-02-04 📖 2 min 📝 569 words ERP BOOT CAMP

> **Summary:** <p class="">Zhang, G., Garrett, D. R., & Luck, S. J. (in press). Optimal filters for ERP research I: A general approach for selecting filter settings. Psychophysiology. https://doi.org/10.1111/psyp.14531 [<a href="https://www...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/2/4/optimal-filters

Q&A with Dr. Richard Carson, Professor of Biomedical Engineering and Radiology & Biomedical Imaging, Yale University and Yale School of Medicine

👤 Adriel Carridice 📅 2025-11-04 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dr-richard-carson-professor-of-biomedical-engineering-and-radiology-biomedical-imaging-yale-university-and-yale-school-of-medicine/

Single-cell RNA sequencing reveals vascular heterogeneity and cell–cell communication remodeling in the glioma blood–tumor barrier

📖 1 min 📝 31 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Zhipeng Zhao, Jiachen Lang, Haiyan Gu, Luodan Wang, Jianchang Luo, Jiawang Lang, Qiongying Yao, Xinyu Qiao, Zida Wang, Ling Yin, Caidi He

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004579?dgcid=rss_sd_all

Subspace Image Synthesis for Style-Invariant Object Recognition

👤 Ono, H., Yamashita, A., Amano, K., Sawayama, M. 📅 2026-10-26 📖 1 min 📝 212 words

BIORXIV NEUROSCIENCE

Summary: Human-like behavior in visual models does not necessarily imply human-like internal computation. When a model recognizes objects robustly across changes in visual style, what internal computations support this behavior, and which are relevant to human perception? We addressed this question by prunin...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.21.759598v1?rss=1

Pre-saccadic suppression and compensatory cortical distortions support predictive visual remapping

Xu, X., Denagamage, S., Nandy, A. S., Jadi, M. P.

2026-10-26 1 min 150 words

BIORXIV NEUROSCIENCE

Summary: Perceptual continuity across saccades relies on the predictive remapping of receptive fields (RFs) of visual neurons, driven by corollary discharge (CD). However, how the brain achieves accurate and coherent remapping within its non-uniform visual representation remains unclear. Here, we present a g...

🔗 **Read full article:**
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Eye movements enhance stereoscopic 3D vision

Li, Y. H., Wang, J. Z., Intoy, J., Victor, J. D., Rucci, M.

2026-10-26 1 min 189 words

BIORXIV NEUROSCIENCE

Summary: In moving organisms, sensory systems must recover spatial properties of the world from signals continually altered by their own movements. These self-generated temporal signals carry useful spatial information across various sensory modalities, but it remains unclear whether they also contribute to ...

🔗 **Read full article:**
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Resource-Rational Encoding of Reward Information in Working Memory During Planning

Ying, Z., Callaway, F., Fox, R., Kiyonaga, A., Mattar, M. G.

2026-10-26 1 min 220 words

BIORXIV NEUROSCIENCE

Summary: Planning requires holding information about potential actions and rewards in mind. This is assumed to rely on limited working memory resources. Although working memory allocation is well-studied in tasks where items are encoded simultaneously and independently, less is known about how memory resourc...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.20.760480v1?rss=1

Characterising sleep spindle emergence in infancy: Effects of prematurity and early locomotor intervention

Dornier, A., Adibpour, P., Biran, V., Pedoux, A., Nasser, H., Conessa, A., Gonzalez Carpinteiro, A., Yrjola, P., Devisscher, L., Gueret, L., Hinnekens, E., Heneau, A., Sibony, O., Hertz Pannier, L., Alison, M., Delorme, R., Barbu Roth, M., Dubois, J., Boutin, A.

2026-10-26 1 min 253 words BIORXIV NEUROSCIENCE

Summary: Study objectives: Sleep spindles are thalamocortical oscillations evolving during early brain maturation. In adults, they support memory consolidation and are organised into rhythmic "trains" thought to support memory reactivation. How this temporal organisation emerges in infancy, and whether prema...

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https://www.biorxiv.org/content/10.64898/2026.10.20.760527v1?rss=1

Walking Produces Oscillations in Orientation and Direction Perception at the Step Rate

👤 Phan, C. K., Rideaux, R., Alais, D.
📅 2026-10-26 📖 1 min 📝 159 words

BIORXIV NEUROSCIENCE

Summary: Walking is one of the most stereotyped oscillatory movement able-bodied individuals produce. It is highly consistent within and across individuals, with people taking ~5000 steps a day on average at approximately 2 Hz. This consistency and its ubiquitous role in daily living make walking an ideal be...

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https://www.biorxiv.org/content/10.64898/2026.10.20.760552v1?rss=1

Joint reconstruction of multi-contrast relaxation-diffusion imaging

👤 Wang, Y., Liu, Q., Consagra, W., Ning, L., Setsompop, K., Rathi, Y.
📅 2026-10-26 📖 1 min 📝 232 words

BIORXIV NEUROSCIENCE

Summary: Diffusion MRI across multiple gradient directions, b-values and echo times (TEs) provides complementary tissue contrast but requires extensive sampling. Joint k-space and q-space undersampling requires reconstruction that preserves diffusion- and TE-dependent signal variation while exploiting shared...

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https://www.biorxiv.org/content/10.64898/2026.10.21.760608v1?rss=1

Functional and Structural Brain Age Gaps as Complementary Predictive Markers for Cognitive Performance

👤 Crocetta, A., Morandini, M., Bittner, N., Caspers, S., Cauda, F., Brovelli, A., Gilson, M.

📅 2026-10-26

📖 1 min 📝 196 words

BIORXIV NEUROSCIENCE

Summary: Cognitive decline is characterized by substantial inter-individual variability across the lifespan, even among clinically healthy adults. However, it is unclear how structural and functional brain organization jointly relate to differences in cognitive performance within the typical populations. We ...

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https://www.biorxiv.org/content/10.64898/2026.10.20.760445v1?rss=1

Hippocampal replay repairs the cognitive map for flexible planning

👤 Bazarjani, A., Mallory, C., Foster, D., Piray, P.

📅 2026-10-26 📖 1 min 📝 192 words

BIORXIV NEUROSCIENCE

Summary: Hippocampal replay is highly structured: prospective replay precedes retrospective replay within a stopping period, each aligns with the animal's future and past paths, with replay sometimes being directed towards goals the animal has abandoned. These regularities have resisted a single explanation....

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https://www.biorxiv.org/content/10.64898/2026.10.20.760493v1?rss=1

RNA processing factors are associated with microglial heterogeneity and intercellular communication in ischemic stroke

📅 2026-10-26 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-76111-w

Nerve activity initiates autoimmune hair loss

📅 2026-10-26 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/d41586-026-03258-x

From aroma cue to feeding behavior: Sichuan pepper essential oil aroma stimulates appetite via cognitive and physiological pathways

📅 2026-10-26 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41538-026-01221-8

A glutamatergic projection from the central medial thalamic nucleus to the red nucleus regulates states of consciousness during sevoflurane anesthesia in mice

📅 2026-10-26 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-11164-1

From pose estimation accuracy to motor biomarker reliability: a reliability-centred evaluation framework for markerless movement analysis

👤 Christine Fernandez-Maloigne 📅 2026-10-01 📖 1 min 📝 246 words

FRONTIERS NEUROINFORMATICS

Summary: Markerless pose estimation has become an essential tool for quantitative movement analysis in neuroscience. However, current evaluation frameworks primarily rely on frame-wise spatial accuracy metrics, which provide only limited information about the reliability of the downstream kinematic biomarker...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1790241

Altered interictal alpha-band organization in childhood absence epilepsy

Justyna Paprocka 2026-10-23 1 min 318 words FRONTIERS NEUROINFORMATICS

Summary: ObjectiveInterictal background EEG in childhood absence epilepsy (CAE) may appear normal. We asked whether drug-naive children with CAE differ from controls in alpha frequency and the regional distribution of phase-lagged synchronization, and how these differences depend on participant-specific surr...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1875377

Short-term synaptic plasticity modulates the outcome of neurodegenerative diseases

Chi Chung Alan Fung 2026-10-23 1 min 264 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionAlzheimer's disease (AD) is characterized by progressive neuronal loss and synaptic dysfunction that disrupt memory-encoding neural circuits. Short-term synaptic plasticity (STP), which governs transient activity-dependent changes in synaptic efficacy on millisecond-to-second timescales,...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1923602

Orientation detection for self-calibrating neuromorphic systems

Elisabetta Chicca
2026-10-23 1 min 183 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Bioinspired sensory architectures help build adaptive touch systems with efficient hardware implementations. Orientation detection is a fundamental capability for manipulating objects in real-world scenarios, with diverse applications in robotic manipulation and manufacturing assessment processes. I...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1878557

A thalamus-centered network model of restless legs syndrome: an integrative hypothesis illustrated by three clinical phenotypes

Szabolcs Várbíró
2026-10-23 1 min 231 words

FRONTIERS HUMAN NEUROSCIENCE

Summary: Restless legs syndrome (RLS) is a clinically heterogeneous sensorimotor disorder whose pathophysiology reflects interacting biological mechanisms rather than a single causal pathway. Established influences—including brain iron biology, dopaminergic regulation, glutamatergic and GABAergic balance, ad...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1977941

A systems-level energetic framework for conscious processing: energetic organization as a constraint on large-scale neural integration

Holly Pollard-Wright 2026-10-23 1 min 259 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Conscious processing is associated with coordinated interactions among distributed neural systems, yet the biological constraints that enable and sustain large-scale neural integration remain incompletely understood. Contemporary theories emphasize computational, informational, and network mechanism...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1966072

Editorial: New horizons in stroke management, volume II

Nasser Kashou 2026-10-23 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.2023091

Cortical responses to lipread and text-based audiovisual stimuli in adults with and without dyslexia

👤 Milene Bonte 📅 2026-10-23 📖 1 min 📝 235 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Phonetic recalibration enables listeners to adjust their perception of ambiguous speech sounds based on visual contextual information. In line with a specific difficulty in forming letter-speech sound associations, adults with dyslexia have been found to show no text-based, but intact lipread-based,...

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Immediate effect of gait training using the hybrid assistive limb in patients in the acute phase of stroke: a prospective pilot study

👤 Yasushi Takagi 📅 2026-10-23 📖 1 min 📝 360 words FRONTIERS NEUROROBOTICS

Summary: IntroductionThe effectiveness of Hybrid Assistive Limb (HAL) gait training in patients in the acute phase of stroke remains unclear. We prospectively assessed the immediate effects of HAL gait training in this patient population.MethodsIn this prospective pilot study, four patients with acute-phase ...

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Comparison of short-term peripheral nerve stimulation, single-target supraorbital nerve pulsed radiofrequency, and dual-target supraorbital nerve–stellate ganglion pulsed radiofrequency for herpes zoster ophthalmicus–associated pain: a retrospective cohort study

👤 Wei Wang 📅 2026-10-23 📖 1 min 📝 326 words FRONTIERS NEUROSCIENCE

Summary: ObjectiveTo compare the clinical outcomes of short-term peripheral nerve stimulation (st-PNS), supraorbital nerve pulsed radiofrequency (SON-PRF), and combined supraorbital nerve and stellate ganglion pulsed radiofrequency (SON + SG-PRF) for herpes zoster ophthalmicus (HZO)-associated pain.MethodsTh...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1976324

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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https://pubmed.ncbi.nlm.nih.gov/41676644/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261026190357&v=2.20.1

Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

> **Summary:** Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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From task-specific AI to multimodal foundation models in PET/CT: a readiness framework for clinical translation

👤 Jia Wu 📅 2026-10-24 📖 1 min 📝 59 words LOW VISION

> **Summary:** Artificial intelligence in PET/CT is evolving from narrowly optimized systems toward reusable pretrained representations and multimodal models. PET/CT is a demanding test case because clinically meaningful interpretation must preserve quantitative molecular information while integrating anatomic CT,...

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Diagnosis-Related Group Payment Systems and Hospital Efficiency, Quality, and Coding Behavior: A Systematic Review and Narrative Synthesis

👤 Wael Alsebaei 📅 2026-10-24 📖 1 min 📝 66 words LOW VISION

Summary: Diagnosis-related group (DRG)-based payment systems are used globally to incentivize hospital efficiency, yet their impact on clinical quality and documentation behavior remains variable. As Saudi Arabia implements the Australian Refined DRGs (AR-DRG) framework under Vision 2030, a synthesis of inte...

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Clinical significance of "complex" multimorbidity in hospitalised adult patients

👤 Gino Roberto Corazza 📅 2026-10-24 📖 1 min 📝 65 words LOW VISION

Summary: The term "complex multimorbidity" lacks universal consensus and its relevance remains unclear. We aimed to assess the prevalence of complex multimorbidity and its association with major healthcare outcomes. This was a prospective, single-centre study (internal medicine ward). We collected sociodemog...

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Outcomes of subthreshold diode micropulse laser and anti-VEGF vs anti-VEGF monotherapy for diabetic macular edema: a meta-analysis

Marko M Popovic 2026-10-26 1 min 41 words LOW VISION

Summary: CONCLUSIONS: In patients with DME, SDMLP and anti-VEGF combination therapy appears comparable, and possibly modestly superior to anti-VEGF monotherapy, with a reduced injection burden over 12 months. BCVA and CMT differences were below clinically important thresholds and supported by low-certainty e...

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Tropospheric Ozone Assessment Report II: Ozone distribution and variability in the oceanic and polar regions

Alfonso Saiz-Lopez 2026-10-26 1 min 76 words LOW VISION

Summary: Oceanic and polar regions are critical for global atmospheric processes, but an evaluation of ozone distributions and trends in these regions has been missing. In this assessment, part of the TOAR-II project, we report lower atmosphere ozone concentrations over open oceans and polar regions: with da...

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Telemedicine-Enabled Medication Optimization for Adults with Polypharmacy: A Systematic Review and Implications for Saudi Vision 2030 Healthcare Transformation

👤 Mark Sim
📅 2026-10-26 📖 1 min 📝 32 words LOW VISION

Summary: CONCLUSIONS: Telemedicine-enabled medication optimization appears feasible, but evidence of clinical effectiveness and cost savings remains uncertain. Saudi implementation should begin with monitored pilots emphasizing safety, equity, Arabic-language accessibility, governance, and prospective evalua...

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Accurate Extraction of Phenotypic Parameters for DUS Testing of Oudemansiella raphanipes Based on Deep Learning

Linyi Li 2026-10-26 1 min 65 words LOW VISION

Summary: To enable the rapid and accurate acquisition of phenotypic parameters during distinctness, uniformity, and stability (DUS) testing of edible mushroom varieties, and to address the low efficiency, strong subjectivity, and insufficient standardization of conventional manual measurement methods, this s...

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Understanding decision making factors influencing referrals to vision rehabilitation services at a tertiary medical center

Christina L Thomas-Virnig 2026-10-26 1 min 74 words LOW VISION

Summary: Functional vision loss at any stage affects quality of life even before clinically quantified to be moderate or severe. Eye care providers as the gatekeepers to vision rehabilitation services often underestimate how small variations in functional vision loss can have a significant impact in a patien...

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Comparative longitudinal evaluation of DIMS, progressive addition and single-vision lenses on myopia progression, binocular and visual performance in Spanish children

José R Jiménez 2026-10-26 1 min 58 words LOW VISION

Summary: An 18-month prospective, non-randomized comparative observational study evaluated the effects of different optical corrections on myopia progression, binocular vision, and visual performance in Spanish children. Forty-five myopic participants aged 9-14 years were allocated to single-vision lenses (S...

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Will Artificial Intelligence Augment Global Health Inequalities?

Matthew J Burton 2026-10-26 1 min 66 words LOW VISION

Summary: AbstractArtificial intelligence (AI) has been promoted as a tool to reduce global health inequities by addressing workforce shortages and expanding access to specialist care. However, if AI development and validation are concentrated in high-income countries, innovation may disproportionately benefi...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux 📅 2026-07-18 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

Zhifei Xie
2026-10-24 1 min 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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In vitro carbapenem activity in Neisseria gonorrhoeae

Béatrice Berçot 2026-09-04 1 min 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan
📅 2026-09-09 📖 1 min 📝 57 words

BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Neuromodulatory interventions for Alzheimer's disease

Wandong Zhang 2026-10-24 1 min 60 words TDCS TACS TRNS

Summary: Alzheimer's disease (AD) and mild cognitive impairment (MCI) represent an escalating global public health challenge with major socioeconomic consequences. Although anti-amyloid monoclonal antibodies have introduced disease-modifying treatment options for early symptomatic AD, their benefits remain m...

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Disrupting objectification: transcranial Direct Current Stimulation (tDCS) enhances the inversion effect toward sexualized women

Arwen Stephens
2026-10-24 1 min 68 words TDCS TACS TRNS

Summary: Sexual objectification is associated with viewing women in an object-like manner, often indexed perceptually by a reduced inversion effect for sexualized women relative to men. Prior research has shown that this index can be attenuated through social manipulations; here we investigated whether it ca...

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Efficacy and Safety of Perioperative Transcranial Electrical Stimulation for Postoperative Pain and Related Outcomes: A Systematic Review and Meta-Analysis With Trial Sequential and Subgroup Analyses

👤 Qing Zhong 📅 2026-10-24 📖 1 min 📝 64 words TDCS TACS TRNS

Summary: CONCLUSIONS: Current evidence suggests that tES, as an adjuvant perioperative intervention, may alleviate postoperative pain, reduce opioid consumption, and decrease the incidence of postoperative delirium. However, the effect of tES on postoperative quality of recovery remains unclear. Considering ...

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Accelerated high-capacity 6 mA transcranial direct current stimulation for major depression

👤 Mark S George 📅 2026-10-25 📖 1 min 📝 14 words TDCS TACS TRNS

Summary: CONCLUSIONS: Accelerated 6 mA HC-tDCS was feasible, well tolerated, and warrants a sham-controlled trial.

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Rapid and robust effects of high-dose intermittent Transcranial Brain Stimulation (iTBS) on depression and suicidal ideation in adolescents with acute major depression: a 10-day randomised, controlled, single-blind trial

👤 Yutao Lan
📅 2026-10-26 📖 1 min 📝 56 words

TDCS TACS TRNS

Summary: CONCLUSION: Current research indicates that a 10-day course of high-dose iTBS is effective in alleviating depressive symptoms and suicidal ideation in adolescents with depression, and that the therapeutic effects persist for up to one month after the end of treatment. We also found that high-dose iT...

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Comparative Efficacy of Noninvasive Neuromodulation in Patients with Disorders of Consciousness: A Systematic Review and Network Meta-Analysis of Randomized Controlled Trials

👤 Quan Wei
📅 2026-10-26
📖 1 min
📝 64 words
TDCS TACS TRNS

Summary: CONCLUSIONS: HF-rTMS-M1 and RMNS plus HF-rTMS-DLPFC, RMNS, HF-rTMS-DLPFC, and tDCS have shown positive signals in patients with DoC. The duration of the disorder and underlying etiology may influence the effectiveness of different noninvasive neuromodulation techniques in DoC. Given the marked heter...

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Associations of Combined Transcranial Magnetic Stimulation and Transcranial Direct Current Stimulation With Psychological and Functional Outcomes After Stroke: A Retrospective Cohort Study

👤 Meng Huang
📅 2026-10-26 📖 1 min 📝 46 words

TDCS TACS TRNS

Summary: CONCLUSIONS: In this retrospective cohort, combined TMS and tDCS with standard rehabilitation was associated with lower depression and anxiety symptom scores and more favorable neurological and motor outcome trajectories than standard rehabilitation. Non-random treatment allocation and residual conf...

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A tool to evaluate the impact of lived experience involvement in research: the Brain and Genomics Hub Impact Log protocol

Sophie E Legge 2026-10-26 1 min 74 words TDCS TACS TRNS

Summary: CONCLUSION: The potential value of this research is to implement a novel tool and framework for facilitating, recording and evaluating LEI in complex mental health research, which can be adapted for wider use. Strengths in design are LE leadership and cross-cutting LE research integration, incorpora...

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Multimodal premotor stimulation during action observation transiently boosts corticospinal excitability and motor performance

Ishan Mahajan 2026-10-26 1 min 2 words TDCS TACS TRNS

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Combining Cognitive Training and Brain Stimulation: A Novel Approach to Address Disordered Eating and Antipsychotic-Induced Weight Gain in People With Severe Mental Illness

👤 Ulrike Schmidt 📅 2026-10-26 📖 1 min 📝 69 words TDCS TACS TRNS

Summary: CONCLUSION: The initiation of antipsychotic therapy is a critical clinical window for early intervention to mitigate the long-term metabolic and psychiatric consequences of antipsychotic-induced weight gain. Combining ABMT + tDCS is acceptable and may help reduce binge eating in individuals on antip...

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Local and network-level prefrontal reactivity to intermittent theta burst stimulation: a randomized, sham-controlled iTBS/fNIRS study

👤 Georg S Kranz 📅 2026-10-23 📖 1 min 📝 2 words FNIRS

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Correction-stage information leakage and target-input asymmetry can distort comparisons of fNIRS short-separation channel correction strategies

Esra Suzen 2026-10-23 1 min 16 words FNIRS

Summary: OBJECTIVE: We examined how evaluation design affects comparisons of functional near-infrared spectroscopy (fNIRS) short-separation correction strategies.

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The test-retest reliability of cortical activation and behavioral metrics in adolescent basketball players during dual-task postural control: an fNIRS-based assessment

Lulu Song 2026-10-24 1 min 59 words FNIRS

Summary: CONCLUSION: These findings provide preliminary test-retest reliability estimates for selected multimodal neurobehavioral measures in male adolescent basketball players although reliability alone does not establish clinical utility and estimates with wide confidence intervals require confirmation in ...

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How School-Based Interventions Shape the Developing Brain during 3-12 Years: A Systematic Review and Exploratory Meta-analysis

👤 Hui Li 📅 2026-10-24 📖 1 min 📝 56 words FNIRS

Summary: The effects of school-based interventions (SBIs) on children's neurodevelopment are poorly synthesized. This systematic review integrates neuroscientific evidence on how structured educational activities influence brain development in children aged 3-12 years. Following PRISMA guidelines, we identif...

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The Walking Brain: Mapping feature-specific cortical responses to naturally unfolding sensory experience in the real world

Meryem A Yücel 2026-10-25 1 min 58 words FNIRS

Summary: Understanding how the brain processes the everyday world is fundamental, yet laboratory paradigms exclude self-generated movement, unscripted events, and a continuously evolving multisensory environment. Wearable neuroimaging enables real-world measurement, but linking neural activity to uncontrolle...

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Functional near infrared spectroscopy-based dual-stream adaptive language mapping in adults with and without post-stroke aphasia

👤 Meryem Yücel
📅 2026-10-25 📖 1 min 📝 67 words FNIRS

Summary: In aphasia, variability in lesion location and post-stroke network reorganization complicates efforts to identify the regions that support semantic versus phonological processing. In prior work, researchers have often examined these domains in separate patient cohorts, further limiting direct compar...

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How Task Load Modulates the Effect of State Anxiety on Working Memory and Its Neural Mechanisms: Evidence from Verbal and Spatial Modalities

Senqing Qi 2026-10-25 1 min 73 words FNIRS

Summary: The impact of state anxiety on working memory (WM) and its neural mechanisms, especially the role of task load, remain uncertain. To address this issue, we conducted two experiments to examine how state anxiety affects WM under different task loads and whether these effects are consistent across ver...

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Prefrontal dysfunction and impulsive decision-making in female methamphetamine use disorder: a fNIRS study

Peng Lei 2026-10-26 1 min 67 words FNIRS

Summary: CONCLUSIONS: Women with MUD showed altered intertemporal choice and reduced prefrontal hemodynamic responses in regions implicated in value evaluation and cognitive control. These findings support the relevance of prefrontal dysfunction to impulsive decision-making in female MUD while also indicatin...

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Melanin Bias in Cerebral fNIRS: A Monte Carlo Study Toward Equitable Optical Brain Sensing

👤 Murad Althobaiti
📅 2026-10-26 📖 1 min 📝 63 words FNIRS

Summary: Functional near-infrared spectroscopy (fNIRS) is an optical biosensing modality that transduces cortical hemodynamics into a measurable near-infrared signal, underpinning wearable brain monitors and brain-computer interfaces (BCIs). Because epidermal melanin is a strong near-infrared absorber, devic...

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Professional Experience Moderates Behavioral Performance and Sensorimotor-Related Cortical Activity During Scoring Decision-Making in Aerobic Gymnastics Judges: An fNIRS Study

👤 Bin Shao 📅 2026-10-26 📖 1 min 📝 58 words FNIRS

Summary: Understanding how professional experience shapes cognitive and neural resource allocation across different task demands is important for elucidating expert decision-making. This study examines whether professional experience influences behavioral performance and sensorimotor-related cortical activit...

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Case Report: An MDT-guided multimodal protocol for large-area stroke plateau - integrating neuromodulation with multimodal assessment

Xiao-Xia Du 2026-10-24 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In this patient with large-area stroke who showed limited response to conventional rehabilitation, an MDT-guided, dynamically adjusted multimodal intervention was temporally associated with concurrent recovery across cognitive, language, motor, and daily living domains. The concomitant n...

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Correction: Perioperative management for invasive brain-computer interface implantation under surgical navigation in a patient with traumatic spinal cord injury: a case report

Wenyao Cui 2026-10-24 1 min 6 words BRAIN COMPUTER INTERFACE

Summary: [This corrects the article DOI: 10.3389/fnhum.2026.1892331.].

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Limits of Trial-Adaptive Neural-Language Fusion Across Large Language Models in P300 Brain-Computer Interfaces

Yiftach Barash 2026-10-24 1 min 21 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: A tuned fixed weight is a difficult-to-beat default across the tested scale range; reliability estimation gave no deployable adaptive advantage.

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Creating online cognitive assessment platform for children's literacy and math skills: a case study illustrated with the ACE kids platform

U Maurer 2026-10-24 1 min 70 words BRAIN COMPUTER INTERFACE

Summary: Computerized versions of paper-and-pencil tests made behavioral assessments of children more automatic and precise. The COVID-19 pandemic further demonstrated the need for assessments to be offered online to allow for remote data collection. However, the design and implementation of computerized tes...

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Combined neuromodulation techniques via a central-peripheral-central strategy facilitate functional recovery from AIS A to AIS C in a patient with complete spinal cord injury: A case report

Liansheng Zhong
2026-10-24 1 min 48 words

BRAIN COMPUTER INTERFACE

Summary: CONTEXT: Complete spinal cord injury (SCI) has traditionally been considered irreversible. However, accumulating evidence suggests that subclinical neural pathways may persist. Multimodal combined neuromodulation targeting brain-spinal-peripheral circuits holds promise to unlock latent neural plasti...

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Fabrication Strategies and Performance Optimization Strategies of Flexible Brain-Computer Interface Electrodes

Jianguo Tang 2026-10-25 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) provides a direct communication pathway between the brain and external devices and has been applied in neurological disease treatment, motor rehabilitation, and human-computer interaction. Electrodes play essential roles in recording and modulating neural signals, and ...

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Melanin Bias in Cerebral fNIRS: A Monte Carlo Study Toward Equitable Optical Brain Sensing

Murad Althobaiti 2026-10-26 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Functional near-infrared spectroscopy (fNIRS) is an optical biosensing modality that transduces cortical hemodynamics into a measurable near-infrared signal, underpinning wearable brain monitors and brain-computer interfaces (BCIs). Because epidermal melanin is a strong near-infrared absorber, devic...

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A Complementary Dual-Branch Receptive Field Network for Cross-Channel-Density Motor Imagery EEG Decoding

Zuguang Rao 2026-10-26 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) enable direct communication between the brain and external devices, providing a bio-inspired link between humans and artificial systems. However, electroencephalography (EEG)-based motor imagery (MI) decoding continues to pose challenges, due to limited temporal expl...

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A Biodegradable Fish-Derived Gelatin-Chitosan Sponge as a Novel Hemostatic Agent in a Rat Tooth Extraction Model

Dwi Ariawan 2026-10-26 1 min 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: These findings demonstrate the potential of gelatin-rich fish-derived gelatin-chitosan composite sponges as biodegradable hemostatic biomaterials for oral surgical applications.

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Beyond Overlap: Development and Validation of the Inclusion of AI in Self (IAIS) Scale and Its Moderating Role in the Relationship Between AI Identity and AI Threat

Xiang Yao 2026-10-26 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Artificial intelligence (AI) is becoming increasingly integrated into people's self-concepts, yet a psychometrically robust multi-item pictorial instrument for assessing the inclusion of AI in self (IAIS) is lacking. Drawing on self-expansion theory and the AI-mediated communication (AI-MC) model, w...

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The sound of manufactured music: Reviewing the role of artificial stimuli in music cognition research.

2024-01-22 1 min 259 words PSYCHOMUSICOLOGY

Summary: Having participants listen and react to musical stimuli is one of music cognition’s foundational methods. Whereas most researchers have used stimuli adapted from existing musical traditions in such work, others have incorporated artificial stimuli (i.e., stimuli generated specifically for research t...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000304

Music-evoked nostalgia and charitable giving: A cross-cultural study in the United States and Mexico.

📅 2024-01-22 📖 1 min 📝 192 words PSYCHOMUSICOLOGY

Summary: Nostalgia, a past-oriented emotion characterized by complex affective responses, is a pervasive and fundamental human experience. Prior research has demonstrated that nostalgia serves various socioemotional functions, such as promoting a sense of belonging, enhancing one's perception of meaning in l...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000302

Preferred music listening does not affect cognitive inhibition in young and older adults.

📅 2023-10-12 📖 1 min 📝 227 words PSYCHOMUSICOLOGY

Summary: Previous literature has found links between music listening and cognitive performance. Specifically, background music may play a role in modulating cognitive inhibition. However, determining what type of background music affects cognitive inhibition throughout the lifespan has not been studied. The ...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000300

Absolute pitch: A literature review of underlying factors, with special regard to music pedagogy.

📅 2023-07-10 📖 1 min 📝 202 words PSYCHOMUSICOLOGY

Summary: Absolute pitch (AP) is a fairly rare and special phenomenon that has relevance for musicology, psychology, genetics, and neuroscience. AP possessors are able to identify the pitch of an isolated sound or to produce that sound without a reference point. The authors' aim is to review the literature on...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000298

Capturing coordination and intentionality in joint musical improvisation.

📅 2023-08-03 📖 1 min 📝 217 words PSYCHOMUSICOLOGY

Summary: Humans collaborate with each other on a wide variety of tasks that are often largely improvised and unscripted. In this study, we investigated the dynamics of coordination in a joint musical improvisation task, what the effect of intentions is on coordination, and how musicians propagate these inten...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000299

Continuous affect responses to a large diverse set of unfamiliar music: Bayesian time-series and cluster analyses.

📅 2023-04-20 📖 1 min 📝 252 words PSYCHOMUSICOLOGY

Summary: Sixty-nine participants made continuous response judgments of perceived arousal and valence while listening to 30-s extracts of 100 unfamiliar pieces within a novel recommender system. Our purpose was to take advantage of the relatively large number of participants and pieces studied (compared with ...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000295

Monthly Updates [July]

📅 2026-07-01 📖 3 min 📝 640 words FMHY

Summary: <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   <p>Added FMHY Plugin to <a href="https://www.flowlau...

🔗 **Read full article:**
https://fmhy.net/posts/july-2026

US designates helium as a critical material

👤 rbanffy 📅 2026-10-26 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://apnews.com/article/energy-critical-materials-minerals-helium-43da160adc1bad389fba0b9ffbd14c3c Comments URL: <a href="https://news.ycombinator.com/item?id...

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The EPA Is Quietly Killing Its Web Archive

👤 robtherobber 📅 2026-10-26 📖 1 min 📝 13 words HACKER NEWS

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Magic state cultivation on a superconducting quantum processor

👤 kheal 📅 2026-10-26 📖 1 min 📝 145 words HACKER NEWS

> **Summary:** Abstract: "Fault-tolerant quantum computing requires a universal gate set, but the necessary non-Clifford gates represent a significant resource cost for most quantum error correction architectures. Magic state cultivation offers an efficient alternative to resource-intensive distillation protoco...

🔗 **Read full article:**
https://arxiv.org/abs/2512.13908

CorridorKey 2.0

👤 ravenical 📅 2026-10-26 📖 1 min 📝 13 words HACKER NEWS

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https://corridorkey.com/

Working with academic communities to build scientific understanding

👤 jonluca 📅 2026-10-26 📖 1 min 📝 13 words HACKER NEWS

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https://openai.com/index/building-scientific-understanding-with-academic-communities/

Cross-request KV cache exposure in the GLM-5.3 Workers AI model (155 responses)

👤 wondersky 📅 2026-10-26 📖 1 min 📝 13 words HACKER NEWS

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Evaluating Dr. Cuddy’s Claim that the Debunking of Power Posing is a Myth

noreply@blogger.com (Daniel Lakens)
2026-05-25
30 min
6008 words

TWENTY PERCENT STATISTICIAN

Summary: In this blog post I will analyse the arguments that Dr. Amy Cuddy provided in a LinkedIn post “The "Power Posing Was Debunked" Myth: What the Research Actually Shows — and Why Scientific Discourse Matters” on February 26. You can find the LinkedIn post here: <span lang="...

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http://daniellakens.blogspot.com/2026/05/evaluating-dr-cuddys-claim-that.html

Episode 29: Q&A with Dr. Anahita Kyani, Director of Neuroscience, Data Science and Digital Health at Johnson & Johnson Innovative Medicine

Adriel Carridice
2026-06-22
1 min
0 words

BRAIN

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Q&A with Felix Schwock, PhD student at University of Washington and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

👤 ieeebrain 📅 2026-06-22 📖 1 min 📝 0 words BRAIN

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Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor in Chief Search

👤 Deidre Artis 📅 2026-07-17 📖 1 min 📝 34 words EMBS

> **Summary:** Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor-in-Chief Search Submit your Application The post <a href="https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/">Transactions on Neural Syst...

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State-specific dynamic functional connectivity disturbances in spinal cord injury-related neuropathic pain

📖 1 min 📝 29 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Chuchu Sun, Haotian Xin, Ling Wang, Qian Chen, Yu Wang, Beining Yang, Xin Chen, Fang Li, Jie Lu, Nan Chen

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https://www.sciencedirect.com/science/article/pii/S0006899326004658?dgcid=rss_sd_all

EcoHIV Treated Primary Microglia Exhibit a Highly Specialized and Sex-Dependent Transcriptional State

👤 Veenstra, B. J., Nelson, K., Elder, T. R., Turner, J. R., McLaurin, K. A. 📅 2026-10-26 📖 1 min

📝 199 words BIORXIV NEUROSCIENCE

Summary: The pathogenesis of HIV-1-associated neurocognitive disorders (HAND) is partly driven by microglial dysregulation. Indeed, HIV-1-induced microglial dyshomeostasis is characterized by a sustained pro-inflammatory state and exacerbated microglia proliferation. Herein, in vitro studies were conducted i...

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A clinically identified Kirrel3 variant disrupts trans-synaptic circuit assembly

Shennib, O., Johnson, A., Raines, O., Williams, M. E.
2026-10-26 1 min 144 words

BIORXIV NEUROSCIENCE

Summary: Trans-synaptic adhesion molecules guide selective connections between neurons to establish circuit specificity. Yet, there are gaps in the understanding of how many synaptic adhesion molecules function in vivo and if clinically identified variants perturb their function. Using cell-type-specific del...

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Adeno-associated virus serotype tropism and transduction in murine dorsal root ganglia following direct injection.

ODonnell, M., Fontaine, A., Caldwell, J., Weir, R.
2026-10-26 1 min 196 words

BIORXIV NEUROSCIENCE

Summary: The peripheral nervous system (PNS) is highly heterogeneous, comprised of diverse subclasses of motor and sensory neurons. Optogenetics offers a high-resolution approach for interrogating neural circuits; however, optogenetic proteins must be selectively delivered to defined neuronal circuits in ord...

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Postsynaptic TAK1-GluN2B axis supports hippocampal synaptic integrity and cognitive function during aging

Liu, Y., Huang, M.-Q., Zhang, Y.-C., Jia, Z.-R., Min, X., Zhou, Z.-Y., Chen, J.-Y., Zong, F.-J., Gao, J., Xu, Y.-W., Wang, T., Zhang, Y.-Y., He, K.-W.

2026-10-26 1 min 225 words BIORXIV NEUROSCIENCE

Summary: Transforming growth factor {beta}-activated kinase 1 (TAK1) is a serine/threonine kinase best known for its roles in immune and stress-response signaling, yet its physiological function in mature neurons remains poorly understood. Here we identify neuronal TAK1 as a regulator of excitatory synaptic ...

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A molecularly defined recurrent CA3 circuit regulates network dynamics and spatial working memory

Cui, W., Li, H., Guo, W., Wang, D. Y., Bai, E., Jiang, Y.-Q., Sun, Q.

2026-10-26 1 min 145 words BIORXIV NEUROSCIENCE

Summary: The hippocampal CA3 region supports coordinated network activity and memory processes, but how molecularly distinct CA3 neuronal populations contribute to recurrent circuit function remains poorly understood. Here we identify St18 as a molecular marker of a distinct population of deep-layer, burst-f...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.25.761711v1?rss=1

Repetitive transcranial magnetic stimulation targeting motor cortical region induces cell-type-specific transcriptional reprogramming and behavioral recovery in non-human primate model of Rett syndrome

Li, W., Sun, W., Su, J., Yang, F., Qiu, X., Zhuo, Y., Yang, Z., Wang, Z., Chen, Z., Huang, T., Wang, H., Xing, H., Ji, W., Yuan, T.-F., Chen, Y., Wang, J.

2026-10-26 1 min 195 words BIORXIV NEUROSCIENCE

Summary: Rett syndrome (RTT), a severe neurodevelopmental disorder caused by mutations in the MECP2 gene, currently lacks effective treatments. The motor cortex is a pivotal hub in RTT and autism spectrum disorders, and its impairments contribute to motor, social, and cognitive deficits. Here we examined the...

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Unsupervised neuronal cell typing from connectivity by minimum description length

Schwartzman, G.

2026-10-26 1 min 149 words BIORXIV NEUROSCIENCE

Summary: Connectomes are being reconstructed at rapidly increasing scale, creating a growing need for automated methods to identify neuronal cell types. Existing approaches rely heavily on expert annotation or on reference connectomes, and reliable unsupervised typing remains an open problem. We present a fu...

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Women might have invented language — but not alone

📅 2026-10-26 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/d41586-026-03392-6

Targeting central thalamus with transcranial ultrasound reduces arousal in healthy volunteers

📅 2026-10-26 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-10637-7

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

Julia-Katharina Pfarr 2026-02-12 1 min 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

Robert Oostenveld 2026-02-12 1 min 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Tele-optometry for eye disease screening: a scoping review of context-dependent diagnostic performance and limitations for large-scale implementation

Chigozie Ekenze 2026-10-24 1 min 50 words LOW VISION

Summary: CONCLUSION: Tele-diagnostic screening shows promise for high-prevalence conditions with straightforward criteria, such as DR and refractive error, but is less reliable for complex diseases like glaucoma. These conclusions are constrained by limited downstream-outcome reporting and geographic concent...

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Early changes in the central vision and the macula in diabetic retinal disease: associations with serum biomarkers

Yoko Ozawa 2026-10-24 1 min 62 words LOW VISION

Summary: CONCLUSION: This study links subtle macular neural and vascular changes with systemic biomarkers in diabetic patients without visible macular lesions. Impaired central vision was associated with shorter PROS, lower RPE volume, and reduced vascular parameters. These findings improve understanding of ...

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Retinal Artificial Intelligence Reclassifies SCORE2-Based Cardiovascular Risk Across the ESC/EAS Statin-Indication Boundary: A UK Biobank Prospective Cohort Study

👤 Sunjin Hwang 📅 2026-10-24 📖 1 min 📝 28 words LOW VISION

Summary: CONCLUSION: Adding the retinal AI risk score to SCORE2 produced modest bidirectional risk reclassification across the ESC/EAS statin-indication boundary; these findings warrant prospective validation.Abstract word count: 250 words.

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Long-Term Visual, Refractive, and Patient-Reported Satisfaction of the Tel-Aviv Protocol (ePRK-CXL) for Progressive Keratoconus

👤 Igor Kaiserman 📅 2026-10-24 📖 1 min 📝 41 words LOW VISION

Summary: CONCLUSIONS: At a mean follow-up of ~8 years, the Tel-Aviv Protocol provided durable visual and refractive improvement, stabilized keratometry without progression, and yielded high patient satisfaction with low symptom burden. These results support ePRK-CXL as a tissue-sparing option for progressive...

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From task-specific AI to multimodal foundation models in PET/CT: a readiness framework for clinical translation

👤 Jia Wu 📅 2026-10-24 📖 1 min 📝 59 words LOW VISION

Summary: Artificial intelligence in PET/CT is evolving from narrowly optimized systems toward reusable pretrained representations and multimodal models. PET/CT is a demanding test case because clinically meaningful interpretation must preserve quantitative molecular information while integrating anatomic CT,...

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Diagnosis-Related Group Payment Systems and Hospital Efficiency, Quality, and Coding Behavior: A Systematic Review and Narrative Synthesis

👤 Wael Alsebaei 📅 2026-10-24 📖 1 min 📝 66 words LOW VISION

Summary: Diagnosis-related group (DRG)-based payment systems are used globally to incentivize hospital efficiency, yet their impact on clinical quality and documentation behavior remains variable. As Saudi Arabia implements the Australian Refined DRGs (AR-DRG) framework under Vision 2030, a synthesis of inte...

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Clinical significance of "complex" multimorbidity in hospitalised adult patients

Gino Roberto Corazza 2026-10-24 1 min 65 words LOW VISION

Summary: The term "complex multimorbidity" lacks universal consensus and its relevance remains unclear. We aimed to assess the prevalence of complex multimorbidity and its association with major healthcare outcomes. This was a prospective, single-centre study (internal medicine ward). We collected sociodemog...

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Outcomes of subthreshold diode micropulse laser and anti-VEGF vs anti-VEGF monotherapy for diabetic macular edema: a meta-analysis

Marko M Popovic 2026-10-26 1 min 41 words LOW VISION

Summary: CONCLUSIONS: In patients with DME, SDMLP and anti-VEGF combination therapy appears comparable, and possibly modestly superior to anti-VEGF monotherapy, with a reduced injection burden over 12 months. BCVA and CMT differences were below clinically important thresholds and supported by low-certainty e...

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Tropospheric Ozone Assessment Report II: Ozone distribution and variability in the oceanic and polar regions

👤 Alfonso Saiz-Lopez 📅 2026-10-26 📖 1 min 📝 76 words LOW VISION

Summary: Oceanic and polar regions are critical for global atmospheric processes, but an evaluation of ozone distributions and trends in these regions has been missing. In this assessment, part of the TOAR-II project, we report lower atmosphere ozone concentrations over open oceans and polar regions: with da...

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Telemedicine-Enabled Medication Optimization for Adults with Polypharmacy: A Systematic Review and Implications for Saudi Vision 2030 Healthcare Transformation

👤 Mark Sim 📅 2026-10-26 📖 1 min 📝 32 words LOW VISION

Summary: CONCLUSIONS: Telemedicine-enabled medication optimization appears feasible, but evidence of clinical effectiveness and cost savings remains uncertain. Saudi implementation should begin with monitored pilots emphasizing safety, equity, Arabic-language accessibility, governance, and prospective evalua...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux 📅 2026-07-18 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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In vitro carbapenem activity in Neisseria gonorrhoeae

Béatrice Berçot 2026-09-04 1 min 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan
📅 2026-09-09 📖 1 min 📝 57 words

BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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